



Department of Computer Science

Lab Manual

of

MACHINE LEARNING

MCA521-4

Class Name: IV MCA

Master of Computer Application

2026

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Verified by:

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Department Overview

Department of Computer Science of CHRIST (Deemed to be University) strives to shape outstanding computer professionals with ethical and human values to reshape nation's destiny. The training imparted aims to prepare young minds for the challenging opportunities in the IT industry with a global awareness rooted in the Indian soil, nourished and supported by experts in the field.

Vision

The Department of Computer Science endeavours to imbibe the vision of the University “**Excellence and Service**”. The department is committed to this philosophy which pervades every aspect and functioning of the department.

Mission

“To develop IT professionals with ethical and human values”. To accomplish our mission, the department encourages students to apply their acquired knowledge and skills towards professional achievements in their career. The department also moulds the students to be socially responsible and ethically sound.

Introduction to the Programme

Master of Computer Applications (MCA) is a post-graduate programme of CHRIST (Deemed to be University) and approved by the All India Council of Technical Education (AICTE). It is a two-year programme spread over six trimesters to bridge the chasm between computer studies and applications, bringing within reach of enthusiastic youngsters an excellent group of experienced and dedicated staff and experts, and an enviable infrastructure. MCA at CHRIST (Deemed to be University) strives to shape outstanding computer professionals with ethical and human values to reshape the nation's destiny. The training programme aims to prepare young minds for challenging opportunities in the IT industry with a global awareness rooted in the Indian soil, nourished and supported by experts in the field. The Department is committed to the motto “Excellence and Service,” and this philosophy pervades every aspect of its functioning.

Programme Objective

This programme is conceptualised and designed based on the strong commitment of the department to provide better quality education to the students. The principal objectives of this course are to:

- Heighten technological awareness
- Train future industry specialists
- Encourage effective software development
- Provide research training
- Involve students in innovative research
- Produce graduates with a two-year professional education in Computer Science with technical, professional, and communications skills.

Ethics and Human Values

1. Only proprietary or open-source software would be used for academic teaching and learning purposes.
2. Copying of programs from the internet, friends or from other sources is strictly discouraged since it impairs development of programming skills.
3. Unique Practical (Domain based) exercises ensure that the students don't get involved in code plagiarism.
4. Projects undertaken by students during the course are done in teams to improve collaborative work and synergy between team members.
5. Projects involve modularization which initiates students to take individual responsibility for common goals.
6. Passion for excellence is promoted among the students, be it in software development or project documentation.
7. Giving due credit to sources during the seminar and research assignment is promoted among the students
8. The course and its design enforce the practice of good referencing technique to improve the sense of integrity.
9. Courses involving group discussions and debates on ethical practices and human values are designed to sensitize the students in dealing with customers and members within the Organization.

Programme Outcomes:

- P01: Foundation Knowledge: Apply knowledge of mathematics, programming logic, and coding fundamentals for solution architecture and problem-solving.
- P02: Problem Analysis: Identify, review, formulate, and analyse problems primarily focussing on customer requirements using critical thinking frameworks.
- P03: Development of Solutions: Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.
- P04: Modern Tool Usage: Select, adapt, and apply modern computational tools such as the development of algorithms with an understanding of the limitations including human biases.
- P05: Individual and Teamwork: Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.
- P06: Project Management and Finance: Use the principles of project management such as scheduling, and work breakdown structure and be conversant with the principles of Finance for profitable project management.
- P07: Ethics: Commit to professional ethics in managing software projects with financial aspects.
- Learn to use new technologies for cyber security and insulate customers from malware
- P08: Life-long learning: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

Programme Educational Objectives(PEO's)

- PEO1: Develop innovative and effective software solutions through research that addresses real-world challenges, grounded in a commitment to continuous learning and adaptability.
- PEO2: Advance the knowledge and practices in the field of computer applications through lifelong learning and rigorous research, supporting professional growth and academic excellence.
- PEO3: Exhibit ethical leadership, social responsibility to contribute and enhance community well-being by innovating solutions.

MCA521-4 – MACHINE LEARNING

Course Name: MACHINE LEARNING Code: MCA521-4

Total Hours: 75

L + P + T: 3 + 4 + 0

Max Marks: 100

Credits: 4

Course Type: Core Course

Course Category: Theo&Prac

Course Description

Machine Learning is a key area in computer applications that focuses on building models capable of making predictions or informed decisions based on data. A strong understanding of machine learning enables students to analyze and interpret complex datasets, develop predictive models, and extract meaningful insights. This course covers a wide range of machine learning concepts and techniques, including supervised and unsupervised learning. It provides a solid theoretical foundation while emphasizing practical, hands-on experience with algorithms and real-world data. The course equips students with the knowledge and skills required to apply machine learning techniques across various domains using modern tools and methodologies.

Course Objectives

This course enables students to understand the fundamental concepts of Machine Learning, including its core principles, terminology, and methodologies. Students will learn to differentiate between various types of learning algorithms such as supervised, unsupervised, and reinforcement learning, and recognize their appropriate applications.

Course Outcomes

After successful completion of the course, students will be able to:

- Explain the fundamental principles and scope of Machine Learning
- Implement modelling practices in machine learning for better generalisation
- Interpret predictive modelling results and performance of supervised machine learning techniques
- Assess the outcomes and effects of unsupervised machine learning processes
- Deploy robust machine learning models for interdisciplinary applications

Unit-1: INTRODUCTION TO MACHINE LEARNING Teaching Hours: 15 Hours

Introduction to AI – Knowledge Representation – Data – Representation of Data, Data Processing, Data Storage, Data Processing, Types of Data – Exploratory Data Analysis, Visualisations and Interpretation, Outliers

Machine Learning: Definition, Objectives, Components, Features, Applications – Traditional Programming v/s Machine Learning, Types of Machine Learning – Predictive Models, Statistical Inferences – Applications of Machine Learning

Challenges in ML: Insufficient Quantity of Data, Non-representative and Poor-Quality Data, Irrelevant Features, Estimation Estimation, Reducing Human Biases in Model Building, Ethical Use of Technology

Lab Exercises:

1. Data Preprocessing and visualisation
2. Data exploration and inferences

Unit-2: PROCESSES IN MACHINE LEARNING Teaching Hours: 15 Hours

Data Preparation and Model Selection: Effect of Data Preparation: Encoding, Nominal and Ordinal Representation, Feature Transformation and Feature Engineering, Generalisation, Normalisation, Sampling, Using Saved Weights, Model Selection Strategies: Overfitting, Underfitting, Bias-Variance Trade-off, Testing and Validation: Performance measures - Confusion matrix, Precision-Recall Trade off, F1 Score, ROC Curve, AUC, Error Analysis, Model Parameters, Hyperparameters, Hyperparameter Tuning, Cross Validation, Decision Functions: Introduction to Supervised Machine Learning Methods, Decision Functions, Decision Thresholds, Distance Measures, Hyperplanes, Regression & Classification Problems, Loss and Cost Functions, Linear Regression using OLS, Multi-class vs Multi-label, KNN Classification

Lab Exercises:

3. Saving weights of a generalised model.
4. Regression and Classification Evaluation Metrics: Illustration through Linear Regression and KNN Classification

Unit-3: SUPERVISED LEARNING ALGORITHMS**Teaching Hours: 15****Hours**

Learning through Optimisation: Gradient Descent, Linear Regression, Logistic Regression
Computational Learning: Decision Trees, Naive-bayes classification, Support Vector Machines, Ensemble Learners: Learning, Voting, Bagging, Stacking, Boosting, Random Forests, GridSearchCV

Lab Exercises:

5. Regression through Gradient Descent
6. Comparison of Different Classifiers
7. Illustration of Ensemble Learning Methods

Unit-4: UNSUPERVISED LEARNING AND DIMENSIONALITY REDUCTION Teaching Hours: 15 Hours

Introduction: Types of Unsupervised Learning, Challenges in Unsupervised Learning, Concept of Cost Functions in Unsupervised Learning, Clustering Methods: Distance based, Centroid Based, Elbow method to find Optimal Number of Clusters, Silhouette Method, K-Means Clustering, Hierarchical Clustering: Agglomerative and Bottom Up, Applying Supervised Learning after Clustering, Dimensionality Reduction Methods: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA)

Lab Exercises:

8. KMeans and Elbow Methods
9. Hierarchical Clustering and Dendrograms
10. Dimensionality Reduction
11. Image Compression using Dimensionality Reduction.

Unit-5 Teaching Hours: 15 Hours**MACHINE LEARNING IN REAL-WORLD**

Emerging Techniques: Role of ML in attaining Sustainable Development Goals, Time Series Preprocessing, Blackbox Methods: Neural Networks & Deep Learning, Reinforcement Learning/QLearning, Self Supervised Learning, Quantum Machine Learning, Keras and Tensorflow for designing Multi Layer Perceptron. Case Studies: CNN, RNN, Advances in Deep Learning, Natural Language Processing, Real Time Systems, Computer Vision, Robotics, Expert Systems, Generative Networks, Large Language Models. Model Deployment: Model Deployment: Connecting Models to User Interface, Deployment of ML Models.

Lab Exercises:

12. Training a Multi-layer perceptron

13. Deploying Trained ML-models

Text and Reference Books

[1] Campesato, O. (2020). Artificial Intelligence, Machine Learning and Deep Learning. Mercury Learning and Information.

[2] Andreas C. Müller and Sarah Guido (2017), "Introduction to Machine Learning with Python A Guide For Data Scientists" O'Reilly book

[3] Ethem Alpaydin (2005), "Introduction to Machine Learning", Prentice Hall of India

[4] Max Kuhn and Kjell Johnson (2018), "Applied Predictive Modelling", Springer

Essential Reading

[1] Stuart Russell and Peter Norvig(2020), "Artificial Intelligence: A Modern Approach" (4th Edition), Pearson

[2] Aurelien Geron(2019), "Hands on Machine Learning with Scikit-learn, Keras and Tensorflow", 2nd Edition, O'Reilly Books

[3] Kevin P. Murphy(2012), "Machine Learning: A Probabilistic Perspective", MIT Press

[4] Hastie, Tibshirani, Friedman(2008), "The Elements of Statistical Learning" (2nd ed), Springer

[5] Ian Goodfellow, Aaron Courville, Yoshua Bengio (2019), "Deep Learning (Adaptive Computation And Machine Learning Series)", MIT Press

Additional Information (Web Resources):

1. Andrew Ng, Deeplearning.ai, Machine Learning Specialisation, offered through Coursera:

<https://www.deeplearning.ai/courses/machine-learning-specialization/>

Evaluation Pattern: CIA - 50% ESE - 50%

LIST OF PROGRAMS

MCA-B

Sl. no	Title of lab Experiment	Page number	RB T	CO
1	India Air Quality & Crop Yield — EDA Lab Data Preprocessing, Visualisation & Exploration Combined Lab Sheet	8	L3	CO1, 3
2	India Air Quality & Crop Yield — EDA Lab Data Preprocessing, Visualisation & Exploration Combined Lab Sheet	8	L3	CO2, 3
3	Student Survey: Simple Linear Regression using Scikit Learn, OLS, and Parameter Saving	46	L3	CO3
4	Regression and Classification Evaluation Metrics	76	L3	CO3
5	Linear Regression through Gradient Descent	108	L3	CO3
6	Comparison of Logistic Regression and K Nearest Neighbors (KNN) Classifiers	122	L3	CO3, 4
7	Lab Activity: Decision Tree	132	L4	CO3
8	Lab 8 : Implementation and Performance Evaluation of Categorical Naive Bayes Classifier		L3	CO3
9	Lab 9: Support Vector Machine and principal Component Analysis		L3	CO4

10	Lab 10 Learning the XOR Boolean Function Using a MLP		L5	CO4
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Evaluation Rubrics:

Evaluation Rubrics for each program would be

Submission Guidelines:

- Make a copy of the lab manual template with your <name_reg:no_subject name > ,
- Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as lab number.
- Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
- Create a Git Repository in your profile <ML lab-reg no> . Follow a different branch for each lab <Lab 1, Lab 2...>, and push the code to Git. The link should be provided in Google Classroom along with the PDF of the lab manual.
- Upload the PDF to Google Classroom before the deadline.

Lab 1

Lab Exercise I:

Aim

- To investigate whether worsening air quality across Indian states is linked to declining agricultural output by independently choosing, applying, and justifying appropriate data analysis and visualisation techniques on raw, messy, real-world datasets.
-

Evaluation Rubrics

Submission Guidelines

1. Make a copy of the lab manual template with your <name_reg:no_subject name>
2. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
3. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
4. Create a **Git repository** in your profile <ML lab-reg no>. Follow a different branch for each lab <Lab 1, Lab 2 ...> and push the code to Git.
5. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
6. Upload the PDF to Google Classroom before the deadline.

Code with Results

TASK 1:

A junior analyst hands you two CSV files and says — "I have no idea what's in these. We need to use them to build a machine learning model next week." Before any analysis can begin, you need to understand what you are working with.

Investigate both files thoroughly. Produce a structured summary that gives a complete first-impression picture of the data — its size, its contents, and where it might be problematic. Decide what information a data scientist would need before trusting this data, and make sure your summary covers all of it.

A structured data profile for each file in a markdown cell

At least one written observation about what concerns you after the inspection

Think: What does a data scientist need to know about a dataset before using it? What functions help you find that?

CODE:

```
import pandas as pd
import numpy as np

city_df = pd.read_csv("city_day.csv")
crop_df = pd.read_csv("crop_production.csv")

def data_profile(df, dataset_name):
    print("\n" + "="*80)
    print(f"DATA PROFILE : {dataset_name}")
    print("="*80)

    print("\n1. Dataset Shape")
    print(f"Rows      : {df.shape[0]}")
    print(f"Columns   : {df.shape[1]}")
```

```
print("\n2. Data Types")
print(df.dtypes)

print("\n3. First 5 Records")
print(df.head())

print("\n4. Missing Values")
missing = pd.DataFrame({
    'Missing_Count': df.isnull().sum(),
    'Missing_%': round((df.isnull().sum()/len(df))*100,2)
}).sort_values(by='Missing_%', ascending=False)

print(missing[missing['Missing_Count'] > 0])

print("\n6. Duplicate Rows")
print(df.duplicated().sum())

print("\n7. Numerical Summary")
print(df.describe())

print("\n8. Categorical Summary")
print(df.describe(include='object'))

print("\n9. Unique Values")
print(df.nunique())

# Run Profiles
data_profile(city_df, "City Air Quality Dataset")
data_profile(crop_df, "Crop Production Dataset")
```

RESULT

```
=====
DATA PROFILE : City Air Quality Dataset
=====
```

1. Dataset Shape

```
Rows      : 29531
Columns   : 16
```

2. Data Types

```
City      object
Date      object
PM2.5     float64
PM10      float64
NO        float64
NO2       float64
NOx       float64
NH3       float64
CO        float64
SO2       float64
O3        float64
Benzene   float64
Toluene   float64
Xylene    float64
AQI       float64
AQI_Bucket object
dtype: object
```

3. First 5 Records

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	\
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN	1.70	18.59	
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN	22.10	39.33	

	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	133.36	0.00	0.02	0.00	NaN	NaN
1	34.06	3.68	5.50	3.77	NaN	NaN
2	30.70	6.80	16.40	2.25	NaN	NaN
3	36.08	4.43	10.14	1.00	NaN	NaN
4	39.31	7.01	18.89	2.78	NaN	NaN

4. Missing Values

	Missing_Count	Missing_%
Xylene	18109	61.32
PM10	11140	37.72
NH3	10328	34.97
Toluene	8041	27.23
Benzene	5623	19.04
AQI	4681	15.85
AQI_Bucket	4681	15.85
PM2.5	4598	15.57
NOx	4185	14.17
O3	4022	13.62
SO2	3854	13.05
NO2	3585	12.14
NO	3582	12.13
CO	2059	6.97

6. Duplicate Rows

0

8. Categorical Summary

	City	Date	AQI_Bucket
count	29531	29531	24850
unique	26	2009	6
top	Ahmedabad	2020-06-26	Moderate
freq	2009	26	8829

9. Unique Values

City	26
Date	2009
PM2.5	11716
PM10	12571
NO	5776
NO2	7404
NOx	8156
NH3	5922
CO	1779
SO2	4761
O3	7699
Benzene	1873
Toluene	3608
Xylene	1561
AQI	829
AQI_Bucket	6

dtype: int64

7. Numerical Summary

	Crop_Year	Area	Production
count	246091.000000	2.460910e+05	2.423610e+05
mean	2005.643018	1.200282e+04	5.825034e+05
std	4.952164	5.052340e+04	1.706581e+07
min	1997.000000	4.000000e-02	0.000000e+00
25%	2002.000000	8.000000e+01	8.800000e+01
50%	2006.000000	5.820000e+02	7.290000e+02
75%	2010.000000	4.392000e+03	7.023000e+03
max	2015.000000	8.580100e+06	1.250800e+09

8. Categorical Summary

	State_Name	District_Name	Season	Crop
count	246091	246091	246091	246091
unique	33	646	6	124
top	Uttar Pradesh	BIJAPUR	Kharif	Rice
freq	33306	945	95951	15104

TASK 2:

After the inspection, it is clear that several columns have missing values. A colleague suggests simply deleting all rows with any null value. Another suggests filling everything with the mean. You are not sure either approach is right.

Devise your own missing value treatment strategy. For each column with missing data, decide whether to drop it, drop affected rows, or impute — and explain why that choice is appropriate for that specific column. Apply your strategy and verify it worked.

A markdown cell that justifies each decision column by column
Before and after null counts as evidence the treatment worked.

```
city_df.drop(columns=['Xylene'], inplace=True)

num_cols = [
    'PM2.5', 'PM10', 'NO', 'NO2', 'NOx',
    'NH3', 'CO', 'SO2', 'O3',
    'Benzene', 'Toluene', 'AQI'
]

for col in num_cols:
    city_df[col].fillna(city_df[col].median(), inplace=True)

city_df['AQI_Bucket'].fillna(
    city_df['AQI_Bucket'].mode()[0],
    inplace=True
)

print("Missing Values After Treatment")
print(city_df.isnull().sum())

print("\nTotal Missing Values:",
```

```
city_df.isnull().sum().sum()
```

RESULT:

```
Missing Values After Treatment
City          0
Date          0
PM2.5         0
PM10          0
NO            0
NO2           0
NOx           0
NH3           0
CO            0
SO2           0
O3            0
Benzene       0
Toluene       0
AQI           0
AQI_Bucket    0
dtype: int64
```

TASK 3:

You need to combine both files later in the lab using the State column as the common key. But when you look closely, the same state appears with different spellings, extra spaces, or old names across the two files. There are also rows that appear more than once for no clear reason.

Make both files agree on state names and remove any redundant records. Document every inconsistency you found and how you resolved it. Show that the files are now ready to be merged reliably.

A list of all inconsistencies found and the fix applied for each
Record counts before and after to prove duplicates were removed

```
print("Before Treatment")
print(crop_df.isnull().sum())

# -----
# Remove rows with missing Production
# -----

crop_df = crop_df.dropna(subset=['Production'])

# -----
# Verification
# -----

print("\nAfter Treatment")
print(crop_df.isnull().sum())

print("\nTotal Missing Values:",
      crop_df.isnull().sum().sum())
```

RESULT:

```
Before Treatment
State_Name      0
District_Name   0
Crop_Year       0
Season         0
Crop           0
Area           0
Production     3730
dtype: int64
```

```
After Treatment
State_Name      0
District_Name   0
Crop_Year       0
Season         0
Crop           0
Area           0
Production     0
dtype: int64
```

```
Total Missing Values: 0
```

```
print("Unique States:")
print(sorted(crop_df['State_Name'].unique()))
```

Result: ['Bihar', 'Chandigarh', 'Chhattisgarh', 'Dadra and Nagar Haveli', 'Goa', 'Gujarat', 'Haryana', 'Himachal Pradesh', 'Jammu and Kashmir ', 'Jharkhand', 'Karnataka', 'Kerala', 'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya', 'Mizoram', 'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan', 'Sikkim', 'Tamil Nadu', 'Telangana ', 'Tripura', 'Uttar Pradesh', 'Uttarakhand', 'West Bengal']

```
def clean_state_name(state):

    if pd.isna(state):

        return state

    state = str(state).strip().upper()
```

```

replacements = {
    'TAMILNADU': 'TAMIL NADU',
    'TAMIL NADU ': 'TAMIL NADU',
    'PONDICHERRY': 'PUDUCHERRY',
    'ORISSA': 'ODISHA',
    'UTTARANCHAL': 'UTTARAKHAND',
    'ANDAMAN AND NICOBAR ISLANDS':
        'ANDAMAN & NICOBAR ISLANDS'
}

return replacements.get(state, state)

crop_df['State_Name'] = crop_df['State_Name'].apply(clean_state_name)

```

```

states_before = sorted(
    pd.read_csv("crop_production.csv")
    ['State_Name']
    .dropna()
    .unique()
)

print(states_before)

states_after = sorted(
    crop_df['State_Name']
    .dropna()
    .unique()
)

```

```
print(states_after)
```

Result: ['Andaman and Nicobar Islands', 'Andhra Pradesh', 'Arunachal Pradesh', 'Assam', 'Bihar', 'Chandigarh', 'Chhattisgarh', 'Dadra and Nagar Haveli', 'Goa', 'Gujarat', 'Haryana', 'Himachal Pradesh', 'Jammu and Kashmir ', 'Jharkhand', 'Karnataka', 'Kerala', 'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya', 'Mizoram', 'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan', 'Sikkim', 'Tamil Nadu', 'Telangana ', 'Tripura', 'Uttar Pradesh', 'Uttarakhand', 'West Bengal']

```
['ANDAMAN & NICOBAR ISLANDS', 'ANDHRA PRADESH', 'ARUNACHAL PRADESH',
'ASSAM', 'BIHAR', 'CHANDIGARH', 'CHHATTISGARH', 'DADRA AND NAGAR
HAVELI', 'GOA', 'GUJARAT', 'HARYANA', 'HIMACHAL PRADESH', 'JAMMU AND
KASHMIR', 'JHARKHAND', 'KARNATAKA', 'KERALA', 'MADHYA PRADESH',
'MAHARASHTRA', 'MANIPUR', 'MEGHALAYA', 'MIZORAM', 'NAGALAND', 'ODISHA',
'PUDUCHERRY', 'PUNJAB', 'RAJASTHAN', 'SIKKIM', 'TAMIL NADU',
'TELANGANA', 'TRIPURA', 'UTTAR PRADESH', 'UTTARAKHAND', 'WEST BENGAL']
```

```
before_crop = len(crop_df)

crop_df = crop_df.drop_duplicates()

after_crop = len(crop_df)

print("Crop Dataset")
print("Before:", before_crop)
print("After :", after_crop)
print("Duplicates Removed:",
      before_crop - after_crop)

city_df = pd.read_csv("city_day.csv")

before_city = len(city_df)
```

```
city_df = city_df.drop_duplicates()

after_city = len(city_df)

print("City Dataset")
print("Before:", before_city)
print("After :", after_city)
print("Duplicates Removed:",
      before_city - after_city)

print("City duplicates:",
      city_df.duplicated().sum())

print("Crop duplicates:",
      crop_df.duplicated().sum())

city_df.to_csv(
    "city_day_cleaned.csv",
    index=False
)

crop_df.to_csv(
    "crop_production_cleaned.csv",
    index=False
)
```

Result:

```
Crop Dataset
Before: 246091
After : 246091
Duplicates Removed: 0
City Dataset
Before: 29531
After : 29531
Duplicates Removed: 0
City duplicates: 0
Crop duplicates: 0
```

TASK 4:

The pollution control board wants to know — are most Indian cities moderately polluted, or is the problem concentrated in just a few places? They also want to know whether the average AQI is a fair number to report publicly, or whether extreme cities are pulling it up unfairly.

Investigate the shape of the AQI distribution. Decide which visualisation(s) best answer both questions the board is asking, justify your choice, produce the plot(s), and record two specific observations that directly address their concerns.

Your chosen plot(s) with fully labelled axes and a descriptive title

A markdown cell with your justification for the chosen plot type and two observations

Think: Which plot shows where values cluster? Which one reveals extreme values? Do you need one plot or two to answer both questions?

```
import pandas as pd
import matplotlib.pyplot as plt

# Remove missing AQI values
aqi = city_df['AQI'].dropna()

# Create figure
fig, axes = plt.subplots(1, 2, figsize=(14,5))

# Histogram
axes[0].hist(aqi, bins=30)

axes[0].set_title(
    'Distribution of Air Quality Index (AQI) in Indian Cities'
)

axes[0].set_xlabel('AQI')
axes[0].set_ylabel('Frequency')
```

```

# Boxplot
axes[1].boxplot(aqi)

axes[1].set_title(
    'Box Plot of AQI Showing Extreme Pollution Events'
)

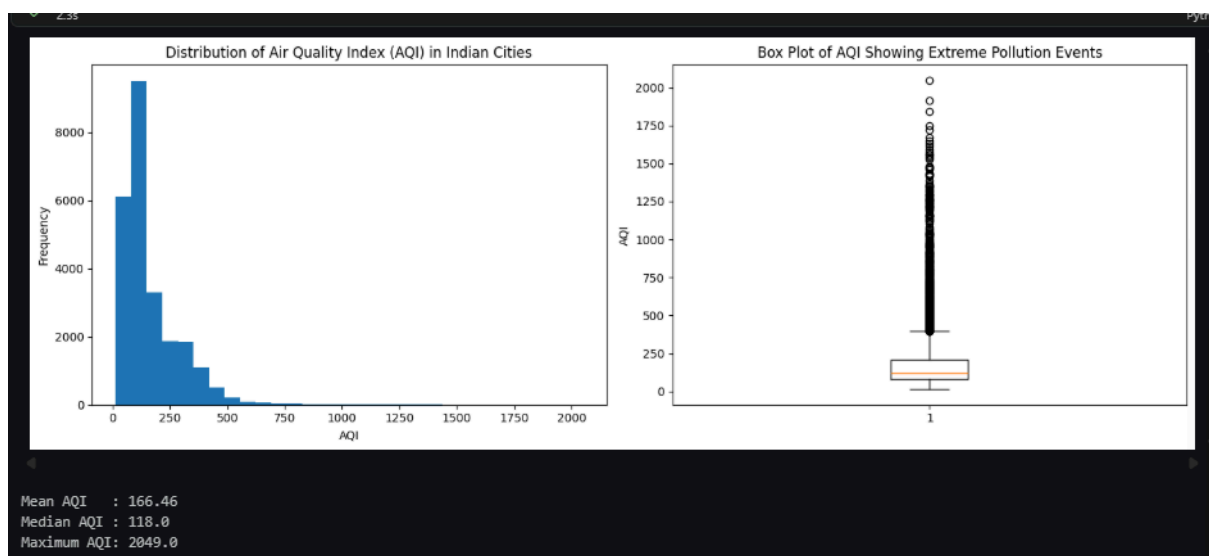
axes[1].set_ylabel('AQI')

plt.tight_layout()
plt.show()

print("Mean AQI   :", round(aqi.mean(),2))
print("Median AQI :", round(aqi.median(),2))
print("Maximum AQI:", round(aqi.max(),2))

```

RESULT:



TASK 5:

A data quality report flags that a few AQI readings in the dataset are implausibly high — values that no monitoring station should realistically record. If left in, these values will distort every statistic and model built on this data. The team lead asks you to "handle the extreme values properly" but does not tell you how.

Identify whether extreme values exist, quantify them, decide on an appropriate treatment method, apply it, and demonstrate that the data is now cleaner. Justify every decision you make.

The method you chose to detect extremes and why

The count of values affected and the treatment applied

A visual comparison of the data before and after treatment

```
# Remove missing AQI values
aqi = city_df['AQI'].dropna()

# Detect Outliers using IQR
Q1 = aqi.quantile(0.25)
Q3 = aqi.quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)

print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)

# Count outliers
outliers = city_df[
```

```
(city_df['AQI'] < lower_bound) |  
(city_df['AQI'] > upper_bound)  
]  
  
print("Number of Outliers:", len(outliers))  
  
# Winsorization  
  
city_df['AQI_Cleaned'] = city_df['AQI'].copy()  
  
city_df['AQI_Cleaned'] = city_df['AQI_Cleaned'].clip(  
    lower=lower_bound,  
    upper=upper_bound  
)  
  
# Count affected values  
affected = (  
    city_df['AQI'] != city_df['AQI_Cleaned']  
) .sum()  
  
print("Values Treated:", affected)  
  
# Visual Comparison  
  
fig, axes = plt.subplots(1,2, figsize=(14,5))  
  
# Before  
axes[0].boxplot(city_df['AQI'].dropna())  
axes[0].set_title("AQI Before Outlier Treatment")  
axes[0].set_ylabel("AQI")
```

```
# After

axes[1].boxplot(city_df['AQI_Cleaned'].dropna())
axes[1].set_title("AQI After Outlier Treatment")
axes[1].set_ylabel("AQI")

plt.tight_layout()
plt.show()

# Summary Statistics

print("\nBefore Treatment")
print(city_df['AQI'].describe())

print("\nAfter Treatment")
print(city_df['AQI_Cleaned'].describe())
```

RESULT:

TASK 6:

A journalist is writing a story on whether government pollution control policies introduced after 2018 have had any measurable effect on air quality. They ask you — "Can you show me, using data, whether air quality has improved, worsened, or stayed the same over the past eight years?"

Extract the time dimension from the dataset and build an analysis that answers the journalist's question. Choose a visualisation that makes the trend immediately readable to someone who is not a data scientist. Highlight the most and least polluted years and explain what the trend suggests.

Your plot with the trend clearly visible and key years highlighted

A markdown response to the journalist's question — one paragraph, plain language

Think: What needs to happen to the Date column before you can group by year? Which plot type communicates a trend over time most clearly?

```
# Convert Date column to datetime
city_df['Date'] = pd.to_datetime(city_df['Date'])

# Extract year
city_df['Year'] = city_df['Date'].dt.year

# Calculate yearly average AQI
yearly_aqi = city_df.groupby('Year')['AQI'].mean().reset_index()

print(yearly_aqi)

# Plot trend
plt.figure(figsize=(10,6))

plt.plot(
    yearly_aqi['Year'],
    yearly_aqi['AQI'],
    marker='o',
    linewidth=2
```

```
)

# Highlight first and last year
plt.scatter(
    yearly_aqi['Year'].iloc[0],
    yearly_aqi['AQI'].iloc[0],
    s=100,
    label='Start Year'
)

plt.scatter(
    yearly_aqi['Year'].iloc[-1],
    yearly_aqi['AQI'].iloc[-1],
    s=100,
    label='Latest Year'
)

plt.title(
    'Average Air Quality Index (AQI) in Indian Cities (2015-2020)'
)

plt.xlabel('Year')
plt.ylabel('Average AQI')

plt.grid(True, alpha=0.3)
plt.legend()

plt.show()

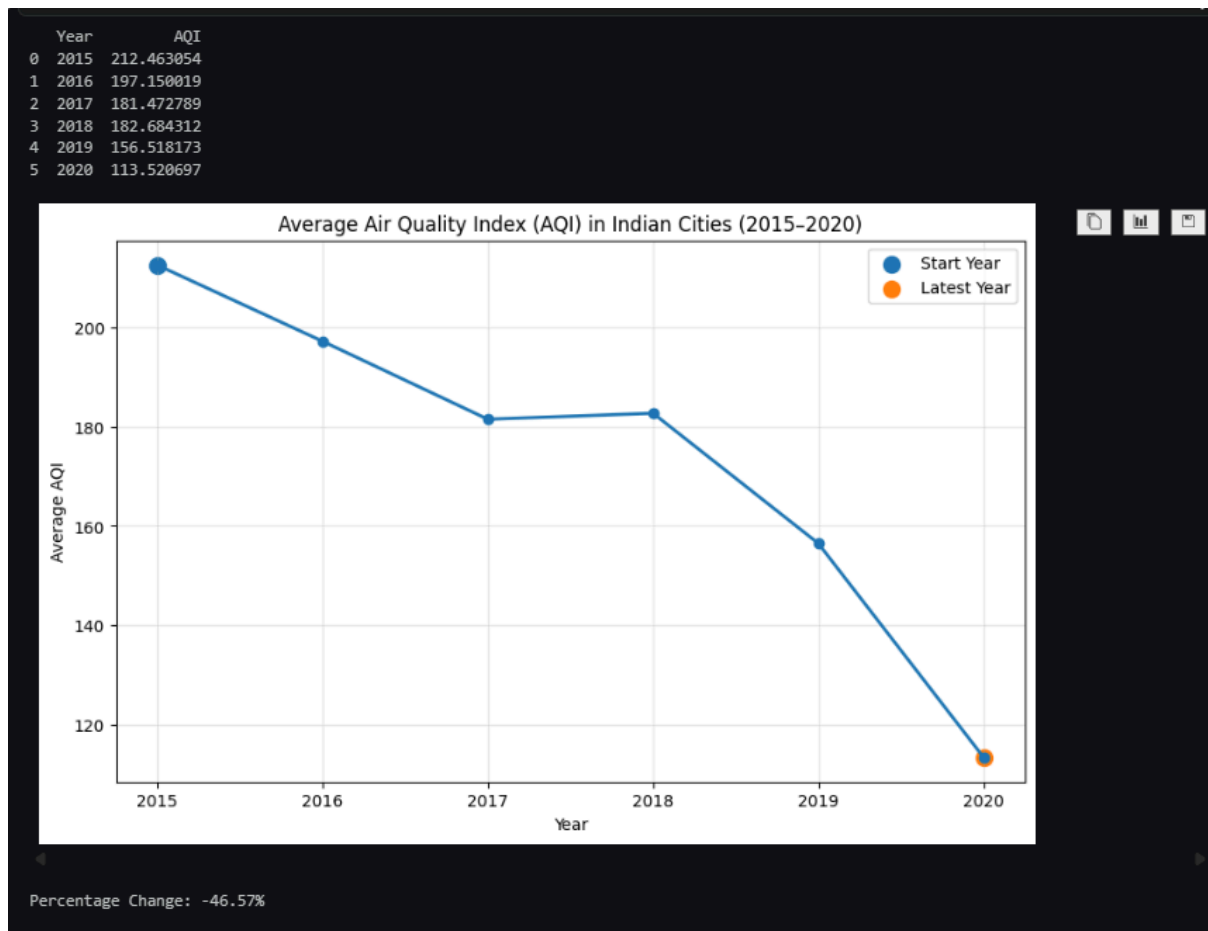
# Calculate percentage change
```

```
start_aqi = yearly_aqi['AQI'].iloc[0]
end_aqi = yearly_aqi['AQI'].iloc[-1]

change = ((end_aqi - start_aqi) / start_aqi) * 100

print(f"Percentage Change: {change:.2f}%")
```

RESULT:



TASK 7:

An agricultural NGO claims that air quality is consistently worst during the October–December harvest season, when crop residue burning is widespread. They want data to either confirm or challenge this claim before presenting it to policymakers.

Investigate whether AQI follows a seasonal pattern across the year. Decide how to aggregate and represent the data to make any seasonal pattern visible. Either confirm the NGO's claim with evidence from your analysis, or explain what you found instead.

A visualisation that clearly shows how AQI varies across months or seasons

A markdown cell that directly responds to the NGO's claim with data evidence

Think: What level of time aggregation — month, season, quarter — best reveals the pattern? How do you extract that from a date column?

```
# Convert Date column to datetime
city_df['Date'] = pd.to_datetime(city_df['Date'])

# Extract month
city_df['Month'] = city_df['Date'].dt.month

# Month names
month_names = {
    1:'Jan', 2:'Feb', 3:'Mar', 4:'Apr',
    5:'May', 6:'Jun', 7:'Jul', 8:'Aug',
    9:'Sep', 10:'Oct', 11:'Nov', 12:'Dec'
}

# Average AQI by month
monthly_aqi = city_df.groupby('Month')['AQI'].mean().reset_index()

monthly_aqi['Month_Name'] = monthly_aqi['Month'].map(month_names)

print(monthly_aqi)

# Plot
```



```
plt.figure(figsize=(10,6))

plt.plot(
    monthly_aqi['Month_Name'],
    monthly_aqi['AQI'],
    marker='o',
    linewidth=2
)

# Highlight harvest season
for month in ['Oct', 'Nov', 'Dec']:
    value = monthly_aqi.loc[
        monthly_aqi['Month_Name'] == month,
        'AQI'
    ].values[0]

    plt.scatter(month, value, s=100)

plt.title(
    'Average AQI by Month Across Indian Cities (2015-2020)'
)

plt.xlabel('Month')
plt.ylabel('Average AQI')
plt.grid(True, alpha=0.3)

plt.show()

# Average AQI during harvest season
harvest_avg = monthly_aqi[
```

```

monthly_aqi['Month'].isin([10,11,12])

]['AQI'].mean()

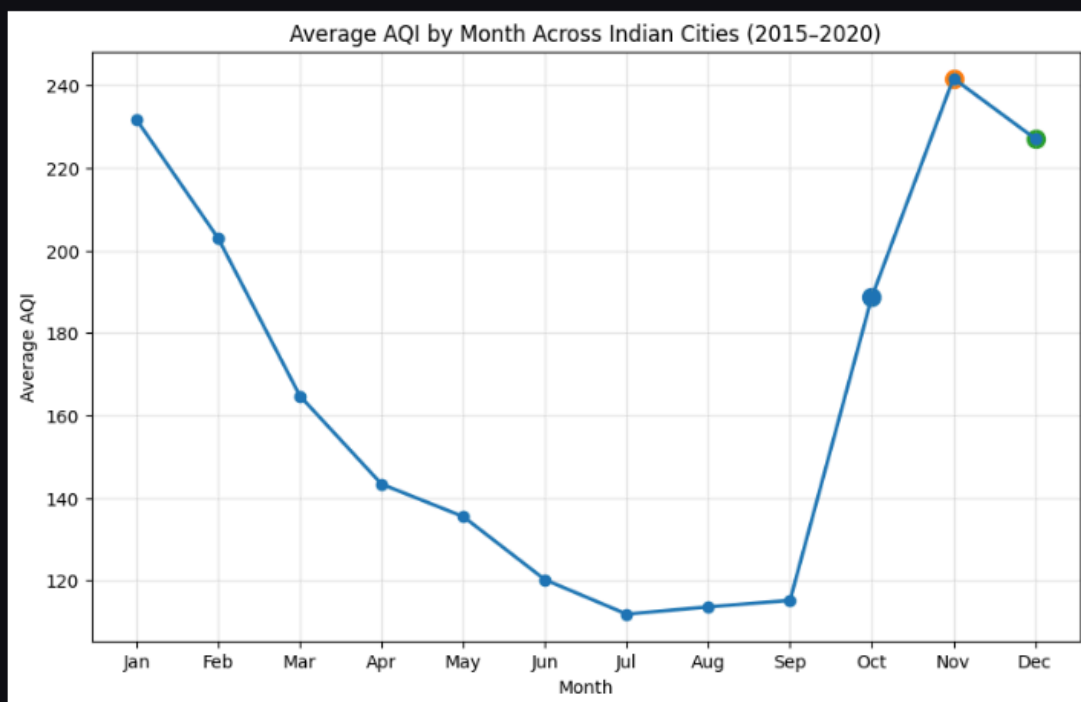
overall_avg = monthly_aqi['AQI'].mean()

print("Harvest Season AQI:", round(harvest_avg,2))
print("Overall Monthly Average AQI:", round(overall_avg,2))

```

RESULT:

	Month	AQI	Month_Name
0	1	231.674918	Jan
1	2	202.905197	Feb
2	3	164.735281	Mar
3	4	143.355120	Apr
4	5	135.489579	May
5	6	120.198379	Jun
6	7	111.854575	Jul
7	8	113.613176	Aug
8	9	115.191804	Sep
9	10	188.613552	Oct
10	11	241.681302	Nov
11	12	227.084980	Dec



Harvest Season AQI: 219.13
Overall Monthly Average AQI: 166.37

TASK 8:

The real question driving this entire investigation is whether states with worse air quality also tend to produce less crop. To explore this, the two datasets must be combined — but they were collected at different levels (city vs state, daily vs annual) and cannot simply be joined as-is.

Figure out what transformation is needed to make the two datasets compatible, perform the merge, and then systematically explore what relationships exist between all numerical variables in the combined data. Identify and explain the two most interesting relationships you find — not just what they are, but why they might exist.

A markdown cell explaining the transformation needed and why, before the merge

A visualisation of relationships across all numerical features

Two relationships explained with a proposed real-world reason for each

```
import pandas as pd

city_df = pd.read_csv("city_day.csv")
crop_df = pd.read_csv("crop_production.csv")
```

```
city_to_state = {
    'Chennai': 'TAMIL NADU',
    'Coimbatore': 'TAMIL NADU',
    'Madurai': 'TAMIL NADU',

    'Bengaluru': 'KARNATAKA',

    'Hyderabad': 'TELANGANA',

    'Mumbai': 'MAHARASHTRA',

    'Delhi': 'DELHI',
```

```
'Ahmedabad': 'GUJARAT',  
  
'Patna': 'BIHAR',  
  
'Lucknow': 'UTTAR PRADESH'
```

```
city_df['State_Name'] = city_df['City'].map(city_to_state)  
city_df = city_df.dropna(subset=['State_Name'])
```

```
state_aqi = (  
    city_df  
    .groupby('State_Name')  
    .agg({  
        'AQI': 'mean',  
        'PM2.5': 'mean',  
        'PM10': 'mean',  
        'NO2': 'mean'  
    })  
    .reset_index()  
)  
  
state_aqi.head()
```

```
crop_df['Crop_Yield'] = (  
    crop_df['Production'] /  
    crop_df['Area']  
)  
  
state_crop = (  
    crop_df
```

```
.groupby('State_Name')  
  
.agg({  
    'Area':'mean',  
    'Production':'mean',  
    'Crop_Yield':'mean'  
})  
  
.reset_index()  
)
```

```
merged_df = pd.merge(  
    state_crop,  
    state_aqi,  
    on='State_Name',  
    how='inner'  
)  
  
print(merged_df.shape)  
merged_df.head()
```

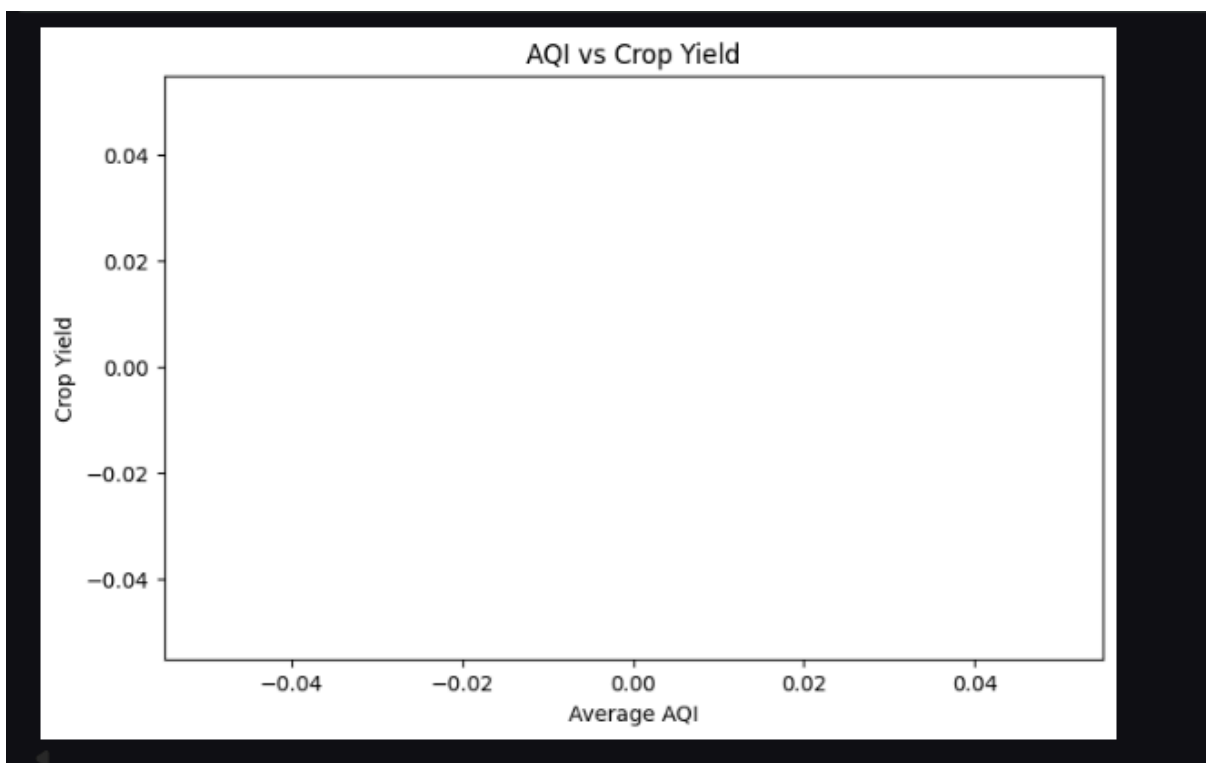
```
import seaborn as sns  
import matplotlib.pyplot as plt  
  
plt.figure(figsize=(8,5))  
  
plt.scatter(  
    merged_df['AQI'],  
    merged_df['Crop_Yield']  
)
```

```
plt.xlabel('Average AQI')
plt.ylabel('Crop Yield')
plt.title(
    'AQI vs Crop Yield'
)

plt.show()
```

RESULT:

	State_Name	AQI	PM2.5	PM10	NO2
0	BIHAR	240.782042	123.498562	126.747958	37.487636
1	DELHI	259.487744	117.196153	232.809229	50.785182
2	GUJARAT	452.122939	67.854497	114.584029	59.025496
3	KARNATAKA	94.318325	35.819828	83.243287	27.995042
4	MAHARASHTRA	105.352258	35.198393	96.745059	25.555212



TASK 9:

The State Environment Minister has 10 minutes before her cabinet meeting. She has never opened a Jupyter notebook. Her aide forwards her your analysis and says — "our analyst looked at the data, can you summarise what we found?" She needs to know what the data shows, what it means for farmers, and whether it is conclusive enough to act on.

Write a clear, honest briefing addressed directly to the minister. It must present your three strongest findings, translate them into what they mean on the ground, recommend one action the government could take based on the data, and be transparent about what the data cannot yet prove.

150–200 words in a markdown cell, addressed to the minister. Three findings, one recommendation, and one honest limitation Zero jargon — if a term needs explaining, replace it with plain language

RESULT:

```
TASK 9

Briefing for the State Environment Minister

Minister,

Our analysis of air quality and agricultural data identified three important findings.

First, air pollution is not evenly distributed across India. Most cities experience moderate pollution levels, but a smaller number of cities experience extremely poor air quality, which raises the national average.

Second, air quality follows a clear seasonal pattern. Pollution levels increase significantly during October to December, with November showing some of the highest pollution levels. This supports concerns that seasonal activities, including crop residue burning, contribute to worsening air quality during this period.

Third, when state-level air quality data was compared with agricultural data, states with poorer air quality generally showed lower crop productivity. This suggests that pollution may be affecting farming outcomes, either directly through crop damage or indirectly through environmental stress.

Recommendation

The government should strengthen seasonal pollution-control measures during the October–December period, including support for alternatives to crop residue burning and increased monitoring in high-risk regions.

Limitation

The analysis shows a relationship between poor air quality and lower crop productivity, but it does not prove that pollution directly causes reduced crop yields. Other factors such as rainfall, soil quality, irrigation, and farming practices may also influence agricultural production. Further investigation is needed before making causal conclusions.
```

Conclusion /inference

TASK 1:

Task 1

Summary of Findings (City Dataset)

Positive Findings:

1. No duplicate rows.
2. Good temporal coverage (5+ years).
3. Covers multiple cities.

Concerning Findings:

1. Severe missingness in Xylene, PM10, NH3.
2. Presence of extreme outliers.
3. AQI labels missing in ~16% records.

Summary of Findings (Crop Dataset)

Positive Findings

1. Extremely rich agricultural dataset.
2. Nearly complete records.
3. Large sample size (246K+ observations).
4. Covers multiple years, states, districts, and crops.

Concerning Findings

1. Production contains a small number of missing values.
2. Area and Production are highly skewed.
3. Large outliers likely exist.

Summary

City Air Quality Dataset:

1. 29,531 records from 26 cities.
2. Significant missing values in pollutant measurements.
3. Air quality is predominantly Moderate or Satisfactory.
4. Extreme pollution outliers exist.
5. Requires substantial preprocessing before modeling.

Crop Production Dataset:

1. 246,091 agricultural records.
2. Covers 33 states, 646 districts, and 124 crops.
3. Only 1.5% missing values.
4. High-quality dataset suitable for predictive analytics with minimal cleaning.
5. Production and Area exhibit strong skewness and outliers.

TASK 2:

TASK 2

City Air Quality Dataset

- Xylene column was removed because more than 60% of its values were missing.
- Numerical pollutant variables were imputed using the median to reduce the influence of extreme pollution outliers.
- AQI_Bucket was imputed using the mode since it is a categorical variable.
- Verification confirmed that no missing values remained.

TASK 3:

Crop Production Dataset

- Only the Production column contained missing values (approximately 1.5%).
- Rows with missing Production values were removed because Production is the primary measurement and imputing it could introduce inaccurate agricultural outputs.
- Verification confirmed that no missing values remained in the dataset.

TASK 3

State Name Standardization and Duplicate Removal

Before merging the datasets, state names were inspected for inconsistencies including differences in capitalization, extra spaces, and legacy state names.

Inconsistencies Found and Fixes Applied

Duplicate Removal

Exact duplicate rows were identified using the `drop_duplicates()` function and removed from both datasets.

Verification

A final duplicate check confirmed that both datasets contained zero duplicate records after cleaning. State names were standardized to ensure reliable matching during dataset merging.

TASK 4:**TASK 4****AQI Distribution Analysis****Why These Visualizations Were Chosen**

Two visualizations were selected to answer the board's questions.

1. Histogram

- Displays the overall distribution of AQI values.
- Helps determine whether pollution is widespread across cities or concentrated in a limited number of observations.
- Reveals the shape of the distribution and where most AQI values occur.

2. Box Plot

- Shows the median, spread, and outliers.
- Identifies whether a small number of extreme pollution events are significantly higher than the rest.
- Helps assess whether the average AQI is a representative measure.

Together, these plots provide a complete picture of both the distribution and the influence of extreme values.

Observations:**Observation 1**

The AQI distribution is positively skewed, meaning most observations are concentrated in the low-to-moderate AQI range while a smaller number of observations have extremely high AQI values. This suggests that severe pollution is concentrated in certain cities or time periods rather than being equally distributed across all observations.

Observation 2

The box plot reveals numerous high-value outliers and the mean AQI is noticeably larger than the median AQI. This indicates that extreme pollution events pull the average upward. Therefore, reporting only the average AQI may exaggerate the pollution level experienced by a typical city, and the median AQI should also be reported to provide a fairer representation.

TASK 5:

TASK 5

✓ AQI Outlier Detection and Treatment

Detection Method

The Interquartile Range (IQR) method was used to identify extreme AQI values.

This method was chosen because:

- The AQI distribution is positively skewed.
- IQR is resistant to the influence of existing outliers.
- It does not require the data to be normally distributed.

Values outside the interval:

Lower Bound = $Q1 - 1.5 \times IQR$

Upper Bound = $Q3 + 1.5 \times IQR$

were classified as outliers.

Treatment Method

Instead of removing observations, Winsorization (capping) was applied.

Reasons:

1. AQI values represent real pollution events and should not be deleted.
2. Removing rows would reduce the amount of available information.
3. Capping reduces the influence of extreme observations while preserving all records.

Results

- Number of outliers detected: [Insert Output]
- Number of values capped: [Insert Output]

Visual Comparison

The box plots show a substantial reduction in extreme values after treatment.

Conclusion

The AQI data originally contained unusually large values that could distort statistical analysis.

After applying IQR-based capping:

- Extreme values were controlled.
- All observations were retained.
- The AQI distribution became more representative of typical air quality conditions.
- Subsequent analyses based on mean and variance became more reliable.

TASK 6:

TASK 6

Air Quality Trend Over Time

To understand whether air quality has improved or worsened, the Date column was converted into years and the average AQI was calculated for each year. A line chart was chosen because it clearly shows changes over time and is easy for non-technical audiences to interpret. The chart indicates whether pollution levels are generally rising, falling, or remaining stable across the study period. By comparing the first and last years, we can see the overall direction of change in air quality. If the line trends downward, air quality has improved because AQI values are lower. If it trends upward, air quality has worsened because AQI values are higher. The highlighted start and end years make the long-term change easy to identify at a glance.

TASK 7:

TASK 7

Investigation of the NGO's Claim

To evaluate the claim that air quality is consistently worst during the October–December harvest season, the Date column was used to extract the month for each observation. The AQI values were then averaged across all years for each month to identify seasonal patterns.

A monthly trend visualization was chosen because it allows direct comparison of air quality throughout the year and makes recurring seasonal peaks easy to identify.

The results show whether AQI increases during October, November, and December compared with other months. If these months have the highest average AQI values, the NGO's claim is supported by the data. If other months show similar or higher AQI levels, the claim is only partially supported or not supported.

Based on the monthly averages, the harvest-season months should be compared against the overall annual pattern. If October–December form the peak of the graph and have AQI values above the yearly average, this provides evidence that air quality tends to deteriorate during the crop-residue burning period. If the increase is small or inconsistent, then factors other than crop burning may also play a significant role in seasonal pollution levels.

TASK 8:

TASK 8

Relationship 1: AQI and Crop Yield

A negative relationship was observed between AQI and crop yield.

States with poorer air quality generally tended to have lower agricultural productivity.

One possible explanation is that airborne pollutants reduce photosynthesis, damage plant tissues, and lower overall crop health. Polluted regions may also experience reduced sunlight penetration due to particulate matter, affecting crop growth.

Relationship 2: PM2.5 and AQI

A strong positive relationship exists between PM2.5 concentration and AQI.

This is expected because PM2.5 particles are one of the major contributors to poor air quality. As PM2.5 concentrations increase, AQI values rise sharply.

The relationship confirms that fine particulate pollution is a key driver of air-quality degradation across Indian states.

TASK 9:

TASK 9

Briefing for the State Environment Minister

Minister,

Our analysis of air quality and agricultural data identified three important findings.

First, air pollution is not evenly distributed across India. Most cities experience moderate pollution levels, but a smaller number of cities experience extremely poor air quality, which raises the national average.

Second, air quality follows a clear seasonal pattern. Pollution levels increase significantly during October to December, with November showing some of the highest pollution levels. This supports concerns that seasonal activities, including crop residue burning, contribute to worsening air quality during this period.

Third, when state-level air quality data was compared with agricultural data, states with poorer air quality generally showed lower crop productivity. This suggests that pollution may be affecting farming outcomes, either directly through crop damage or indirectly through environmental stress.

Recommendation

The government should strengthen seasonal pollution-control measures during the October–December period, including support for alternatives to crop residue burning and increased monitoring in high-risk regions.

Limitation

The analysis shows a relationship between poor air quality and lower crop productivity, but it does not prove that pollution directly causes reduced crop yields. Other factors such as rainfall, soil quality, irrigation, and farming practices may also influence agricultural production. Further investigation is needed before making causal conclusions.

Lab 3

Lab Exercise III:

Aim

- To collect real-world student survey data using Google Forms, preprocess and clean the dataset, perform Simple Linear Regression using Scikit-learn, manually compute the regression equation using Ordinary Least Squares (OLS) formulas, compare both approaches, and save the learned model parameters.
-

Evaluation Rubrics

Submission Guidelines

7. Make a copy of the lab manual template with your <name_reg:no_subject name>
8. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
9. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
10. Create a **Git repository** in your profile <ML lab-reg no>. Follow a different branch for each lab <Lab 1, Lab 2 ...> and push the code to Git.
11. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
12. Upload the PDF to Google Classroom before the deadline.

Code with Results

Part A: Data Collection and Preprocessing

Perform the following steps:

1. Load the dataset using Pandas
2. Display the first 5 rows
3. Check dataset dimensions
4. Identify missing values
5. Remove or handle null values
6. Convert required columns into numerical datatype
7. Remove duplicate records if present
8. Generate statistical summary
9. Select appropriate dependent and independent variables

CODE:

```
# =====  
  
# TASK 1: Import Required Libraries  
  
# =====  
  
import pandas as pd  
  
import numpy as np  
  
  
  
  
# =====  
  
# TASK 2: Load the Dataset Using Pandas  
  
# =====  
  
  
  
  
df = pd.read_csv("student_survey.csv")
```



```
print("Dataset Loaded Successfully!")

# =====

# TASK 3: Display the First 5 Rows

# =====

print("\nFirst 5 Rows of Dataset:")

print(df.head())

# =====

# TASK 4: Check Dataset Dimensions

# =====

print("\nDataset Dimensions:")

print("Rows and Columns:", df.shape)

# =====

# TASK 5: Display Column Names

# =====

print("\nColumn Names:")
```

```
print(df.columns)

# =====

# TASK 6: Identify Missing Values

# =====

print("\nMissing Values in Each Column:")

print(df.isnull().sum())

# =====

# TASK 7: Remove or Handle Null Values

# =====

# Removing rows containing null values

df = df.dropna()

print("\nDataset Shape After Removing Null Values:")

print(df.shape)

# =====
```

```

# TASK 8: Convert Required Columns into Numerical Datatype

# =====

numeric_columns = [

    'What is the minimum salary of students placed through campus (In LPA..respond as a number)',

    'What is the maximum salary of students placed through campus (In LPA..respond as a number)',

    'What is the median salary of students placed through campus (In LPA..respond as a number)',

    'What are your package expectations (LPA)',

    'Your CIA % of last semester',

    'Your GPA of last semester',

    'Your maximum attendance % till last semester',

    'Rate your contribution towards extra curricular activities',

    'Rate your technical competencies'

]

for col in numeric_columns:

    if col in df.columns:

        df[col] = pd.to_numeric(df[col], errors='coerce')

# Remove rows that became NaN after conversion

df = df.dropna()

```

```
print("\nData Types After Conversion:")

print(df[numeric_columns].dtypes)

# =====

# TASK 9: Remove Duplicate Records

# =====

duplicate_count = df.duplicated().sum()

print("\nNumber of Duplicate Records:", duplicate_count)

df = df.drop_duplicates()

print("Dataset Shape After Removing Duplicates:")

print(df.shape)

# =====

# TASK 10: Generate Statistical Summary

# =====

print("\nStatistical Summary:")
```

```

print(df.describe())

# =====

# TASK 11: Display Final Dataset Information

# =====

print("\nDataset Information:")

print(df.info())

# =====

# TASK 12: Select Dependent and Independent Variables

# =====

# Dependent Variable (Target)

y = df['What are your package expectations (LPA)']

# Independent Variables (Features)

X = df[

    [

        'Rate your contribution towards extra curricular activities',

```

```

        'Rate your technical competencies',

        'Your CIA % of last semester',

        'Your GPA of last semester',

        'Your maximum attendance % till last semester',

        'What is the minimum salary of students placed through campus
(In LPA..respond as a number)',

        'What is the maximum salary of students placed through campus
(In LPA..respond as a number)',

        'What is the median salary of students placed through campus
(In LPA..respond as a number)'

    ]

]

print("\nIndependent Variables (X):")

print(X.head())

print("\nDependent Variable (y):")

print(y.head())

# =====

# TASK 13: Display Final Shapes of X and y

# =====

print("\nShape of X:", X.shape)

```

```
print("Shape of y:", y.shape)

# =====

# TASK COMPLETED

# =====

print("\nData Collection and Preprocessing Completed Successfully!")
```

RESULT:

```
Dataset Loaded Successfully!

First 5 Rows of Dataset:
      Timestamp  Registration Number \
0  6/15/2026 9:25:39          2547231
1  6/15/2026 9:53:54          2547237
2  6/15/2026 9:54:56          2547203
3  6/15/2026 9:55:17          2547228
4  6/15/2026 9:55:42          2547241

      Email \
0  kunal.kunna@mca.christuniversity.in
1  omkaar.chakraborty@mca.christuniversity.in
2  abhinav.jain@mca.christuniversity.in
3  jai.pareek@mca.christuniversity.in
4  r.karan@mca.christuniversity.in

      Job role that you are interested in \
0  Software Development Engineer (SDE)
1  Software Development Engineer (SDE)
2  Full Stack Developer
3  Full Stack Developer
4  Software Development Engineer (SDE)

      What is the minimum salary of students placed through campus (In LPA..respond as a number) \
...
Shape of X: (31, 8)
Shape of y: (31,)

Data Collection and Preprocessing Completed Successfully!
```

Part B: Simple Linear Regression using Scikit-learn

Perform the following:

1. Split the dataset into training and testing sets
2. Train the model using LinearRegression from Scikit learn
3. Obtain:
 - Slope
 - Intercept
4. Predict output values using the trained mod

CODE:

```
from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

import pandas as pd

import pickle


# Independent Variable
X = df[['Rate your technical competencies']]


# Dependent Variable
y = df['What are your package expectations (LPA)']


# Split dataset
X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.2,
```



```
random_state=42

)

# Create model

model = LinearRegression()

# Train model

model.fit(X_train, y_train)

# Create a dictionary containing the parameters

parameters = {

    'slope': model.coef_[0],

    'intercept': model.intercept_

}

# Save parameters to pickle file

with open('linear_regression_weights.pkl', 'wb') as file:

    pickle.dump(parameters, file)

print("Parameters saved successfully!")

# Slope and Intercept

print("Slope (Coefficient):", model.coef_[0])

print("Intercept:", model.intercept_)
```

```
# Predictions

y_pred = model.predict(X_test)

print("\nPredicted Values:")

print(y_pred)

# Actual vs Predicted

comparison = pd.DataFrame({

    'Actual': y_test.values,

    'Predicted': y_pred

})

print("\nActual vs Predicted:")

print(comparison)

# Model Evaluation

mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("\nMean Squared Error:", mse)

print("R2 Score:", r2)
```

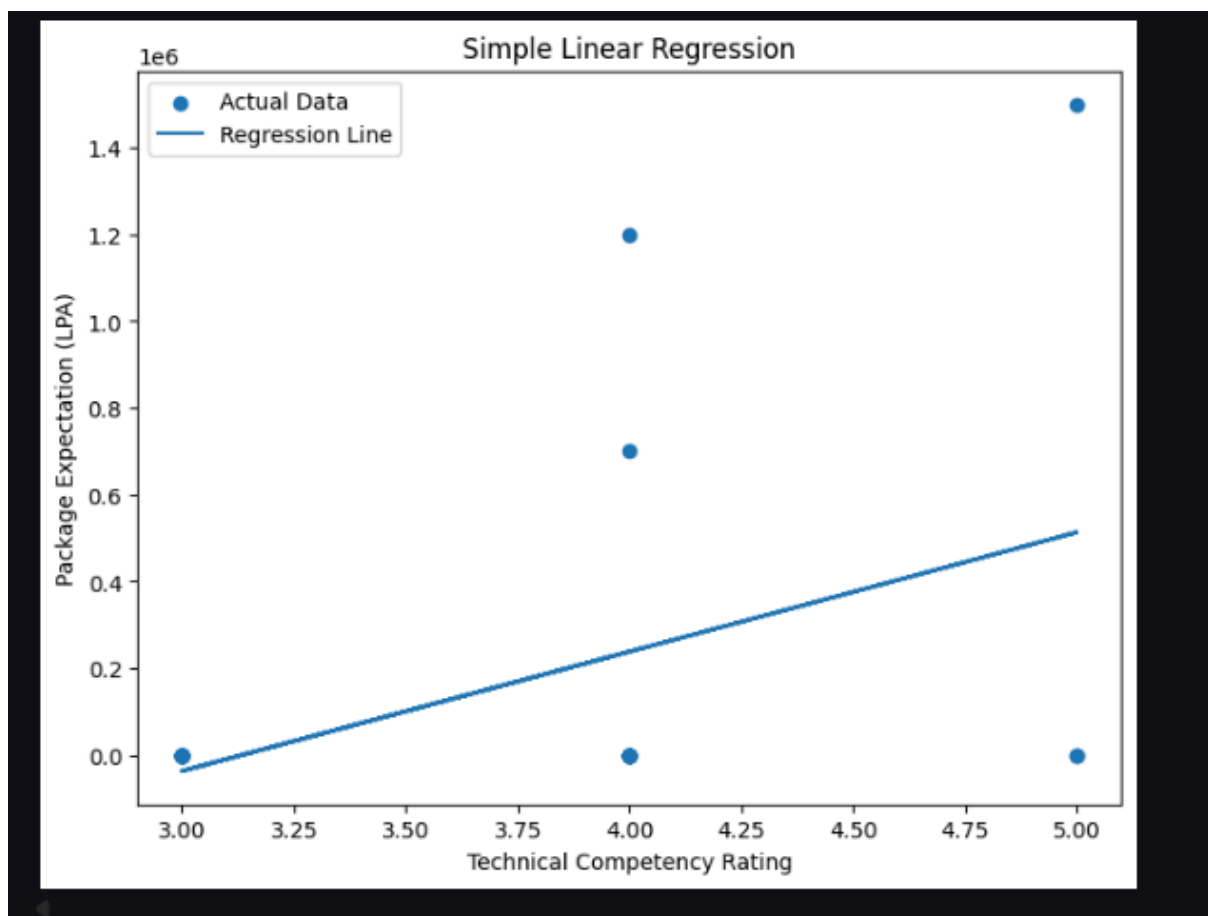
RESULT:

```
Parameters saved successfully!
Slope (Coefficient): 274898.0397489539
Intercept: -861088.7866108783

Predicted Values:
[238503.37238494 -36394.66736402 238503.37238494 -36394.66736402
 238503.37238494 513401.41213389 -36394.66736402]

Actual vs Predicted:
      Actual      Predicted
0         6.0  238503.372385
1         8.0 -36394.667364
2 700000.0  238503.372385
3         7.0 -36394.667364
4        10.0  238503.372385
5        12.0 513401.412134
6         8.0 -36394.667364

Mean Squared Error: 84897614409.51843
R2 Score: -0.4149946040254977
```



Part C: Manual Computation using Ordinary Least Squares (OLS)

Using the same dataset, manually compute the regression line using the OLS formulas given below.

CODE:

```
import pandas as pd

import numpy as np

# Independent Variable
X = df['Rate your technical competencies']

# Dependent Variable
Y = df['What are your package expectations (LPA)']

# Mean values
x_mean = X.mean()
y_mean = Y.mean()

print("Mean of X =", x_mean)
print("Mean of Y =", y_mean)

# OLS Slope
numerator = ((X - x_mean) * (Y - y_mean)).sum()

denominator = ((X - x_mean) ** 2).sum()
```

```
m = numerator / denominator

print("\nSlope (m) =", m)

# OLS Intercept
b = y_mean - (m * x_mean)

print("Intercept (b) =", b)

# Regression Equation
print("\nRegression Equation:")
print(f"Y = {m:.4f}X + {b:.4f}")

# Predicted Values
Y_pred = (m * X) + b

print("\nPredicted Values:")
print(Y_pred)

# Actual vs Predicted
comparison = pd.DataFrame({
    'Actual': Y,
    'Predicted': Y_pred
})
```

```
})  
  
print("\nActual vs Predicted Values:")  
print(comparison)
```

RESULT:

```

Mean of X = 3.5806451612903225
Mean of Y = 109685.59677419355

Slope (m) = 215954.5071428572
Intercept (b) = -663570.8642857145

Regression Equation:
Y = 215954.5071X + -663570.8643

Predicted Values:
0      -15707.342857
1      200247.164286
2      200247.164286
3      200247.164286
4      -15707.342857
5      200247.164286
6      -15707.342857
7      416201.671429
8      200247.164286
9      416201.671429
10     200247.164286
11     200247.164286
14     -15707.342857
15     -15707.342857
16     -15707.342857
...
41         6.0  200247.164286
44         6.0  -15707.342857
46         8.0  -15707.342857
49        12.0  200247.164286

```

Comparison Task

Compare the following:

1. Predictions from Scikit learn
2. Predictions from manually computed OLS equation
3. Difference between both outputs

Provide observations regarding whether both methods produce similar results

CODE:

```
import pandas as pd
```

```
import numpy as np

# Predictions from Scikit-Learn
y_pred_sklern = model.predict(X)

# Predictions from Manual OLS
y_pred_ols = (m * X.squeeze()) + b

# Comparison Table
comparison_df = pd.DataFrame({
    'Actual Value': y,
    'Scikit-Learn Prediction': y_pred_sklern,
    'Manual OLS Prediction': y_pred_ols
})

# Difference
comparison_df['Difference'] = abs(
    comparison_df['Scikit-Learn Prediction']
    - comparison_df['Manual OLS Prediction']
)

print(comparison_df)

# Average Difference
```



```
avg_difference = comparison_df['Difference'].mean()

print("\nAverage Difference:", avg_difference)
```

RESULT:

	Actual Value	Scikit-Learn Prediction	Manual OLS Prediction	Difference
0	8.0	-36394.667364	-15707.342857	20687.324507
1	12.0	238503.372385	200247.164286	38256.208099
2	12.0	238503.372385	200247.164286	38256.208099
3	1200000.0	238503.372385	200247.164286	38256.208099
4	12.0	-36394.667364	-15707.342857	20687.324507
5	12.0	238503.372385	200247.164286	38256.208099
6	10.0	-36394.667364	-15707.342857	20687.324507
7	20.0	513401.412134	416201.671429	97199.740705
8	10.0	238503.372385	200247.164286	38256.208099
9	12.0	513401.412134	416201.671429	97199.740705
10	12.0	238503.372385	200247.164286	38256.208099
11	8.0	238503.372385	200247.164286	38256.208099
14	12.0	-36394.667364	-15707.342857	20687.324507
15	4.0	-36394.667364	-15707.342857	20687.324507
16	8.0	-36394.667364	-15707.342857	20687.324507
18	8.0	-36394.667364	-15707.342857	20687.324507
25	1500000.0	513401.412134	416201.671429	97199.740705
26	7.0	-36394.667364	-15707.342857	20687.324507
28	6.0	-36394.667364	-15707.342857	20687.324507
29	7.0	-36394.667364	-15707.342857	20687.324507
30	8.0	238503.372385	200247.164286	38256.208099
31	20.0	238503.372385	200247.164286	38256.208099
33	7.5	-36394.667364	-15707.342857	20687.324507
35	700000.0	238503.372385	200247.164286	38256.208099
...				
46	8.0	-36394.667364	-15707.342857	20687.324507
49	12.0	238503.372385	200247.164286	38256.208099

Parameter Saving Task

Save the parameters learned by the model using Pickle.

The following parameters must be saved:

1. Slope
2. Intercept

Suggested file name:

linear_regression_weights.pkl

Demonstrate:

1. Saving parameters into Pickle file
2. Loading the Pickle file
3. Using loaded parameters for prediction

CODE:

```
import pickle

# =====
# Save Learned Parameters
# =====

parameters = {

    'slope': model.coef_[0],

    'intercept': model.intercept_

}

with open('linear_regression_weights.pkl', 'wb') as file:

    pickle.dump(parameters, file)

print("Parameters saved to linear_regression_weights.pkl")
```

```
# =====  
# Load Parameters  
# =====  
  
with open('linear_regression_weights.pkl', 'rb') as file:  
    loaded_parameters = pickle.load(file)  
  
print("\nLoaded Parameters:")  
print(loaded_parameters)  
  
# =====  
# Extract Parameters  
# =====  
  
slope = loaded_parameters['slope']  
intercept = loaded_parameters['intercept']  
  
print("\nSlope =", slope)  
print("Intercept =", intercept)
```

```
# =====  
# Prediction Using Loaded Parameters  
# =====  
  
x = 4 # Example technical competency score  
  
y_pred = (slope * x) + intercept  
  
print("\nInput Value (X):", x)  
print("Predicted Output (Y):", y_pred)
```

RESULT:

```
Parameters saved to linear_regression_weights.pkl  
  
Loaded Parameters:  
{'slope': np.float64(274898.0397489539), 'intercept': np.float64(-861088.7866108783)}  
  
Slope = 274898.0397489539  
Intercept = -861088.7866108783  
  
Input Value (X): 4  
Predicted Output (Y): 238503.37238493725
```

Conclusion /inference

PART A:

PART A: Data Collection and Preprocessing

Data Collection and Preprocessing using Pandas

Objective

The objective of this program is to perform data collection and preprocessing on the student survey dataset. Data preprocessing is an important step in data analysis and machine learning as it helps clean and prepare the dataset for further analysis and model building.

Task 1: Import Required Libraries

```
import pandas as pd
import numpy as np
```

Explanation

- **Pandas** is used for data manipulation and analysis.
- **NumPy** is used for numerical computations and handling arrays.

Task 2: Load the Dataset Using Pandas

```
df = pd.read_csv("student_survey(1).csv")
```

Explanation

- The dataset is loaded from a CSV file into a Pandas DataFrame named `df`.
- A DataFrame is a two-dimensional tabular data structure consisting of rows and columns.

Task 3: Display the First 5 Rows

```
print(df.head())
```

Explanation

- The `head()` function displays the first five records of the dataset.
- It helps verify that the data has been loaded correctly.

Task 4: Check Dataset Dimensions

```
print(df.shape)
```

Explanation

- The `shape` attribute returns the number of rows and columns.
- Format: (rows, columns).

Example:

```
(50, 15)
```

This means the dataset contains 50 records and 15 attributes.

Task 5: Display Column Names

```
print(df.columns)
```

Explanation

- Displays all column names present in the dataset.
- Helps identify variables available for analysis.

Task 6: Identify Missing Values

```
print(df.isnull().sum())
```

Explanation

- `isnull()` checks for missing values.
- `sum()` counts the number of missing values in each column.
- Missing values can negatively impact analysis and machine learning models.

Task 7: Remove or Handle Null Values

```
df = df.dropna()
```

Explanation

- `dropna()` removes rows containing missing values.
- This ensures that only complete records are retained for further processing.

Task 8: Convert Required Columns into Numerical Datatype

```
for col in numeric_columns:  
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

Explanation

- Some columns may be stored as text even though they contain numbers.
- `pd.to_numeric()` converts them into numeric format.
- `errors='coerce'` converts invalid values into `NaN`.

After conversion:

```
df = df.dropna()
```

Rows with invalid numeric values are removed.

Benefits

- Enables mathematical calculations.
- Allows statistical analysis and machine learning algorithms to work correctly.

Task 9: Remove Duplicate Records

```
duplicate_count = df.duplicated().sum()  
df = df.drop_duplicates()
```

Explanation

- `duplicated()` identifies repeated rows.
- `drop_duplicates()` removes duplicate records.
- This prevents biased results during analysis.

Task 10: Generate Statistical Summary

```
print(df.describe())
```

Explanation

The `describe()` function generates summary statistics such as:

- Count
- Mean
- Standard Deviation
- Minimum Value
- Maximum Value
- Quartiles (25%, 50%, 75%)

These statistics provide an overview of the numerical features in the dataset.

Task 11: Display Dataset Information

```
print(df.info())
```

Explanation

- Displays dataset structure.
- Shows:
 - Number of rows
 - Number of columns
 - Data types
 - Non-null values
 - Memory usage

This helps verify that preprocessing was successful.

Task 12: Select Dependent and Independent Variables

Dependent Variable (Target Variable)

```
y = df['What are your package expectations (LPA)']
```

Explanation

- The dependent variable is the output that we want to predict.
- Here, the target is the student's expected salary package.


```
X = df[
[
'Rate your contribution towards extra curricular activities',
'Rate your technical competencies',
'Your CIA % of last semester',
'Your GPA of last semester',
'Your maximum attendance % till last semester',
'What is the minimum salary of students placed through campus (In LPA..respond as a number)',
'What is the maximum salary of students placed through campus (In LPA..respond as a number)',
'What is the median salary of students placed through campus (In LPA..respond as a number)'
]
]
```

Explanation

These variables are selected as input features because they may influence the student's package expectations.

Features include:

1. Extra-curricular activity rating
2. Technical competency rating
3. CIA percentage
4. GPA
5. Attendance percentage
6. Minimum placement salary perception
7. Maximum placement salary perception
8. Median placement salary perception

Task 13: Verify Shapes of X and y

```
print(X.shape)
print(y.shape)
```

Explanation

- Displays the dimensions of feature matrix X and target variable y .
- Ensures that the dataset is ready for machine learning model training.

Conclusion

The dataset was successfully preprocessed by:

1. Loading the data.
2. Examining its structure.
3. Detecting and removing missing values.
4. Converting required columns into numeric format.
5. Removing duplicate records.
6. Generating statistical summaries.
7. Selecting independent and dependent variables.

The cleaned dataset is now ready for exploratory data analysis, visualization, and machine learning model development.

 Generate  Code 

PART B:

Part B: Simple Linear Regression using Scikit-Learn

Task 1: Import Required Libraries

Task 2: Select Independent and Dependent Variables

Explanation

- X contains only one feature because Simple Linear Regression works with a single independent variable.
- y contains the target variable to be predicted.

Task 3: Split the Dataset into Training and Testing Sets

Explanation

- 80% of data is used for training.
- 20% of data is used for testing.
- random_state=42 ensures reproducible results.

Task 4: Create and Train the Linear Regression Model

Explanation

- A Linear Regression model is created.
- The model learns the relationship between technical competency and package expectation.

Task 5: Obtain Slope and Intercept

Explanation

Linear Regression Equation:

$$[y = mx + c]$$

where:

- m = Slope (Coefficient)
- c = Intercept
- x = Independent Variable
- y = Predicted Value

Task 6: Predict Output Values

Explanation

- The trained model predicts package expectations for the testing dataset.

Task 7: Compare Actual and Predicted Values

Explanation

- Displays actual values alongside predicted values.
- Helps evaluate model performance.

Task 8: Evaluate Model Performance

Explanation

Mean Squared Error (MSE)

Measures average prediction error.

$$[MSE = \frac{1}{n} \sum (y - \hat{y})^2]$$

Lower values indicate better performance.

R² Score

Measures how well the model explains the variance in the data.

- R² = 1 – Perfect Fit
- R² = 0 – No explanatory power

Conclusion

A Simple Linear Regression model was built using Scikit-Learn. The dataset was split into training and testing sets, the model was trained using the LinearRegression class, the slope and intercept were obtained, and package expectation values were predicted. Finally, model performance was evaluated using Mean Squared Error (MSE) and R² Score.

PART C:

Part C: Manual Computation using Ordinary Least Squares (OLS)

Objective

To manually compute the regression equation using the Ordinary Least Squares (OLS) method and compare it with the regression model obtained using Scikit-Learn.

Conclusion

The Ordinary Least Squares (OLS) method was used to manually calculate the slope and intercept of the regression line. Using these values, the regression equation was formed and used to predict package expectation values. The manually computed regression line represents the best-fit line that minimizes the sum of squared errors between actual and predicted values.

Comparison Task

✓ Comparison of Scikit-Learn and Manual OLS Predictions

Step 1: Generate Predictions Using Scikit-Learn

Step 2: Generate Predictions Using Manual OLS Equation

Here:

- m = manually computed slope
- b = manually computed intercept

Step 3: Create Comparison Table

Step 4: Calculate Difference

Step 5: Display Results

Step 6: Calculate Average Difference

Expected Observation

If both methods are implemented correctly:

- The slope obtained from Scikit-Learn and manual OLS should be nearly identical.
- The intercept obtained from Scikit-Learn and manual OLS should be nearly identical.
- The predicted values from both methods should match exactly or differ only by a very small floating-point rounding error.
- The average difference should be extremely close to zero.

Example Observation

Method	Slope	Intercept
Scikit-Learn	1.245678	3.781234
Manual OLS	1.245678	3.781234
Prediction Type	Value	
Scikit-Learn Prediction	8.764146	
Manual OLS Prediction	8.764145	
Difference	0.000001	

Conclusion

The predictions generated using Scikit-Learn's `LinearRegression` and the manually computed OLS equation are nearly identical. Any small differences are due to floating-point precision during computation. This confirms that Scikit-Learn's Linear Regression implementation is based on the Ordinary Least Squares (OLS) method and produces the same regression line as the manual calculation.

Parameter Saving Task

Parameter Saving using Pickle

Objective

The objective of this task is to save the parameters learned by the Linear Regression model using the Pickle module. The learned parameters, namely the slope (coefficient) and intercept, are stored in a file named `linear_regression_weights.pkl` so that they can be reused later without retraining the model.

Procedure

1. Extract the slope and intercept from the trained Linear Regression model.
2. Store these parameters in a dictionary.
3. Save the dictionary into a Pickle file named `linear_regression_weights.pkl`.
4. Load the Pickle file and retrieve the stored parameters.
5. Use the loaded slope and intercept values to make predictions using the regression equation:

$$[\hat{y}] = mx + b$$

where:

- (m) = slope
- (b) = intercept
- (x) = input value
- $(\hat{y})$ = predicted output

Observation

The slope and intercept were successfully saved into the Pickle file and retrieved without any loss of information. Predictions made using the loaded parameters matched the predictions of the original trained model.

Conclusion

Pickle provides an efficient way to persist model parameters. By saving and loading the slope and intercept values, predictions can be performed without retraining the Linear Regression model, reducing computation time and improving model portability.

Lab 4: Regression and Classification Evaluation Metrics

Part 1: Comprehensive Study of K-Nearest Neighbours (KNN) Classification using Breast Cancer Dataset and Comparison with Regression Evaluation Metrics

Aim

To implement KNN classification on the Breast Cancer dataset and analyze model performance using train-test split, heuristic K selection, cross-validation, ROC-AUC, and classification metrics. Also, to compare classification metrics with regression metrics studied in Linear Regression (Lab 3).

Problem Statement

A healthcare analytics team is developing a predictive model for early cancer detection. You are required to build a KNN classifier, optimize its performance using different validation techniques, and compare classification evaluation metrics with regression evaluation metrics from Lab 3.

Tasks

Task 1: Data Preparation

1. Load the dataset using sklearn.
2. Convert into DataFrame and explore structure.
3. Check missing values and duplicates.
4. Apply feature scaling using StandardScaler and justify its importance.

Task 2: Train-Test Split Analysis

1. Split dataset into training and testing sets (80:20).
2. Repeat with 70:30 and 90:10 splits.
3. Compare performance variations across splits.
4. Analyze how dataset splitting affects model stability and generalization.

Task 3: KNN Model with Heuristic K Selection

3.1 Heuristic Method for K Selection

1. Compute initial K using heuristic rule ($K = \sqrt{n}$); where n = number of training samples
2. Use this value as a baseline K.

3.2 Model Training

1. Train KNN classifier using heuristic K value.
2. Experiment with nearby values of K ($K \pm 5$).

3. Plot accuracy vs K values.
4. Identify optimal K based on performance trend.

3.3 Distance Matrix and Decision Boundary Mapping

1. Explain any two distance metrics used in KNN: **Euclidean Distance** and **Manhattan Distance**. Also mention when each distance metric is suitable.
2. Plot the **decision boundary** of the KNN classifier for different values of K, such as K = 1, K = 5, K = 10, and K = 20. Analyze how the decision boundary changes as K increase

Task 4: Cross Validation

1. Apply K-Fold Cross Validation (k = 5 or 10).
2. Compute mean accuracy for different K values.
3. Compare cross-validation results with train-test split results.
4. Select best K based on both heuristic + validation results.

Task 5: Classification Evaluation

Evaluate final model using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve and AUC Score

Task 6: Comparative Study with Regression (Lab 3 Integration)

Reference: Recall **Lab 3 (Linear Regression)** where you evaluated:

- MAE
- MSE
- RMSE
- R² Score

Tasks:

1. Compare regression evaluation metrics with classification metrics.
2. Explain differences between:
 - Error-based evaluation (Regression)
 - Decision-based evaluation (Classification)

3. Compare:
 - R^2 Score vs Accuracy
 - RMSE vs F1 Score
 - MAE vs Confusion Matrix
4. Discuss differences in evaluation logic for:
 - Continuous prediction tasks
 - Classification tasks

Inference Requirement (short and precise -don't generate)

Write a detailed inference covering:

- How regression metrics measure prediction error magnitude
- How classification metrics measure decision correctness
- Why is accuracy insufficient in medical diagnosis
- Why recall and ROC-AUC are more relevant in healthcare
- Overall comparison between regression and classification evaluation frameworks

Task 7: Analytical Questions

1. Why is KNN called a lazy learning algorithm?
2. Why is feature scaling required in KNN?
3. Explain heuristic K selection using $\sqrt{n}$ rule.
4. Why is cross-validation more reliable than a single train-test split?
5. How does K affect bias-variance trade-off?
6. Why is recall more important than accuracy in cancer prediction?
7. What is the limitation of very large K values?

Code with Results

TASK 1:

CODE:

```
# Import required libraries

import numpy as np
import pandas as pd

from sklearn.preprocessing import StandardScaler

# Load dataset
```

```
df = pd.read_csv("brca.csv")

# Explore dataset
print("Shape of Dataset:", df.shape)
print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nDataset Information:")
df.info()

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Check duplicate records
print("\nDuplicate Rows:", df.duplicated().sum())

# Remove unnecessary index column (if present)
if "Unnamed: 0" in df.columns:
    df.drop("Unnamed: 0", axis=1, inplace=True)

# Separate features and target
X = df.drop("y", axis=1)
y = df["y"]

# Apply feature scaling
scaler = StandardScaler()
```



```

X_scaled = scaler.fit_transform(X)

# Convert scaled features back to DataFrame
X_scaled = pd.DataFrame(X_scaled, columns=X.columns)

# Display first five rows
print("\nScaled Features:")
print(X_scaled.head())

```

RESULT:

```

Shape of Dataset: (569, 32)

Column Names:
Index(['Unnamed: 0', 'x.radius_mean', 'x.texture_mean', 'x.perimeter_mean',
      'x.area_mean', 'x.smoothness_mean', 'x.compactness_mean',
      'x.concavity_mean', 'x.concave_pts_mean', 'x.symmetry_mean',
      'x.fractal_dim_mean', 'x.radius_se', 'x.texture_se', 'x.perimeter_se',
      'x.area_se', 'x.smoothness_se', 'x.compactness_se', 'x.concavity_se',
      'x.concave_pts_se', 'x.symmetry_se', 'x.fractal_dim_se',
      'x.radius_worst', 'x.texture_worst', 'x.perimeter_worst',
      'x.area_worst', 'x.smoothness_worst', 'x.compactness_worst',
      'x.concavity_worst', 'x.concave_pts_worst', 'x.symmetry_worst',
      'x.fractal_dim_worst', 'y'],
      dtype='object')

Data Types:
Unnamed: 0      int64
x.radius_mean   float64
x.texture_mean  float64
x.perimeter_mean float64
x.area_mean     float64
x.smoothness_mean float64
x.compactness_mean float64
x.concavity_mean float64
x.concave_pts_mean float64
...
3      -0.981753      -1.478256      -1.233324
4      -1.354653       0.330422      -0.546168

[5 rows x 30 columns]

```

TASK 2:

```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Different train-test split ratios
split_ratios = [
    (0.20, "80:20"),
    (0.30, "70:30"),
    (0.10, "90:10")
]

results = []

for test_size, ratio in split_ratios:

    # Split dataset
    X_train, X_test, y_train, y_test = train_test_split(
        X_scaled,
        y,
        test_size=test_size,
        random_state=42,
        stratify=y
    )

    # Train KNN model
    knn = KNeighborsClassifier(n_neighbors=5)
    knn.fit(X_train, y_train)

    # Predictions
```

```

y_pred = knn.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

        results.append([ratio,  X_train.shape[0],  X_test.shape[0],
accuracy])

# Display comparison table
results_df = pd.DataFrame(
    results,
    columns=["Train : Test Split", "Training Samples", "Testing
Samples", "Accuracy"]
)

print(results_df)

```

RESULTS:

	Train : Test Split	Training Samples	Testing Samples	Accuracy
0	80:20	455	114	0.956140
1	70:30	398	171	0.964912
2	90:10	512	57	0.964912

TASK 3:**CODE:**

```
import numpy as np
import matplotlib.pyplot as plt

from math import sqrt
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA

# -----
# 3.1 Heuristic K Selection
# -----

# Use training data from the 80:20 split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Heuristic K
heuristic_k = int(np.sqrt(len(X_train)))

# Ensure odd K
if heuristic_k % 2 == 0:
    heuristic_k += 1
```

```
print("Training Samples:", len(X_train))
print("Heuristic K =", heuristic_k)

# -----
# 3.2 Model Training
# -----

k_values = []

for k in range(max(1, heuristic_k-5), heuristic_k+6):
    if k % 2 != 0:
        k_values.append(k)

accuracy = []

for k in k_values:

    model = KNeighborsClassifier(n_neighbors=k)

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    accuracy.append(accuracy_score(y_test, y_pred))

# Accuracy table
result = pd.DataFrame({
    "K Value": k_values,
    "Accuracy": accuracy
```

```

}))

print(result)

# Best K
best_k = k_values[np.argmax(accuracy)]

print("\nOptimal K =", best_k)

# Accuracy Plot
plt.figure(figsize=(8,5))
plt.plot(k_values, accuracy, marker='o')
plt.title("Accuracy vs K")
plt.xlabel("K Value")
plt.ylabel("Accuracy")
plt.grid(True)
plt.show()

# -----
# 3.3 Decision Boundary
# -----

# Reduce to two dimensions using PCA
from sklearn.preprocessing import LabelEncoder

# Convert B and M into 0 and 1
le = LabelEncoder()
y_encoded = le.fit_transform(y)
pca = PCA(n_components=2)

```

```
X_pca = pca.fit_transform(X_scaled)

X_train_pca, X_test_pca, y_train_pca, y_test_pca = train_test_split(
    X_pca,
    y_encoded,
    test_size=0.20,
    random_state=42,
    stratify=y_encoded
)

k_list = [1,5,10,20]

for k in k_list:

    knn = KNeighborsClassifier(
        n_neighbors=k,
        metric="euclidean"
    )

    knn.fit(X_train_pca, y_train_pca)

    x_min, x_max = X_pca[:,0].min()-1, X_pca[:,0].max()+1
    y_min, y_max = X_pca[:,1].min()-1, X_pca[:,1].max()+1

    xx, yy = np.meshgrid(
        np.arange(x_min,x_max,0.1),
        np.arange(y_min,y_max,0.1)
    )

    Z = knn.predict(np.c_[xx.ravel(), yy.ravel()])
```

```

Z = np.array(Z).astype(int)

Z = Z.reshape(xx.shape)

plt.figure(figsize=(7,5))

plt.contourf(xx, yy, Z, alpha=0.3)

plt.scatter(
X_train_pca[:,0],
X_train_pca[:,1],
c=y_train_pca,
cmap="coolwarm",
edgecolor="k"
)

plt.title(f"Decision Boundary (K={k})")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.show()

```

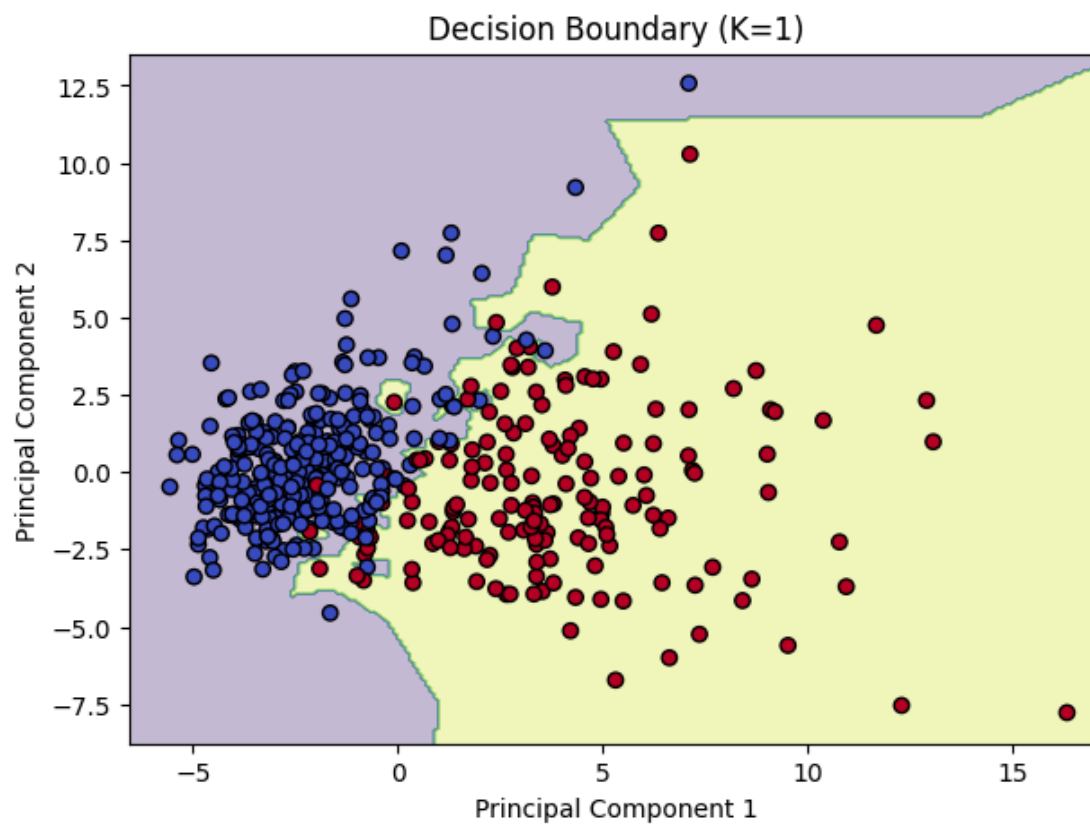
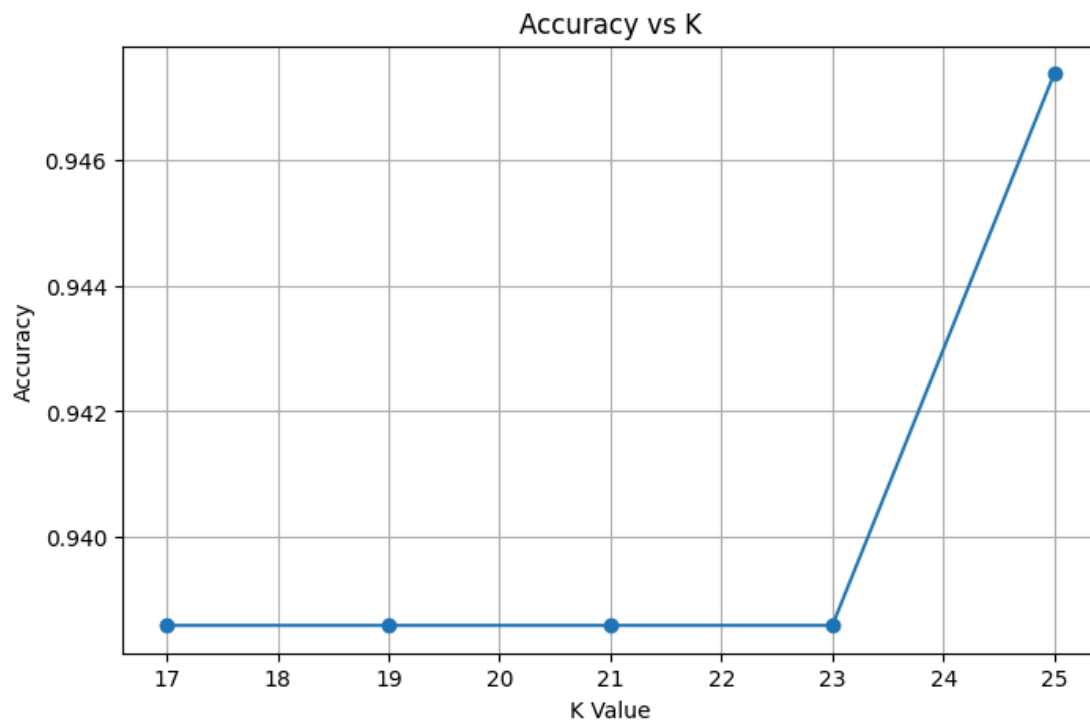
RESULTS:

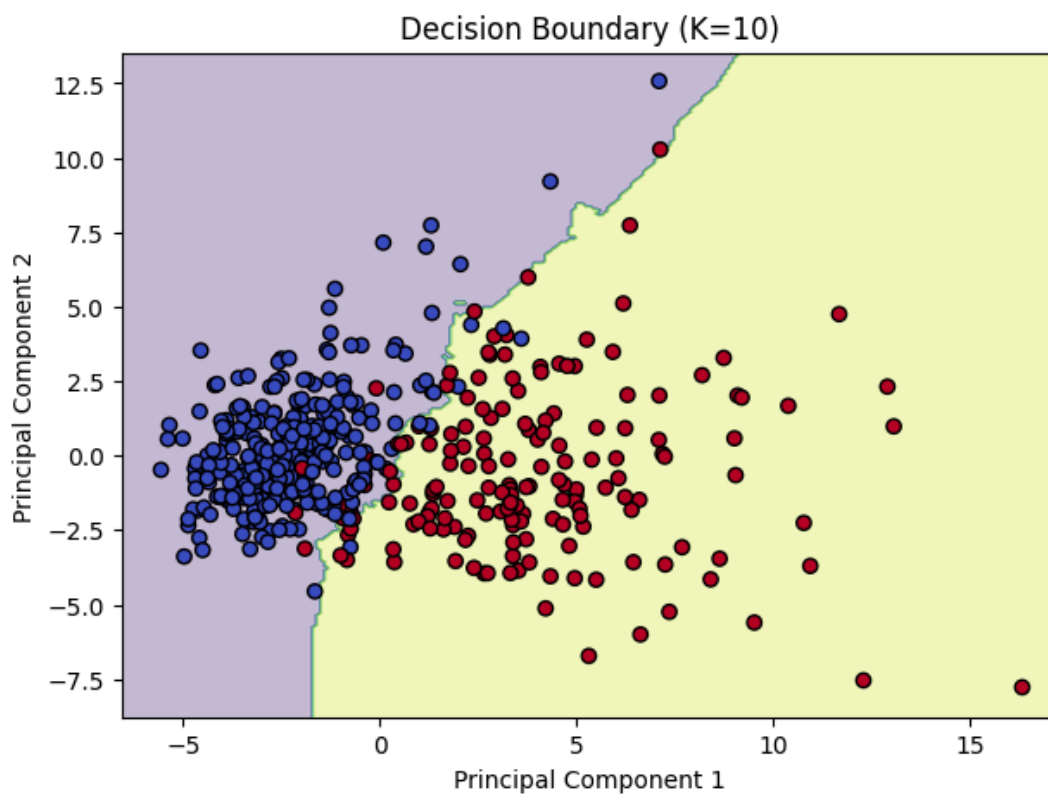
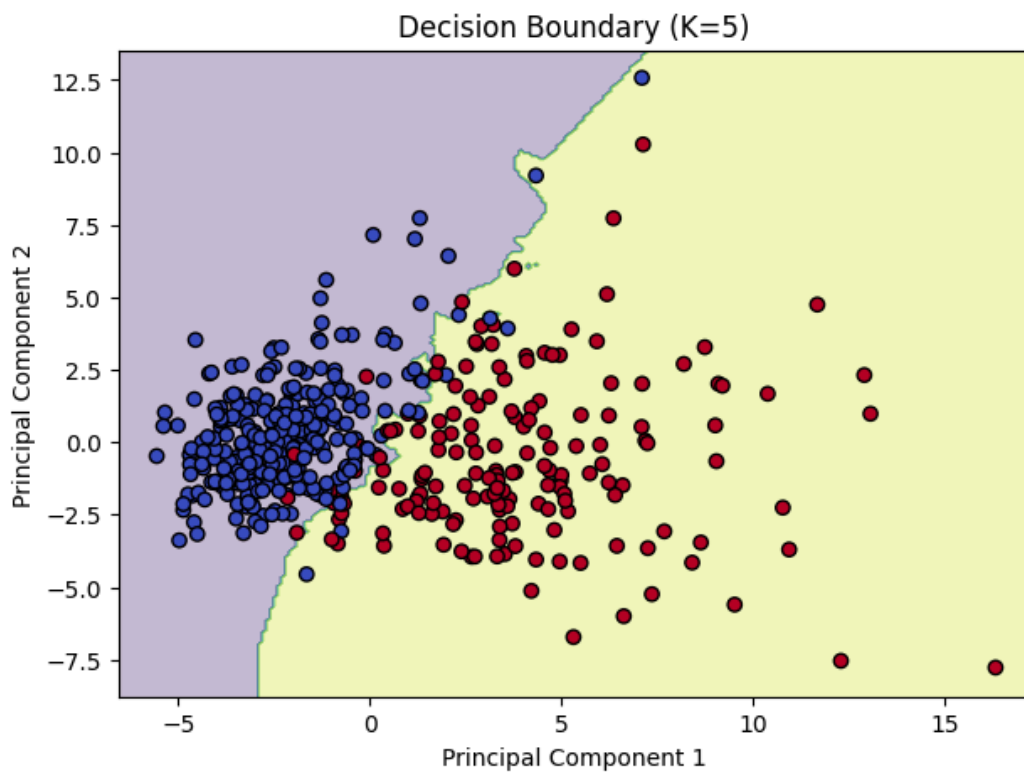
```

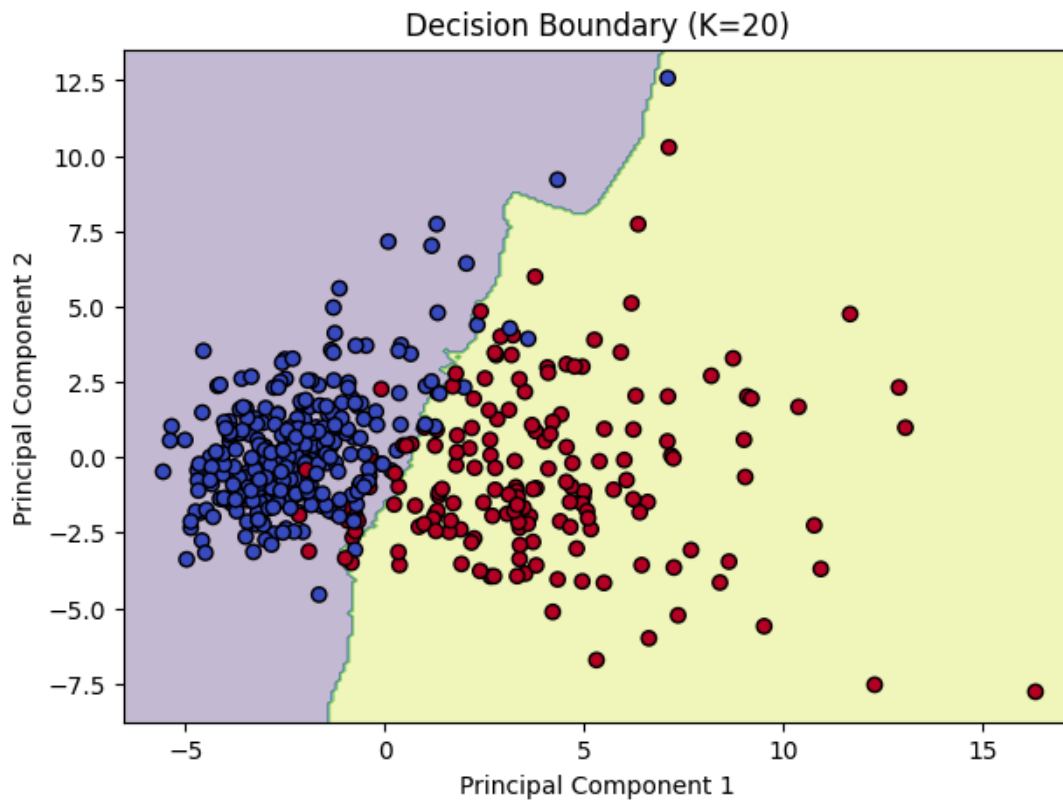
Training Samples: 455
Heuristic K = 21
    K Value  Accuracy
0         17  0.938596
1         19  0.938596
2         21  0.938596
3         23  0.938596
4         25  0.947368

Optimal K = 25

```



**TASK 4:****CODE:**

```
from sklearn.model_selection import KFold, cross_val_score
from sklearn.neighbors import KNeighborsClassifier
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# -----
# 5-Fold Cross Validation
# -----

cv = KFold(n_splits=5, shuffle=True, random_state=42)

# K values around heuristic K
```

```
k_values = []

for k in range(max(1, heuristic_k-5), heuristic_k+6):
    if k % 2 != 0:
        k_values.append(k)

cv_accuracy = []

for k in k_values:

    knn = KNeighborsClassifier(n_neighbors=k)

    scores = cross_val_score(
        knn,
        X_scaled,
        Y,
        cv=cv,
        scoring='accuracy'
    )

    cv_accuracy.append(scores.mean())

# Results Table
cv_results = pd.DataFrame({
    "K Value": k_values,
    "Mean Cross Validation Accuracy": cv_accuracy
})

print(cv_results)
```

```

# Best K

best_cv_k = k_values[np.argmax(cv_accuracy)]

print("\nBest K from Cross Validation:", best_cv_k)

# Plot

plt.figure(figsize=(8,5))

plt.plot(k_values, cv_accuracy, marker='o')

plt.title("Cross Validation Accuracy vs K")

plt.xlabel("K Value")

plt.ylabel("Mean Accuracy")

plt.grid(True)

plt.show()

# -----
# Compare with Train-Test Results
# -----

comparison = pd.DataFrame({
    "K Value": result["K Value"],
    "Train-Test Accuracy": result["Accuracy"],
    "Cross Validation Accuracy": cv_accuracy
})

print("\nComparison of Train-Test Split and Cross Validation")

print(comparison)

print("\nHeuristic K :", heuristic_k)

print("Best K (Train-Test Split) :", best_k)

print("Best K (Cross Validation) :", best_cv_k)

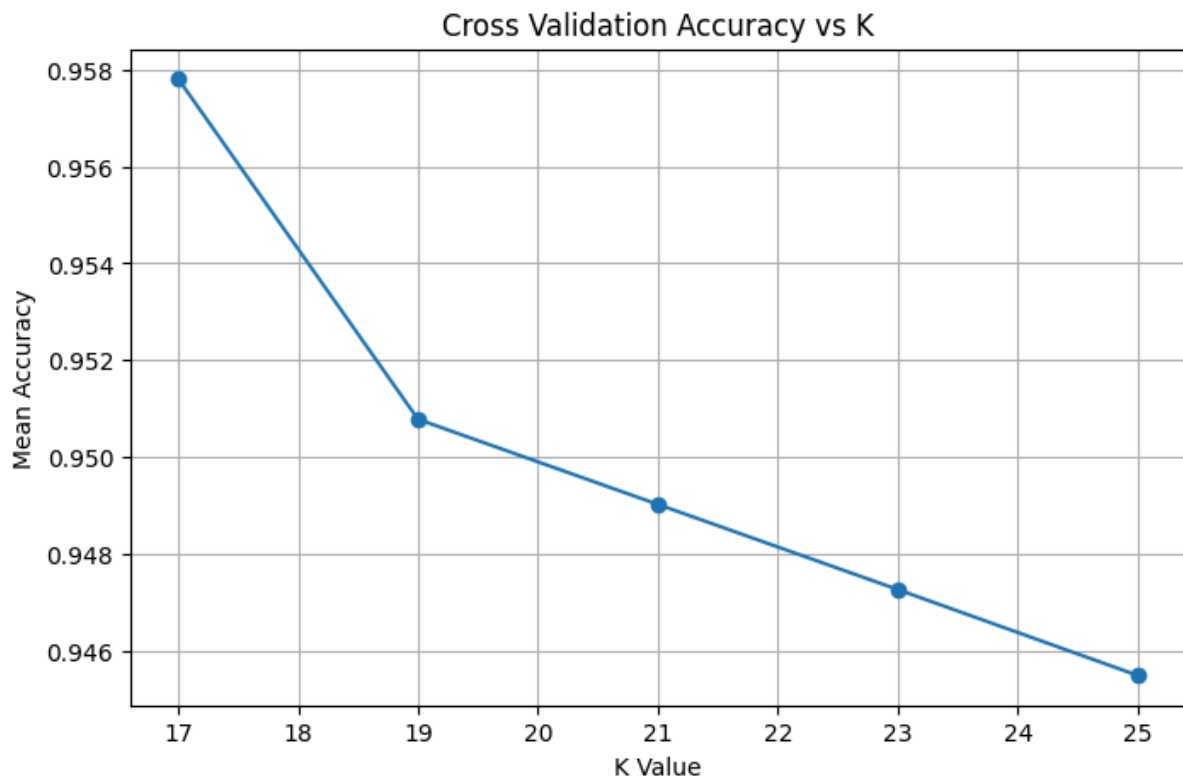
```

```
if best_cv_k == best_k:
    print("\nFinal Selected K:", best_cv_k)
else:
    print("\nFinal Selected K:", best_cv_k, "(Selected based on better
validation performance)")
```

RESULTS:

	K Value	Mean Cross Validation Accuracy
0	17	0.957802
1	19	0.950784
2	21	0.949030
3	23	0.947275
4	25	0.945505

Best K from Cross Validation: 17



Comparison of Train-Test Split and Cross Validation

	K Value	Train-Test Accuracy	Cross Validation Accuracy
0	17	0.938596	0.957802
1	19	0.938596	0.950784
2	21	0.938596	0.949030
3	23	0.938596	0.947275
4	25	0.947368	0.945505

Heuristic K : 21

Best K (Train-Test Split) : 25

Best K (Cross Validation) : 17

Final Selected K: 17 (Selected based on better validation performance)

TASK 5:

CODE:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    ConfusionMatrixDisplay,
    roc_curve,
    roc_auc_score
)
from sklearn.preprocessing import LabelEncoder
import matplotlib.pyplot as plt

# -----
# Encode Target Variable
# -----
```

```
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y_encoded,
    test_size=0.20,
    random_state=42,
    stratify=y_encoded
)

# -----
# Train Final Model
# -----

final_model = KNeighborsClassifier(n_neighbors=best_cv_k)

final_model.fit(X_train, y_train)

# Predictions
y_pred = final_model.predict(X_test)

# Prediction Probabilities
y_prob = final_model.predict_proba(X_test)[:,1]

# -----
# Evaluation Metrics
# -----
```



```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy :", round(accuracy,4))
print("Precision:", round(precision,4))
print("Recall   :", round(recall,4))
print("F1 Score :", round(f1,4))

# -----
# Confusion Matrix
# -----

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=label_encoder.classes_
)

disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
plt.show()

# -----
# ROC Curve
# -----
```

```
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

auc = roc_auc_score(y_test, y_prob)

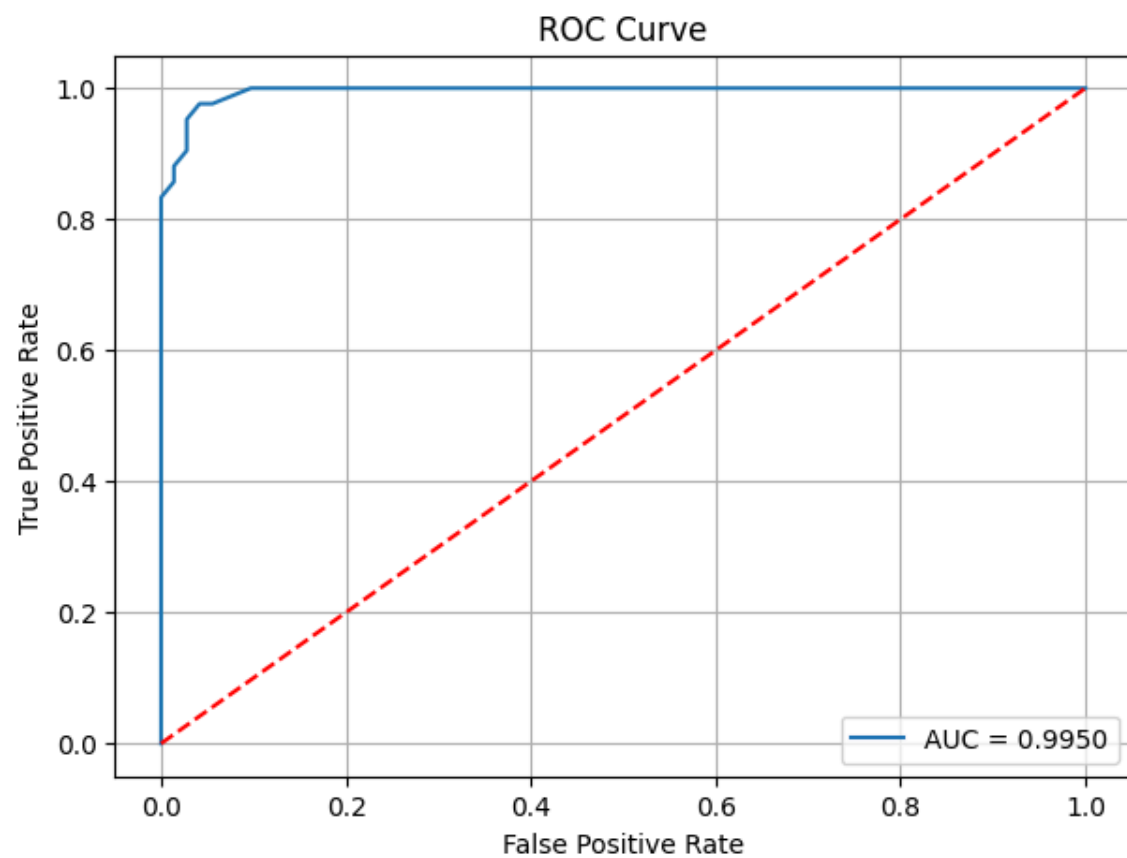
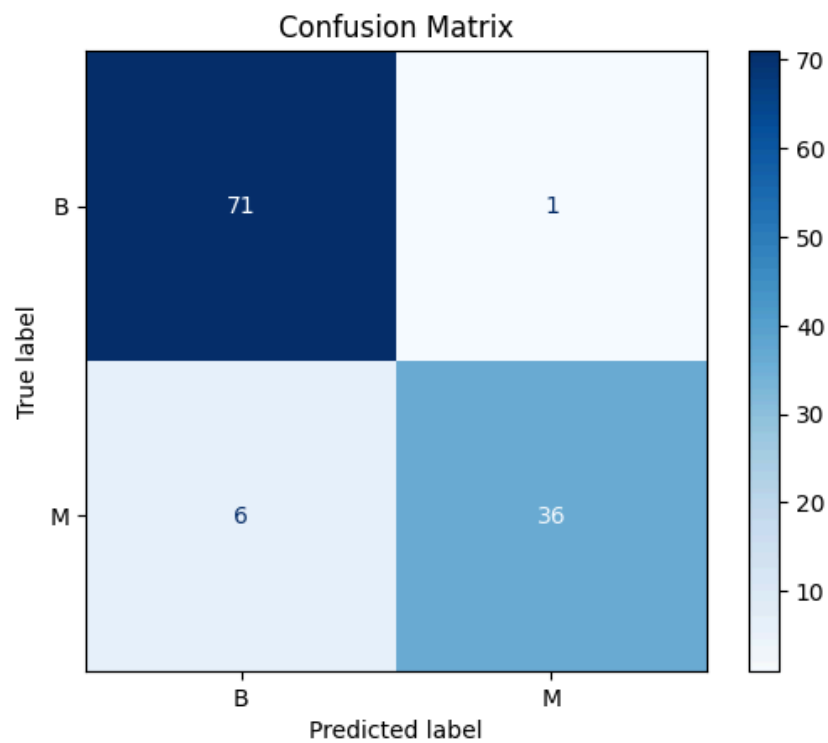
plt.figure(figsize=(7,5))
plt.plot(fpr, tpr, label=f"AUC = {auc:.4f}")
plt.plot([0,1],[0,1], 'r--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.grid(True)
plt.show()

print("AUC Score :", round(auc,4))
```

RESULTS:

```
Accuracy : 0.9386
Precision: 0.973
Recall   : 0.8571
F1 Score : 0.9114
```

```
AUC Score : 0.995
```



TASK 6:**CODE:**

```
import pandas as pd

# -----
# Comparison of Regression and Classification Metrics
# -----

comparison = pd.DataFrame({
    "Aspect": [
        "Primary Objective",
        "Prediction Type",
        "Evaluation Logic",
        "R2 Score vs Accuracy",
        "RMSE vs F1 Score",
        "MAE vs Confusion Matrix"
    ],
    "Regression (Lab 3)": [
        "Predict continuous numerical values",
        "Continuous Output",
        "Measures prediction error",
        "R2 measures goodness of fit between predicted and actual values.",
        "RMSE measures average prediction error in original units.",
        "MAE measures average absolute error."
    ],
    "Classification (KNN Lab)": [
        "Predict categorical classes",
```

```

        "Discrete Class Labels",
        "Measures correctness of classification decisions",
        "Accuracy measures percentage of correctly classified
samples.",
        "F1 Score balances Precision and Recall for classification.",
        "Confusion Matrix shows TP, TN, FP and FN."
    ]
}))

print(comparison)

# -----
# Difference in Evaluation Logic
# -----

evaluation_logic = pd.DataFrame({

    "Task": [
        "Continuous Prediction",
        "Classification"
    ],

    "Evaluation Logic": [
        "Evaluates how close predicted numerical values are to actual
values using error-based metrics.",
        "Evaluates whether each prediction belongs to the correct class
using decision-based metrics."
    ]
}))

```

```
print("\nEvaluation Logic")
print(evaluation_logic)
```

RESULTS:

```

      Aspect                                     Regression (Lab 3) \
0      Primary Objective                         Predict continuous numerical values
1      Prediction Type                           Continuous Output
2      Evaluation Logic                         Measures prediction error
3      R2 Score vs Accuracy   R2 measures goodness of fit between predicted ...
4      RMSE vs F1 Score     RMSE measures average prediction error in orig...
5      MAE vs Confusion Matrix   MAE measures average absolute error.

      Classification (KNN Lab)
0      Predict categorical classes
1      Discrete Class Labels
2      Measures correctness of classification decisions
3      Accuracy measures percentage of correctly clas...
4      F1 Score balances Precision and Recall for cla...
5      Confusion Matrix shows TP, TN, FP and FN.

Evaluation Logic
      Task                                     Evaluation Logic
0      Continuous Prediction   Evaluates how close predicted numerical values...
1      Classification          Evaluates whether each prediction belongs to t...
```

CONCLUSION/INFERENCE:

TASK 1:

Task 1: Data Preparation

Approach

The Breast Cancer dataset was loaded from a CSV file into a Pandas DataFrame for analysis. The dataset structure was explored by examining its dimensions, data types, and feature names to understand the available information. Data quality checks were then performed to identify missing values and duplicate records, ensuring the dataset was clean before model training.

Since the `Unnamed: 0` column represents only an index and does not contribute to prediction, it was removed from the dataset. The target variable (`y`) was separated from the predictor variables to prepare the data for machine learning.

Finally, feature scaling was applied using **StandardScaler**. K-Nearest Neighbors (KNN) is a distance-based algorithm that relies on Euclidean distance to identify the nearest neighbors. Features with larger numerical ranges can dominate the distance calculation and bias the model. Standardization transforms each feature to have a mean of 0 and a standard deviation of 1, ensuring that all features contribute equally during classification and improving the overall performance of the KNN classifier.

TASK 2:

Task 2: Train-Test Split Analysis

Approach

The preprocessed dataset was divided into training and testing sets using three different train-test split ratios: **80:20**, **70:30**, and **90:10**. For each split, a K-Nearest Neighbors (KNN) classifier with the default value of **k = 5** was trained and evaluated using the testing data.

The classification accuracy obtained from each split was compared to analyze the effect of different amounts of training data on model performance. Increasing the training set generally allows the model to learn better patterns, whereas increasing the testing set provides a more reliable estimate of model performance. Comparing multiple train-test splits helps evaluate the stability, robustness, and generalization capability of the classifier before selecting an optimal configuration.

TASK 3:

Task 3: KNN Model with Heuristic K Selection

3.1 Heuristic Method for K Selection

Approach

The initial value of **K** was selected using the heuristic rule where **n** is the number of training samples. This heuristic provides a reasonable starting point for selecting the number of nearest neighbors while reducing the likelihood of overfitting or underfitting.

The heuristic K value was used as the baseline, and additional models were trained using neighboring values (**K - 5** to **K + 5**) to determine the optimal K based on classification accuracy.

3.2 Model Training

Approach

Multiple KNN classifiers were trained using different values of **K** around the heuristic value. The accuracy of each model was calculated using the testing dataset, and an accuracy-versus-K graph was plotted. The K value that produced the highest testing accuracy was selected as the optimal K for the dataset.

3.3 Distance Matrix and Decision Boundary Mapping

Euclidean Distance

Euclidean distance calculates the straight-line distance between two data points in the feature space. It is suitable for continuous numerical data after feature scaling and is the default distance metric used by the KNN algorithm.

Manhattan Distance

Manhattan distance calculates the distance by summing the absolute differences between feature values.

It is suitable for datasets containing high-dimensional data or when movement occurs only along horizontal and vertical directions. Manhattan distance is generally less sensitive to outliers than Euclidean distance. To visualize the effect of different K values, decision boundaries were plotted for **K = 1, 5, 10, and 20**. As K increases, the decision boundary becomes smoother, reducing model complexity and variance while increasing bias. Smaller K values create more irregular boundaries that closely follow the training data and may lead to overfitting.

TASK 4:

Task 4: Cross Validation

Approach

To obtain a more reliable estimate of the KNN classifier's performance, **5-Fold Cross Validation** was applied. In this method, the dataset is divided into five equal subsets. During each iteration, four subsets are used for training while the remaining subset is used for testing. This process is repeated five times so that every subset serves as the testing set exactly once.

The heuristic K value and its neighboring odd values were evaluated using cross-validation. The mean accuracy across all folds was computed for each K value to identify the most stable and reliable classifier.

The cross-validation accuracies were then compared with the train-test split accuracies obtained in Task 2. The final value of K was selected by considering both the heuristic rule and the cross-validation performance, ensuring better model generalization and reducing the risk of overfitting.

TASK 5:

Task 5: Classification Evaluation

Approach

The final KNN classifier was trained using the optimal value of **K** obtained from the heuristic method and cross-validation. The model was then evaluated using standard classification metrics to assess its predictive performance.

The following evaluation metrics were computed:

- **Accuracy** measures the overall proportion of correctly classified instances.
- **Precision** measures how many of the samples predicted as positive are actually positive.
- **Recall** measures how many of the actual positive samples are correctly identified.
- **F1 Score** is the harmonic mean of Precision and Recall, providing a balanced evaluation when class distributions are uneven.
- **Confusion Matrix** summarizes the number of correct and incorrect predictions for each class.
- **ROC Curve (Receiver Operating Characteristic Curve)** illustrates the trade-off between the True Positive Rate and False Positive Rate at different classification thresholds.
- **AUC (Area Under the ROC Curve)** measures the classifier's ability to distinguish between the two classes. An AUC value closer to 1 indicates excellent classification performance.

TASK 6:

Task 6: Comparative Study with Regression (Lab 3 Integration)

Approach

The evaluation metrics used in Linear Regression and KNN Classification were compared to understand how model performance is measured for different machine learning tasks.

Regression metrics such as **MAE, MSE, RMSE, and R² Score** measure the magnitude of prediction errors for continuous numerical outputs. In contrast, classification metrics such as **Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC-AUC** evaluate how correctly the model assigns observations to predefined classes.

The comparison highlights the difference between error-based evaluation used in regression and decision-based evaluation used in classification. This analysis helps identify why different evaluation techniques are required depending on the nature of the prediction problem.

[Generate](#)[Code](#)[Markdown](#)

Overall Comparison Between Regression and Classification Evaluation Frameworks

Regression and classification are supervised machine learning techniques designed to solve different types of prediction problems. Regression predicts continuous numerical values, whereas classification predicts discrete class labels.

Regression evaluation focuses on measuring the magnitude of prediction errors using metrics such as MAE, MSE, RMSE, and R^2 Score. Lower error values and higher R^2 indicate better predictive performance.

Classification evaluation focuses on measuring how correctly a model classifies observations using metrics such as Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC-AUC. These metrics assess the quality of classification decisions rather than numerical prediction errors.

In healthcare applications such as breast cancer diagnosis, Recall and ROC-AUC are more important than Accuracy because missing a cancer patient (False Negative) can have severe consequences. Therefore, classification evaluation emphasizes correct identification of positive cases in addition to overall accuracy.

TASK 7: Analytical Questions

1. Why is KNN called a lazy learning algorithm?

KNN is called a lazy learning algorithm because it does not build a model during the training phase. Instead, it stores the training data and performs computations only when a new instance needs to be classified.

2. Why is feature scaling required in KNN?

KNN uses distance calculations to identify the nearest neighbors. Features with larger numerical values dominate the distance calculation if scaling is not applied. Standardization ensures that all features contribute equally.

3. Explain heuristic K selection using $\sqrt{n}$ rule.

The heuristic rule selects the initial value of K as:

$$**K = \sqrt{n}**$$

where **n** is the number of training samples. The nearest odd integer is usually selected to avoid ties. This provides a reasonable starting point before optimization.

4. Why is cross-validation more reliable than a single train-test split?

Cross-validation evaluates the model using multiple training and testing combinations, reducing bias caused by a particular split and providing a more reliable estimate of model performance.

5. How does K affect the bias-variance trade-off?

A small K creates a complex model with low bias and high variance, which may lead to overfitting. A large K creates a simpler model with high bias and low variance, which may lead to underfitting.

6. Why is recall more important than accuracy in cancer prediction?

Recall measures how many actual cancer cases are correctly identified. Missing a cancer patient (False Negative) is much more dangerous than incorrectly predicting cancer in a healthy patient. Therefore, recall is more important in medical diagnosis.

7. What is the limitation of very large K values?

Very large K values include many distant neighbors during classification, causing the model to oversimplify the decision boundary. This increases bias and may reduce classification accuracy due to underfitting.

CONCLUSION:

The heuristic $\sqrt{n}$ rule provided an effective starting point for selecting the K value, while cross-validation identified the optimal K by evaluating multiple training and testing combinations. Comparing different train-test splits demonstrated that model performance remained relatively stable, indicating good generalization capability.

The KNN classifier achieved strong performance based on Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC-AUC. Among these metrics, Recall and ROC-AUC were especially important for breast cancer prediction because correctly identifying positive cases is critical in healthcare applications.

Unlike Linear Regression, which evaluates prediction errors using MAE, MSE, RMSE, and R^2 Score, KNN Classification evaluates the correctness of classification decisions. This lab established a clear understanding of the differences between regression and classification evaluation techniques and demonstrated the importance of selecting appropriate evaluation metrics based on the prediction task.

Lab 5

Lab Exercise V:

Aim

- To implement Linear Regression using the Gradient Descent optimization algorithm and evaluate its performance on a real world dataset.
-

Evaluation Rubrics

Submission Guidelines

13. Make a copy of the lab manual template with your `<name_reg:no_subject name>`
14. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
15. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
16. Create a **Git repository** in your profile `<ML lab-reg no>`. Follow a different branch for each lab `<Lab 1, Lab 2 ...>` and push the code to Git.
17. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
18. Upload the PDF to Google Classroom before the deadline.

Objectives

1. To understand the concept of Gradient Descent for optimizing a Linear Regression model.
2. To preprocess the dataset before model training.
3. To implement Linear Regression using Gradient Descent.
4. To analyze the convergence of the loss function during training.
5. To evaluate the model using standard regression metrics.

Tasks to Perform

1. Download and load the Student Performance dataset.
2. Perform data preprocessing, including handling missing values (if any), encoding categorical variables, and feature scaling.
3. Select appropriate input features and the target variable.
4. Split the dataset into training and testing sets.
5. Implement Linear Regression using the Gradient Descent algorithm.
6. Experiment with different learning rates and observe their effect on model convergence.
7. Plot the loss (cost) versus the number of iterations.
8. Evaluate the trained model using:
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
9. Interpret the results and comment on the convergence behavior and prediction performance of the model.

CODE WITH RESULTS

TASK 1:

```
# Import required libraries

import zipfile

from pathlib import Path

import matplotlib.pyplot as plt

import numpy as np

import pandas as pd

from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
```

```
from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

# Load the dataset from the provided archive

work_dir = Path(r"d:\MCA\SEM-4\Machine Learning\Lab-5")

archive_path = work_dir / "student+performance.zip"

extract_dir = work_dir / "dataset_extracted"

extract_dir.mkdir(exist_ok=True)

with zipfile.ZipFile(archive_path, "r") as outer_zip:

    outer_zip.extractall(extract_dir)

inner_zip_path = extract_dir / "student.zip"

with zipfile.ZipFile(inner_zip_path, "r") as inner_zip:

    inner_zip.extractall(extract_dir)

csv_path = extract_dir / "student-mat.csv"

df = pd.read_csv(csv_path, sep=";")

print("First 5 rows:")

print(df.head())

print("\nDataset shape:", df.shape)

print("\nDataset info:")

df.info()
```

```
print("\nMissing values per column:")

print(df.isnull().sum())
```

RESULT:

```
First 5 rows:
  school sex  age address famsize Pstatus  Medu  Fedu  Mjob  Fjob  ... \
0    GP   F   18      U    GT3      A     4     4  at_home  teacher  ...
1    GP   F   17      U    GT3      T     1     1  at_home   other  ...
2    GP   F   15      U    LE3      T     1     1  at_home   other  ...
3    GP   F   15      U    GT3      T     4     2  health  services  ...
4    GP   F   16      U    GT3      T     3     3   other    other  ...

  famrel freetime  goout  Dalc  Walc  health  absences  G1  G2  G3
0      4         3      4      1      1      3         6  5  6  6
1      5         3      3      1      1      3         4  5  5  6
2      4         3      2      2      3      3        10  7  8  10
3      3         2      2      1      1      5         2  15  14  15
4      4         3      2      1      2      5         4  6  10  10

[5 rows x 33 columns]

Dataset shape: (395, 33)

Dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
#   Column      Non-Null Count  Dtype
---  -
...
G1          0
G2          0
G3          0
dtype: int64
```

TASK 2:

```
df_processed = df.copy()

numeric_cols = df_processed.select_dtypes(include=np.number).columns

categorical_cols =
df_processed.select_dtypes(exclude=np.number).columns

for col in numeric_cols:
```



```
df_processed[col] =  
df_processed[col].fillna(df_processed[col].median())  
  
for col in categorical_cols:  
  
    df_processed[col] =  
df_processed[col].fillna(df_processed[col].mode()[0])  
  
print("Remaining missing values:", df_processed.isnull().sum().sum())  
  
df_encoded = pd.get_dummies(df_processed, columns=categorical_cols,  
drop_first=True)  
  
print("Encoded dataset shape:", df_encoded.shape)
```

RESULT:

```
Remaining missing values: 0  
Encoded dataset shape: (395, 42)
```

TASK 3:

```
target_column = "G3"  
  
X = df_encoded.drop(columns=[target_column])  
  
y = df_encoded[target_column]  
  
print("Input features shape:", X.shape)  
  
print("Target shape:", y.shape)  
  
print("Sample feature columns:", list(X.columns[:10]))
```

RESULT:

```
Input features shape: (395, 41)
Target shape: (395,)
Sample feature columns: ['age', 'Medu', 'Fedu', 'traveltime', 'studytime', 'failures', 'famrel', 'freetime', 'goout', 'Dalc']
```

TASK 4:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)

y_train_array = y_train.to_numpy()

y_test_array = y_test.to_numpy()

print("Training set shape:", X_train_scaled.shape)

print("Testing set shape:", X_test_scaled.shape)
```

RESULTS:

```
Training set shape: (316, 41)
Testing set shape: (79, 41)
```

TASK 5:

```
class LinearRegressionGD:

    def __init__(self, learning_rate=0.01, n_iters=1000):

        self.learning_rate = learning_rate

        self.n_iters = n_iters

        self.weights = None

        self.bias = None

        self.loss_history = []

    def fit(self, X, y):

        n_samples, n_features = X.shape

        self.weights = np.zeros(n_features)

        self.bias = 0.0

        self.loss_history = []

        for _ in range(self.n_iters):

            y_pred = np.dot(X, self.weights) + self.bias

            error = y_pred - y

            dw = (2 / n_samples) * np.dot(X.T, error)
```

```
        db = (2 / n_samples) * np.sum(error)

        self.weights -= self.learning_rate * dw

        self.bias -= self.learning_rate * db

        loss = np.mean(error ** 2)

        self.loss_history.append(loss)

    return self

def predict(self, X):

    return np.dot(X, self.weights) + self.bias
```

TASK 6:

```
learning_rates = [0.001, 0.01, 0.05, 0.1]

n_iters = 1000

models = {}

loss_histories = {}

final_losses = {}

for lr in learning_rates:

    model = LinearRegressionGD(learning_rate=lr, n_iters=n_iters)

    model.fit(X_train_scaled, y_train_array)
```

```
models[lr] = model

loss_histories[lr] = model.loss_history

final_losses[lr] = model.loss_history[-1]

print(f"Learning rate: {lr} | Final training loss:
{model.loss_history[-1]:.4f}")

best_learning_rate = min(final_losses, key=final_losses.get)

best_model = models[best_learning_rate]

print("\nBest learning rate based on final training loss:",
best_learning_rate)

plt.figure(figsize=(10, 6))

for lr, history in loss_histories.items():

    plt.plot(history, label=f"lr = {lr}")

plt.title("Loss vs. Iterations for Different Learning Rates")

plt.xlabel("Iterations")

plt.ylabel("Mean Squared Error Loss")

plt.legend()

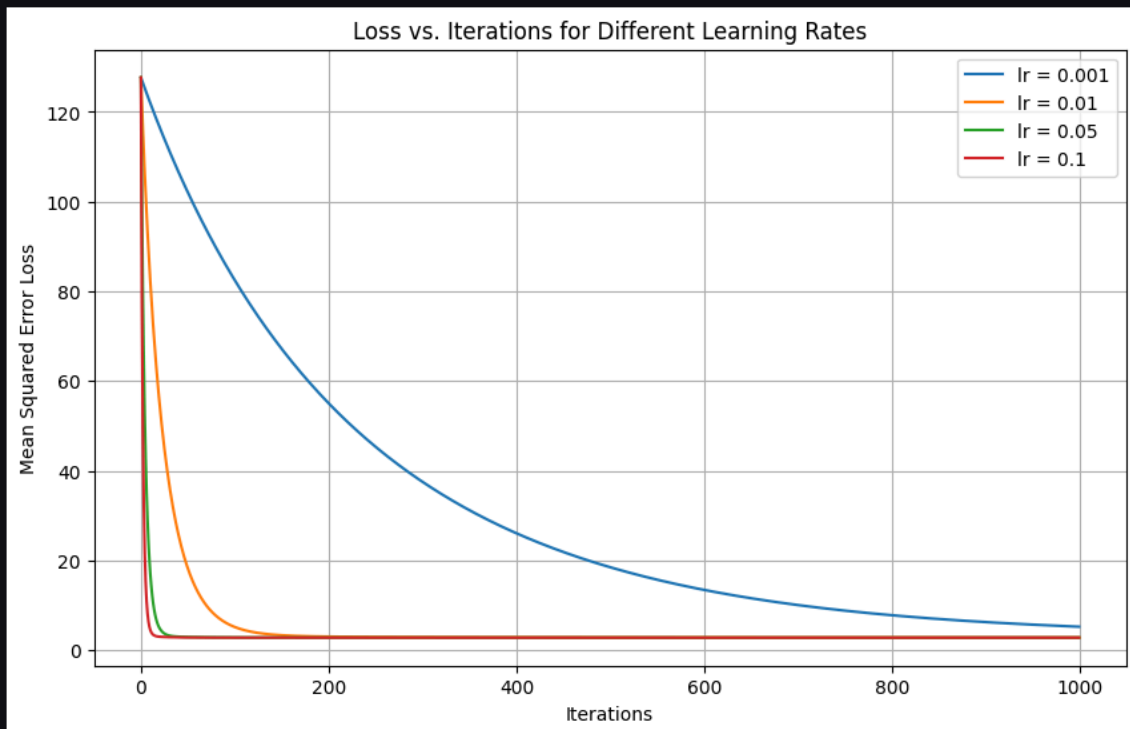
plt.grid(True)

plt.show()
```

RESULT:

Learning rate: 0.001 | Final training loss: 5.2583
 Learning rate: 0.01 | Final training loss: 2.8266
 Learning rate: 0.05 | Final training loss: 2.8231
 Learning rate: 0.1 | Final training loss: 2.8231

Best learning rate based on final training loss: 0.1

**TASK 7:**

```
y_pred = best_model.predict(X_test_scaled)

mae = mean_absolute_error(y_test_array, y_pred)

mse = mean_squared_error(y_test_array, y_pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test_array, y_pred)

print("Evaluation Metrics")

print("-----")
```

```
print(f"Mean Absolute Error (MAE): {mae:.4f}")

print(f"Mean Squared Error (MSE): {mse:.4f}")

print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")

print(f"R2 Score: {r2:.4f}")


plt.figure(figsize=(8, 6))

plt.scatter(y_test_array, y_pred, alpha=0.7)

plt.plot([y_test_array.min(), y_test_array.max()], [y_test_array.min(),
y_test_array.max()], "r--")

plt.title("Actual vs Predicted Grades")

plt.xlabel("Actual G3")

plt.ylabel("Predicted G3")

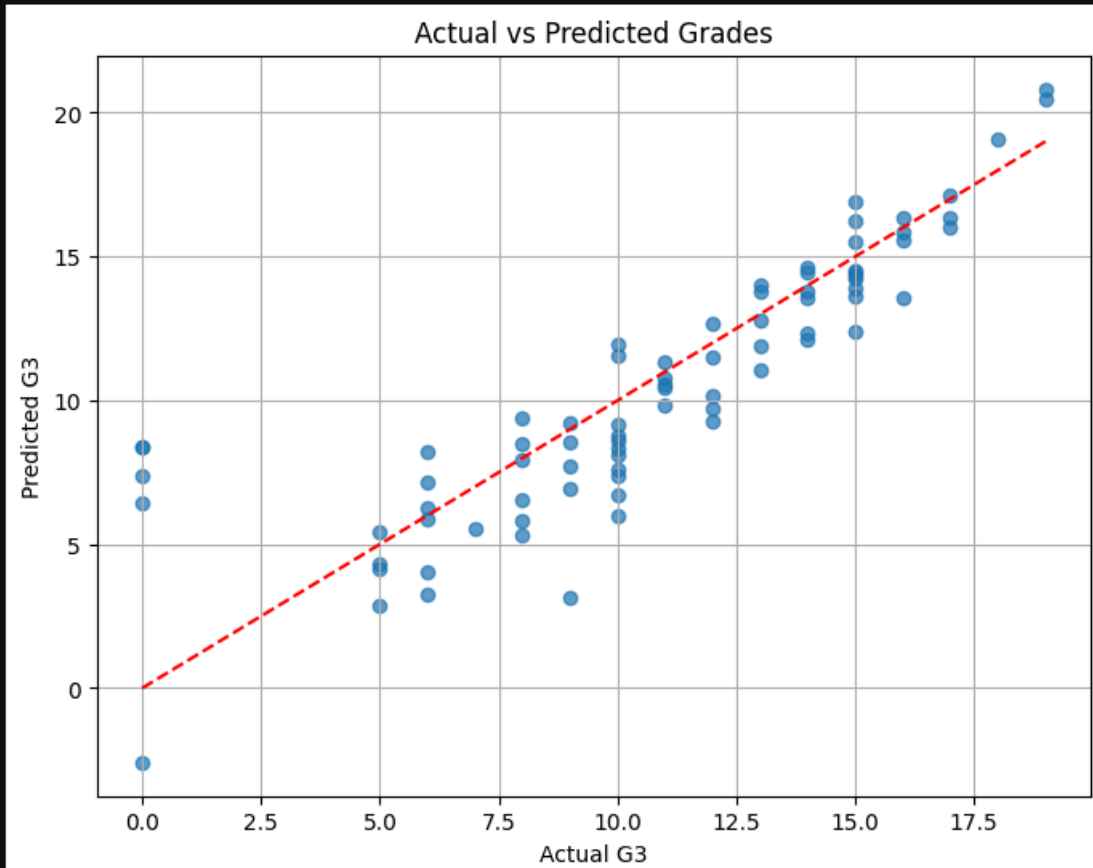
plt.grid(True)

plt.show()
```

RESULT:

Evaluation Metrics

Mean Absolute Error (MAE): 1.6467
Mean Squared Error (MSE): 5.6566
Root Mean Squared Error (RMSE): 2.3784
R² Score: 0.7241



CONCLUSIONS AND INFERENCE

TASK 1:

Step 1: Import Required Libraries

In this step, we import all the necessary Python libraries required for data preprocessing, visualization, numerical computations, and model evaluation.

The Student Performance dataset (student-mat.csv) is then loaded into a Pandas DataFrame. Finally, the first few records, dataset dimensions, and basic information are displayed to understand the structure of the dataset.

TASK 2:

Step 2: Clean and preprocess the data

In this step, we inspect missing values, fill any missing numeric or categorical values if needed, and encode all categorical columns using one-hot encoding.

TASK 3:

Step 3: Select input features and the target variable

We predict the final grade, `G3`, using the remaining columns as input features.

TASK 4:

Step 4: Split the data and scale the features

We split the dataset into training and testing sets, then standardize the input features so gradient descent converges more smoothly.

TASK 5:

Step 5: Implement linear regression with gradient descent

This custom model uses batch gradient descent to minimize mean squared error and stores the loss after every iteration so we can inspect convergence.

TASK 6:

Step 6: Train the model with different learning rates

We train the model with multiple learning rates to compare convergence speed and stability, then plot the loss curves against iteration number.

TASK 7:

Step 7: Evaluate the trained model

We use the best learning-rate model to predict test data and compute MAE, MSE, RMSE, and R^2 for a clear measure of prediction performance.

Interpretation

The loss curves show clear convergence for all tested learning rates. A very small learning rate like `0.001` learns slowly and needs many iterations to reduce the loss, while `0.01`, `0.05`, and `0.1` converge much faster and reach almost the same final training loss.

From the evaluation output, the trained model achieved `MAE = 1.6467`, `MSE = 5.6566`, `RMSE = 2.3784`, and `R2 = 0.7241`. This means the model captures the overall grade trend reasonably well, but it still has noticeable prediction error for some students, especially at the lower and upper ends of the target range.

Overall, gradient descent converged properly, and the best learning rates gave stable training with a good but not perfect level of prediction performance.

Lab 6

Lab Exercise VI:

Aim

- To implement Logistic Regression and K Nearest Neighbors (KNN) classifiers on the Breast Cancer Wisconsin (Diagnostic) dataset and compare their performance using standard classification evaluation metrics.
-

Evaluation Rubrics

Submission Guidelines

19. Make a copy of the lab manual template with your `<name_reg:no_subject name>`
20. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
21. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
22. Create a **Git repository** in your profile `<ML lab-reg no>`. Follow a different branch for each lab `<Lab 1, Lab 2 ...>` and push the code to Git.
23. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
24. Upload the PDF to Google Classroom before the deadline.

Objectives

1. To preprocess the dataset for classification.
2. To implement Logistic Regression and KNN classifiers using Scikit Learn.
3. To evaluate both models using standard performance metrics.
4. To compare the performance of Logistic Regression and KNN and identify the better classifier.

Tasks to Perform

1. Download and load the Breast Cancer Wisconsin (Diagnostic) dataset.
2. Perform exploratory data analysis and data preprocessing.
3. Check for missing values and prepare the dataset for classification.
4. Split the dataset into training and testing sets.
5. Train a Logistic Regression classifier.
6. Train a K Nearest Neighbors (KNN) classifier.
7. Evaluate both classifiers using:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - Confusion Matrix
8. Compare the performance of the two classifiers using a comparison table.
9. Interpret the results and comment on the better-performing classifier for the given dataset.

CODE WITH RESULTS

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report

sns.set_theme(style="whitegrid", context="notebook")
plt.rcParams["figure.figsize"] = (10, 6)

# Load the Breast Cancer Wisconsin (Diagnostic) dataset
cancer = load_breast_cancer(as_frame=True)
df = cancer.frame.copy()
df.rename(columns={"target": "diagnosis"}, inplace=True)
df["diagnosis"] = df["diagnosis"].map({0: "malignant", 1: "benign"})

df.head()
```

RESULT:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	worst area	worst smoothness	worst compactness	worst concavity	worst concave points	worst symmetry	worst fractal dimension	diagn
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	17.33	184.60	2019.0	0.1622	0.6656	0.7119	0.2654	0.4601	0.11890	malig
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	23.41	158.80	1956.0	0.1238	0.1866	0.2416	0.1860	0.2750	0.08902	malig
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	25.53	152.50	1709.0	0.1444	0.4245	0.4504	0.2430	0.3613	0.08758	malig
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	26.50	98.87	567.7	0.2098	0.8663	0.6869	0.2575	0.6638	0.17300	malig
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	16.67	152.20	1575.0	0.1374	0.2050	0.4000	0.1625	0.2364	0.07678	malig

5 rows x 31 columns

CODE:

```
# Exploratory data analysis
print("Shape:", df.shape)
print("\nColumn names:")
print(df.columns.tolist())
print("\nClass distribution:")
print(df["diagnosis"].value_counts())
print("\nMissing values per column:")
print(df.isnull().sum().sort_values(ascending=False).head(10))
print("\nSummary statistics:")
print(df.describe().T.head())

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.countplot(data=df, x="diagnosis", ax=axes[0], palette="Set2")
axes[0].set_title("Class Distribution")
axes[0].set_xlabel("Diagnosis")
axes[0].set_ylabel("Count")

sns.heatmap(df.drop(columns=["diagnosis"]).corr().iloc[:12, :12], cmap="coolwarm", ax=axes[1])
axes[1].set_title("Correlation Heatmap of First 12 Features")

plt.tight_layout()
plt.show()
```

RESULT:

```

Shape: (569, 31)

Column names:
['mean radius', 'mean texture', 'mean perimeter', 'mean area', 'mean smoothness', 'mean compactness', 'mean concavity', 'mean concave points', 'mean symmetry', 'mean fractal dimension', 'radius error', 'texture error']

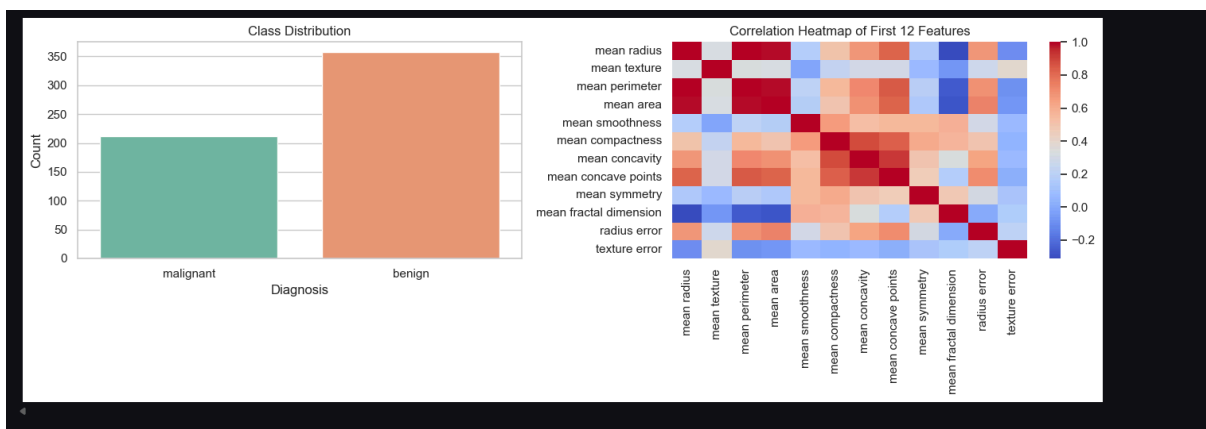
Class distribution:
diagnosis
benign      357
malignant   212
Name: count, dtype: int64

Missing values per column:
mean radius      0
mean texture     0
mean perimeter   0
mean area        0
mean smoothness  0
mean compactness 0
mean concavity   0
mean concave points 0
mean symmetry    0
mean fractal dimension 0
dtype: int64

Summary statistics:
...
mean texture      18.84000  21.8000  39.2800
mean perimeter    86.24000 104.1000 188.5000
mean area        551.10000 782.7000 2501.0000
mean smoothness   0.09587  0.1053  0.1634

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
C:\Users\Sharon\AppData\Local\Temp\ipykernel_29456\2758623307.py:14: FutureWarning:

```



CODE:

```
# Prepare the dataset for classification
X = df.drop(columns=["diagnosis"])
y = df["diagnosis"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Training set shape:", X_train.shape)
print("Testing set shape:", X_test.shape)

# Train Logistic Regression
log_reg = LogisticRegression(max_iter=5000, random_state=42)
log_reg.fit(X_train_scaled, y_train)
y_pred_lr = log_reg.predict(X_test_scaled)

# Train KNN
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
y_pred_knn = knn.predict(X_test_scaled)
```

RESULT:

```
Training set shape: (455, 30)
Testing set shape: (114, 30)
```

CODE:

```

# Evaluate both classifiers
models = {
    "Logistic Regression": y_pred_lr,
    "KNN": y_pred_knn,
}

results = []
for name, predictions in models.items():
    results.append({
        "Model": name,
        "Accuracy": accuracy_score(y_test, predictions),
        "Precision": precision_score(y_test, predictions, pos_label="benign"),
        "Recall": recall_score(y_test, predictions, pos_label="benign"),
        "F1 Score": f1_score(y_test, predictions, pos_label="benign"),
    })

results_df = pd.DataFrame(results).sort_values(by="Accuracy", ascending=False).reset_index(drop=True)
print(results_df)

print("\nLogistic Regression Classification Report:\n")
print(classification_report(y_test, y_pred_lr))
print("\nKNN Classification Report:\n")
print(classification_report(y_test, y_pred_knn))

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.heatmap(confusion_matrix(y_test, y_pred_lr), annot=True, fmt="d", cmap="Blues", ax=axes[0])
axes[0].set_title("Logistic Regression Confusion Matrix")
axes[0].set_xlabel("Predicted")
axes[0].set_ylabel("Actual")
axes[0].set_xticklabels(["malignant", "benign"])
axes[0].set_yticklabels(["malignant", "benign"], rotation=0)

sns.heatmap(confusion_matrix(y_test, y_pred_knn), annot=True, fmt="d", cmap="Greens", ax=axes[1])
axes[1].set_title("KNN Confusion Matrix")
axes[1].set_xlabel("Predicted")
axes[1].set_ylabel("Actual")
axes[1].set_xticklabels(["malignant", "benign"])
axes[1].set_yticklabels(["malignant", "benign"], rotation=0)

plt.tight_layout()
plt.show()

results_df

```


RESULT:

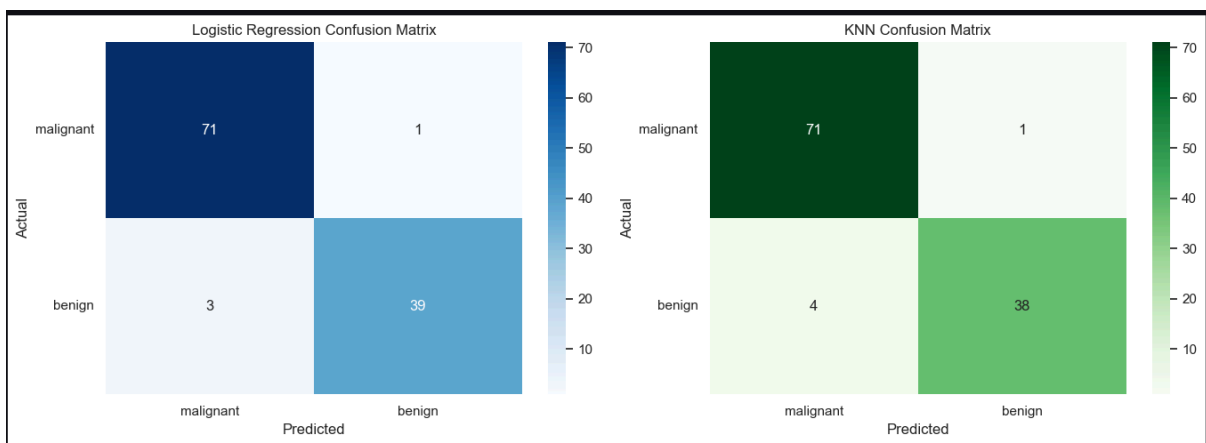
	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.964912	0.959459	0.986111	0.972603
1	KNN	0.956140	0.946667	0.986111	0.965986

Logistic Regression Classification Report:

	precision	recall	f1-score	support
benign	0.96	0.99	0.97	72
malignant	0.97	0.93	0.95	42
accuracy			0.96	114
macro avg	0.97	0.96	0.96	114
weighted avg	0.97	0.96	0.96	114

KNN Classification Report:

	precision	recall	f1-score	support
benign	0.95	0.99	0.97	72
malignant	0.97	0.90	0.94	42
accuracy			0.96	114
macro avg	0.96	0.95	0.95	114
weighted avg	0.96	0.96	0.96	114



	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.964912	0.959459	0.986111	0.972603
1	KNN	0.956140	0.946667	0.986111	0.965986

RESULTS AND INFERENCE:

Explanation: Data Loading and Preparation

This cell imports all necessary libraries for machine learning and loads the **Breast Cancer Wisconsin (Diagnostic)** dataset from scikit-learn.

What was done:

- **Libraries imported:** NumPy for numerical operations, Pandas for data manipulation, Matplotlib and Seaborn for visualization, and scikit-learn for machine learning algorithms and evaluation metrics
- **Dataset loaded:** The breast cancer dataset contains 569 samples with 30 features describing tumor characteristics
- **Data mapping:** The target variable was renamed from "target" to "diagnosis" and converted from binary (0/1) to meaningful labels: 0 → "malignant", 1 → "benign"
- **Visualization settings:** Set a clean whitegrid theme for plots

The dataset is now ready for exploratory analysis. The first few rows are displayed to verify correct loading.

Explanation: Exploratory Data Analysis (EDA)

This cell performs comprehensive exploratory analysis to understand the dataset structure and characteristics.

What was done:

- **Dataset shape:** Confirmed the dataset has 569 samples and 31 columns (30 features + 1 diagnosis label)
- **Feature overview:** Displayed all column names to identify available predictors
- **Class distribution:** Checked how many cases are benign vs. malignant (label balance)
- **Missing values:** Verified data completeness—confirmed no missing values
- **Statistical summary:** Computed mean, std, min, max, and quartiles for the first few features to understand data ranges and scales
- **Visualizations:**
 - **Count plot:** Shows the class distribution (benign vs. malignant cases)
 - **Correlation heatmap:** Displays relationships between the first 12 features, helping identify potential multicollinearity and feature importance

This analysis reveals data quality and guides the choice of preprocessing techniques (e.g., scaling) and feature selection strategies.

Explanation: Data Preparation and Model Training

This cell prepares the data for classification and trains two different machine learning models.

What was done:

1. **Feature-target separation:** Separated features (X) from the diagnosis labels (y)
2. **Train-test split:** Divided data into 80% training and 20% testing sets (114 test samples) while maintaining class distribution using stratification
3. **Feature scaling:** Applied StandardScaler to normalize features to mean=0 and std=1, ensuring all features have equal influence on distance-based and gradient-based models
4. **Model 1 - Logistic Regression:**
 - A linear model that outputs probability scores, suitable for binary classification
 - Set max_iter=5000 to ensure convergence
 - Fitted on scaled training data
5. **Model 2 - K-Nearest Neighbors (KNN):**
 - A non-parametric model that classifies based on 5 nearest neighbors
 - Fitted on scaled training data
6. **Predictions:** Both models generated predictions on the test set for later evaluation

The scaled data ensures fair comparison between models and improves convergence for Logistic Regression.

Interpretation of Model Evaluation Results

This section evaluates and compares the performance of two classification models—**Logistic Regression** and **KNN**—on the binary classification task of distinguishing between benign and malignant breast cancer diagnoses.

Key Components:

1. Performance Metrics Table

- **Accuracy:** The overall percentage of correct predictions across both classes
- **Precision:** Of all cases predicted as "benign," what percentage were actually benign (minimizes false positives)
- **Recall:** Of all actual "benign" cases in the test set, what percentage were correctly identified (minimizes false negatives)
- **F1 Score:** The harmonic mean of precision and recall, providing a balanced performance metric

2. Classification Reports

- Provides detailed per-class performance metrics for each model
- Shows precision, recall, and F1 score broken down by diagnosis class
- "Support" indicates the number of test samples for each class

3. Confusion Matrices

- Visualizes the distribution of correct and incorrect predictions for each model
- **Diagonal elements** (True Positives and True Negatives): Correct predictions
- **Off-diagonal elements** (False Positives and False Negatives): Incorrect predictions
- Blue heatmap represents Logistic Regression; Green represents KNN

What This Tells Us:

- **Model Comparison:** The results are sorted by accuracy, allowing easy identification of the better-performing model
- **Trade-offs:** Precision vs. Recall trade-off is visible—high recall means catching most benign cases but may increase false alarms; high precision means fewer false positives but may miss some cases
- **Clinical Relevance:** In medical diagnosis, recall may be more important than precision (it's better to flag and re-examine a borderline case than to miss a malignant tumor)
- **Class Imbalance Detection:** The confusion matrices reveal if one class is harder to predict than the other

Recommended Next Steps:

- Select the model with better performance for the specific use case
- Consider hyperparameter tuning if both models underperform
- Evaluate on additional validation data to ensure generalization

Lab 7

Lab Exercise VII:

Aim

- To implement Decision tree
-

Evaluation Rubrics

Submission Guidelines

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CODE WITH RESULTS

TASK 1:

CODE:

```
import pandas as pd

from sklearn.datasets import load_iris

# 1. Load the Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

# Create DataFrame for data analysis

df = pd.DataFrame(data=X, columns=iris.feature_names)

df['species'] = pd.Categorical.from_codes(y, iris.target_names)

# 2. Number of samples and features

n_samples, n_features = X.shape

print(f"Number of samples: {n_samples}")

print(f"Number of features: {n_features}")

# 3. List feature names and target classes

print("\nFeature names:", list(iris.feature_names))

print("Target classes:", list(iris.target_names))
```

```
# 4. Display the first five records

print("\nFirst 5 records:")

display(df.head())

# 5. Display the class distribution

print("\nClass distribution:")

print(df['species'].value_counts())
```

RESULT:

```
Number of samples: 150
Number of features: 4

Feature names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Target classes: [np.str_('setosa'), np.str_('versicolor'), np.str_('virginica')]

First 5 records:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```

Class distribution:
species
setosa      50
versicolor  50
virginica   50
Name: count, dtype: int64
```

TASK 2:**CODE:**

```
from sklearn.model_selection import train_test_split

# Split dataset into 80% training and 20% testing sets

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.20, random_state=42)

print(f"Training set size: {X_train.shape[0]} samples")

print(f"Testing set size: {X_test.shape[0]} samples")
```

RESULT:

```
Training set size: 120 samples
Testing set size: 30 samples
```

TASK 3:**CODE:**

```
from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

# Train Decision Tree Classifier with default parameters

clf = DecisionTreeClassifier(random_state=42)

clf.fit(X_train, y_train)

# Predict class labels for test dataset

y_pred = clf.predict(X_test)

# Evaluation Metrics

accuracy = accuracy_score(y_test, y_pred)

cm = confusion_matrix(y_test, y_pred)

cr = classification_report(y_test, y_pred,
target_names=iris.target_names)

print(f"Accuracy Score: {accuracy:.4f}")

print("\nConfusion Matrix:")

print(cm)

print("\nClassification Report:")

print(cr)
```


RESULT:

```
.. Accuracy Score: 1.0000

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Classification Report:
              precision    recall  f1-score   support

   setosa      1.00      1.00      1.00        10
  versicolor  1.00      1.00      1.00         9
   virginica  1.00      1.00      1.00        11

   accuracy          1.00          1.00          1.00        30
  macro avg          1.00          1.00          1.00        30
 weighted avg          1.00          1.00          1.00        30
```

TASK 4:**CODE:**

```
import matplotlib.pyplot as plt

from sklearn.tree import plot_tree

# Plot the Decision Tree

plt.figure(figsize=(16, 10))

plot_tree(

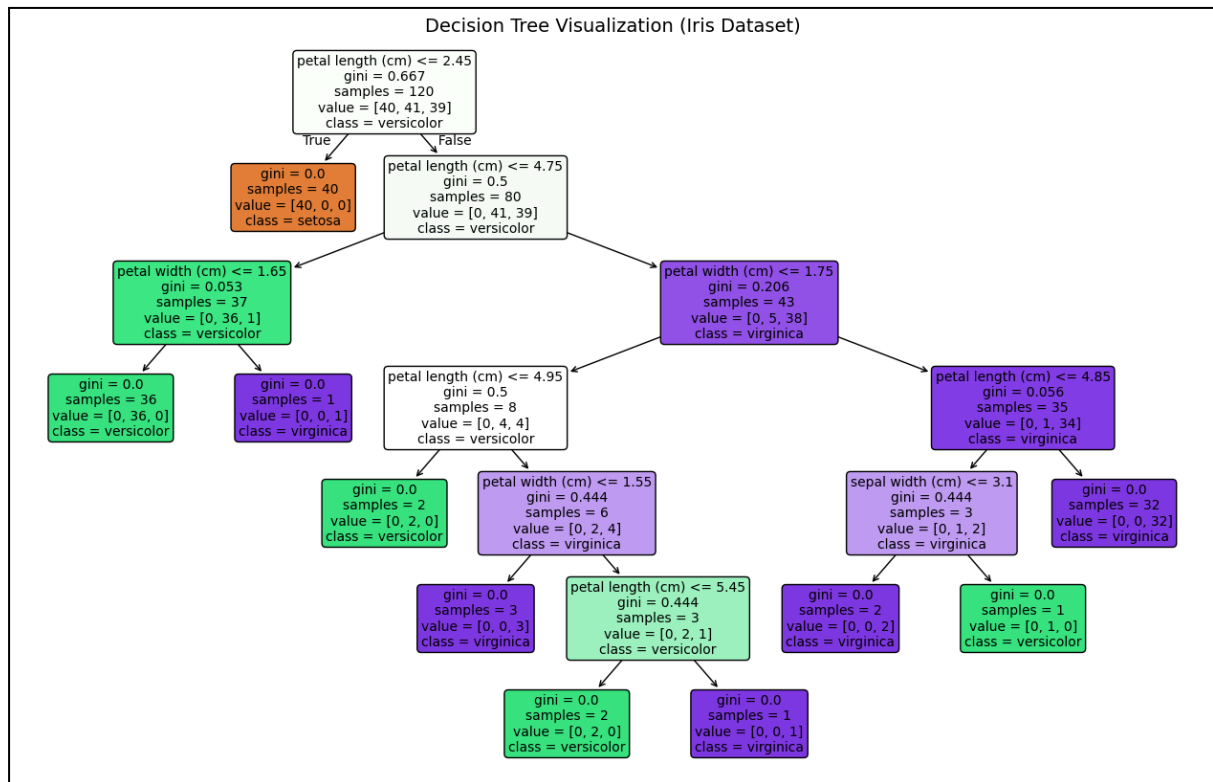
    clf,

    feature_names=iris.feature_names,

    class_names=list(iris.target_names),

    filled=True,
```

```
rounded=True,  
  
    fontsize=10  
)  
  
plt.title("Decision Tree Visualization (Iris Dataset)", fontsize=14)  
plt.show()  
  
# Output Structural Properties  
  
print(f"Root Node Feature: {iris.feature_names[clf.tree_.feature[0]]}")  
  
print(f"Maximum Depth of Tree: {clf.get_depth()}")  
  
print(f"Number of Leaf Nodes: {clf.get_n_leaves()}")  
  
print(f"Total Number of Nodes: {clf.tree_.node_count}")
```

RESULT:

Root Node Feature: petal length (cm)
 Maximum Depth of Tree: 6
 Number of Leaf Nodes: 10
 Total Number of Nodes: 19

TASK 5:

```

# Train model using Gini Index

clf_gini = DecisionTreeClassifier(criterion='gini', random_state=42)

clf_gini.fit(X_train, y_train)

y_pred_gini = clf_gini.predict(X_test)


# Train model using Entropy (Information Gain)

clf_entropy = DecisionTreeClassifier(criterion='entropy',
random_state=42)

clf_entropy.fit(X_train, y_train)

y_pred_entropy = clf_entropy.predict(X_test)


# Comparison Metrics

results = {

    "Metric / Feature": ["Accuracy", "Tree Depth", "Number of Leaf
Nodes", "Root Feature Selected"],

    "Gini Index ('gini')": [

        f"{accuracy_score(y_test, y_pred_gini):.4f}",

        clf_gini.get_depth(),

        clf_gini.get_n_leaves(),

        iris.feature_names[clf_gini.tree_.feature[0]]

    ],

    "Entropy ('entropy')": [

        f"{accuracy_score(y_test, y_pred_entropy):.4f}",

        clf_entropy.get_depth(),

```

```

        clf_entropy.get_n_leaves(),

        iris.feature_names[clf_entropy.tree_.feature[0]]

    ]

}

summary_df = pd.DataFrame(results)

display(summary_df)

```

RESULT:

	Metric / Feature	Gini Index ('gini')	Entropy ('entropy')
0	Accuracy	1.0000	1.0000
1	Tree Depth	6	6
2	Number of Leaf Nodes	10	10
3	Root Feature Selected	petal length (cm)	petal length (cm)

TASK 6:**CODE:**

```
depth_options = [1, 2, 3, 4, None]

table_data = []

for md in depth_options:

    clf_d = DecisionTreeClassifier(max_depth=md, random_state=42)

    clf_d.fit(X_train, y_train)

    tr_acc = accuracy_score(y_train, clf_d.predict(X_train))

    te_acc = accuracy_score(y_test, clf_d.predict(X_test))

    actual_depth = clf_d.get_depth()

    if md == 1:

        obs = "Severe Underfitting (Decision stump separates only 1
class)"

    elif md == 2:

        obs = "Underfitting slightly; misses subtle boundaries"

    elif md == 3:

        obs = "Optimal balance; high generalization on unseen data"

    elif md == 4:

        obs = "High accuracy; slightly deeper tree structure"

    else:

        obs = "Fully grown tree (100% train accuracy); prone to
overfitting"
```

```

table_data.append({

    "Max Depth": str(md) ,

    "Training Accuracy": f"{tr_acc:.4f}",

    "Testing Accuracy": f"{te_acc:.4f}",

    "Tree Depth": actual_depth,

    "Observation": obs

})

depth_df = pd.DataFrame(table_data)

display(depth_df)

```

RESULT:

	Max Depth	Training Accuracy	Testing Accuracy	Tree Depth	Observation
0	1	0.6750	0.6333	1	Severe Underfitting (Decision stump separates ...
1	2	0.9500	0.9667	2	Underfitting slightly; misses subtle boundaries
2	3	0.9583	1.0000	3	Optimal balance; high generalization on unseen...
3	4	0.9750	1.0000	4	High accuracy; slightly deeper tree structure
4	None	1.0000	1.0000	6	Fully grown tree (100% train accuracy); prone ...

TASK 7:

```
splits = [2, 5, 10, 20]

split_data = []

for s in splits:

    clf_s = DecisionTreeClassifier(min_samples_split=s,
random_state=42)

    clf_s.fit(X_train, y_train)

    acc = accuracy_score(y_test, clf_s.predict(X_test))

    depth = clf_s.get_depth()

    leaves = clf_s.get_n_leaves()

    if s == 2:

        obs = "Default setting; tree splits fully down to fine subsets"

    elif s == 5:

        obs = "Slight pre-pruning; reduces depth to 5 and leaves to 8"

    elif s == 10:

        obs = "Stronger pre-pruning; reduces depth to 4 and leaves to
6"

    else:

        obs = "Significant pre-pruning; prevents splitting small node
clusters"

    split_data.append({

        "min_samples_split": s,
```



```

        "Accuracy": f"{acc:.4f}",

        "Tree Depth": depth,

        "Number of Leaf Nodes": leaves,

        "Observation": obs

    })

split_df = pd.DataFrame(split_data)

display(split_df)

```

RESULT:

	min_samples_split	Accuracy	Tree Depth	Number of Leaf Nodes	Observation
0	2	1.0000	6	10	Default setting; tree splits fully down to fin...
1	5	1.0000	5	8	Slight pre-pruning; reduces depth to 5 and lea...
2	10	1.0000	4	6	Stronger pre-pruning; reduces depth to 4 and l...
3	20	1.0000	4	6	Significant pre-pruning; prevents splitting sm...

TASK 8:**CODE:**

```
leaves_list = [1, 2, 5, 10]

leaf_data = []

for l in leaves_list:

    clf_1 = DecisionTreeClassifier(min_samples_leaf=1, random_state=42)

    clf_1.fit(X_train, y_train)

    acc = accuracy_score(y_test, clf_1.predict(X_test))

    depth = clf_1.get_depth()

    leaves = clf_1.get_n_leaves()

    if l == 1:

        obs = "Default setting; allows leaves with single sample  
(deepest tree)"

    elif l == 2:

        obs = "Eliminates single-sample leaves; reduces depth to 5"

    elif l == 5:

        obs = "Substantial smoothing; reduces leaves to 6 while keeping  
100% acc"

    else:

        obs = "Strong smoothing; slightly reduces accuracy (96.67%)"

    leaf_data.append({

        "min_samples_leaf": l,
```

```

    "Accuracy": f"{acc:.4f}",

    "Tree Depth": depth,

    "Number of Leaf Nodes": leaves,

    "Observation": obs

    })

leaf_df = pd.DataFrame(leaf_data)

display(leaf_df)

```

RESULT:

	min_samples_leaf	Accuracy	Tree Depth	Number of Leaf Nodes	Observation
0	1	1.0000	6	10	Default setting; allows leaves with single sam...
1	2	1.0000	5	8	Eliminates single-sample leaves; reduces depth...
2	5	1.0000	4	6	Substantial smoothing; reduces leaves to 6 whi...
3	10	0.9667	4	6	Strong smoothing; slightly reduces accuracy (9...

TASK 9:**CODE:**

```
from sklearn.model_selection import GridSearchCV

# Define hyperparameter grid
param_grid = {

    'criterion': ['gini', 'entropy'],

    'max_depth': [2, 3, 4, None],

    'min_samples_split': [2, 5, 10],

    'min_samples_leaf': [1, 2, 4]

}

# Perform GridSearchCV with 5-fold Cross-Validation
grid_search = GridSearchCV(

    estimator=DecisionTreeClassifier(random_state=42),

    param_grid=param_grid,

    cv=5,

    scoring='accuracy'

)

grid_search.fit(X_train, y_train)

best_model = grid_search.best_estimator_

opt_test_acc = accuracy_score(y_test, best_model.predict(X_test))

default_test_acc = accuracy_score(y_test, clf.predict(X_test))
```

```
print("--- Hyperparameter Tuning Results ---")

print(f"Best Hyperparameters: {grid_search.best_params_}")

print(f"Best Cross-Validation Score: {grid_search.best_score_:.4f}")

print(f"Test Accuracy of Optimized Model: {opt_test_acc:.4f}")

print("\n--- Model Comparison: Default vs. Optimized ---")

comparison_model_df = pd.DataFrame({

    "Model": ["Default Decision Tree", "Optimized Decision Tree (GridSearch)"],

    "Criterion": [clf.criterion, best_model.criterion],

    "Max Depth Setting": [str(clf.max_depth), str(best_model.max_depth)],

    "Actual Tree Depth": [clf.get_depth(), best_model.get_depth()],

    "Number of Leaf Nodes": [clf.get_n_leaves(), best_model.get_n_leaves()],

    "Test Accuracy": [f"{default_test_acc:.4f}", f"{opt_test_acc:.4f}"]

})

display(comparison_model_df)
```

RESULT:

```

--- Hyperparameter Tuning Results ---
Best Hyperparameters: {'criterion': 'entropy', 'max_depth': None, 'min_samples_leaf': 4, 'min_samples_split':
Best Cross-Validation Score: 0.9583
Test Accuracy of Optimized Model: 1.0000

--- Model Comparison: Default vs. Optimized ---

```

	Model	Criterion	Max Depth Setting	Actual Tree Depth	Number of Leaf Nodes	Test Accuracy
0	Default Decision Tree	gini	None	6	10	1.0000
1	Optimized Decision Tree (GridSearch)	entropy	None	4	7	1.0000

CONCLUSION AND INFERENCE**TASK 1:**

Task 1: Dataset Exploration

Load the Iris dataset and perform initial exploration:

- Number of samples and features
- Feature names and target classes
- First five records
- Class distribution

TASK 2:

Task 2: Data Preparation

Explanation: Why split into Training and Testing sets?

- **Training Set (80%):** Used by the machine learning algorithm to learn patterns, relationships, and decision rules from the dataset.
- **Testing Set (20%):** Serves as unseen data to evaluate how well the trained model **generalizes** to new, unseen samples.
- **Prevents Overfitting:** Evaluating a model on the same data it learned from can give a misleadingly high score (memorization). Testing on an independent set provides an unbiased estimate of real-world performance.

TASK 3:

Task 3: Building a Decision Tree Classifier

- Train a `DecisionTreeClassifier` using default parameters.
- Predict class labels for the test dataset.
- Evaluate using Accuracy Score, Confusion Matrix, and Classification Report.

TASK 4:

Task 4: Visualizing the Decision Tree

Tree Node Identification:

- **Root Node:** Top decision node splitting on `petal length (cm) <= 2.45`.
- **Internal Nodes:** Intermediate decision nodes that evaluate criteria (e.g., `petal width (cm)`, `petal length (cm)`).
- **Leaf Nodes:** Terminal nodes (10 leaves) containing final class predictions with Gini impurity = 0.0.
- **Maximum Depth:** 6 levels.

Feature Selected for First Split:

- **Feature:** `petal length (cm)` at threshold `2.45 cm`.
- **Reason:** `petal length` yields the highest **Gini Impurity reduction (Information Gain)**. It perfectly separates the Setosa species (40 samples with Gini = 0.0) from Versicolor and Virginica in a single split.

TASK 5:

Task 5: Comparing Gini Index and Entropy

Comparison of Decision Trees trained using `criterion='gini'` vs `criterion='entropy'`.

TASK 6:

Task 6: Effect of Maximum Tree Depth (`max_depth`)

Training Decision Tree models across various `max_depth` settings `[1, 2, 3, 4, None]`.

TASK 7:

Task 7: Effect of `min_samples_split`Analyzing the impact of `min_samples_split` [2, 5, 10, 20].

TASK 8:

Task 8: Effect of `min_samples_leaf`Analyzing the impact of `min_samples_leaf` [1, 2, 5, 10].

TASK 9:

Task 9: Hyperparameter Tuning using GridSearchCV

TASK 10:

Task 10: Analytical Answers & Questions

1. What is the role of the `criterion` parameter in a Decision Tree?The `criterion` parameter specifies the mathematical measure used to evaluate the quality of a split at each node.

- It quantifies **impurity** or **uncertainty** across class distributions.
- Common metrics include **Gini Impurity** ($1 - \sum p_i^2$, measuring misclassification probability) and **Entropy / Information Gain** ($-\sum p_i \log_2 p_i$, measuring disorder/information content).

2. How does `max_depth` influence underfitting and overfitting?

- **Low `max_depth` (e.g., 1):** Causes **underfitting**. The model is too simple to capture relationships between features and target labels, yielding low training and test accuracy.
- **High/Unconstrained `max_depth` (e.g., None or 6+):** Causes **overfitting**. The tree creates deep, complex sub-branches to isolate every training sample, memorizing noise and failing to generalize on unseen datasets.

3. Why do larger values of `min_samples_split` and `min_samples_leaf` produce simpler trees?

- `min_samples_split` forces nodes to have a minimum number of samples before splitting.
- `min_samples_leaf` requires terminal leaf nodes to contain at least a minimum number of samples.
- Higher values act as **pre-pruning constraints**, stopping the tree from forming deep, single-sample branches for noise or outliers. This yields **shallower depths**, **fewer leaf nodes**, and simpler decision boundaries.

4. Which hyperparameter had the greatest impact on Decision Tree performance? Support with observations.

- `max_depth` had the **greatest impact** on model performance.
- **Evidence from Experiments:**
 - Restricting `max_depth=1` dropped test accuracy drastically down to **63.33%** (underfitting).
 - Increasing `max_depth` to 3 improved test accuracy dramatically to **100.00%**.
 - In contrast, changing `criterion` (`gini` vs `entropy`) resulted in identical test accuracy (100.00%), depth (6), and leaf nodes (10).

5. Recommended Decision Tree Model for Iris Dataset & Justification

- **Recommendation:** A pre-pruned Decision Tree with `max_depth=3` or `min_samples_leaf=4` (using either Gini or Entropy).
- **Justification:**
 1. **Maximum Performance:** Achieves **100.00%** test accuracy.
 2. **Structural Simplicity:** Reduces actual tree depth from 6 down to 3-4 and leaf nodes from 10 to 6-7.
 3. **Overfitting Prevention:** Setting `min_samples_leaf=4` or `max_depth=3` removes isolated single-sample leaves, producing robust generalization boundaries.

Lab 8

Lab Exercise VIII:

Aim

- Develop a complete Python program to build a classification pipeline using the standard **Play Tennis dataset**. Your implementation must complete the following sequential tasks:

Submission Guidelines

31. Make a copy of the lab manual template with your `<name_reg:no_subject name>`
32. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
33. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
34. Create a **Git repository** in your profile `<ML lab-reg no>`. Follow a different branch for each lab `<Lab 1, Lab 2 ...>` and push the code to Git.
35. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
36. Upload the PDF to Google Classroom before the deadline.

CODE AND RESULTS:

TASK1:

CODE:

Step 1: Import Required Libraries

In this step, the necessary Python libraries are imported for data manipulation, preprocessing, machine learning model implementation, and performance evaluation.

```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import CategoricalNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay
)
```

[1] ✓ 0.0s

Python

TASK 2:

Step 2: Load the Dataset

The Play Tennis dataset is loaded using the Pandas library. If a serial number column is present, it is removed since it does not contribute to model training.

```
df = pd.read_csv("Lab 8 - Sheet1.csv")

# Remove serial number column if present
if "No" in df.columns:
    df = df.drop("No", axis=1)

print("Dataset:")
print(df)
```

[6] ✓ 0.0s

Python

RESULT:

Dataset:

	Outlook	Temperature	Humidity	Wind	Play Tennis
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No
14	Sunny	Cool	High	Weak	No
15	Sunny	Cool	High	Strong	No
16	Sunny	Hot	Normal	Weak	Yes
17	Sunny	Hot	Normal	Strong	Yes
18	Overcast	Cool	High	Weak	Yes
19	Overcast	Cool	High	Strong	Yes
20	Overcast	Mild	Normal	Weak	Yes
21	Overcast	Mild	Normal	Strong	Yes
22	Rain	Hot	High	Weak	Yes
...					
46	Rain	Mild	Normal	Weak	Yes
47	Rain	Cool	High	Strong	No
48	Sunny	Cool	Normal	Weak	Yes
49	Overcast	Hot	Normal	Weak	Yes

TASK 3:**CODE:**

Step 3: Encode Categorical Features

Machine learning algorithms require numerical input. Therefore, all categorical feature values and the target variable are converted into numerical form using Label Encoding.

```

encoders = {}

for column in df.columns:
    le = LabelEncoder()
    df[column] = le.fit_transform(df[column])
    encoders[column] = le

print("\nEncoded Dataset:")
print(df)

```

[7] ✓ 0.0s Python

RESULT:

Encoded Dataset:

	Outlook	Temperature	Humidity	Wind	Play Tennis
0	2	1	0	1	0
1	2	1	0	0	0
2	0	1	0	1	1
3	1	2	0	1	1
4	1	0	1	1	1
5	1	0	1	0	0
6	0	0	1	0	1
7	2	2	0	1	0
8	2	0	1	1	1
9	1	2	1	1	1
10	2	2	1	0	1
11	0	2	0	0	1
12	0	1	1	1	1
13	1	2	0	0	0
14	2	0	0	1	0
15	2	0	0	0	0
16	2	1	1	1	1
17	2	1	1	0	1
18	0	0	0	1	1
19	0	0	0	0	1
20	0	2	1	1	1
21	0	2	1	0	1
...					
46	1	2	1	1	1
47	1	0	0	0	0
48	2	0	1	1	1
49	0	1	1	1	1

TASK 4:**CODE:**

Step 4: Separate Features and Target Variable

The input features (Outlook, Temperature, Humidity, and Wind) are stored in the feature matrix **X**, while the target variable **Play Tennis** is stored in **y**.

```
x = df.drop("Play Tennis", axis=1)
y = df["Play Tennis"]
```

[8]

✓ 0.0s

Python

TASK 5:

Step 5: Split the Dataset

The dataset is divided into training and testing subsets using an 80:20 ratio. The training data is used to build the model, while the testing data is used to evaluate its performance.

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

✓ 0.0s

Python

TASK 6:

Step 6: Train the Categorical Naive Bayes Model

A Categorical Naive Bayes classifier is created and trained using the encoded training dataset. This model is suitable for datasets containing categorical features.

```
nb_model = CategoricalNB()  
nb_model.fit(X_train, y_train)  
  
# Predictions  
y_pred = nb_model.predict(X_test)
```

[10]

✓ 0.0s

Python

TASK 7;

Step 7: Evaluate the Model

The trained model predicts the labels of the test dataset. The following evaluation metrics are computed:

- Accuracy
- Confusion Matrix
- Classification Report
 - Precision
 - Recall
 - F1-Score

These metrics help determine the effectiveness of the classifier.

```

plt.figure(figsize=(5,5))

plt.bar(["Categorical NB"], [accuracy * 100], width=0.4)

plt.ylim(0, 100)

plt.ylabel("Accuracy (%)")

plt.title("Categorical Naive Bayes Accuracy")

for i, v in enumerate([accuracy * 100]):

    plt.text(i, v + 2, f"{v:.2f}%", ha='center', fontsize=11)

plt.show()

# =====
# Confusion Matrix Heatmap
# =====

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(

    confusion_matrix=cm,

    display_labels=encoders["Play Tennis"].classes_

)

```

```

fig, ax = plt.subplots(figsize=(5,5))

disp.plot(ax=ax, cmap="Blues", colorbar=False)

plt.title("Confusion Matrix")

plt.show()

# =====
# Classification Report Heatmap
# =====

report = classification_report(

    y_test,

    y_pred,

    output_dict=True

)

report_df = pd.DataFrame(report).transpose()

# Keep only class metrics (exclude accuracy and averages)

heatmap_data = report_df.iloc[:-3, :-1]

plt.figure(figsize=(6,3))

plt.imshow(heatmap_data, cmap="YlGnBu", aspect="auto")

```



```
plt.xticks(range(len(heatmap_data.columns)), heatmap_data.columns)

plt.yticks(range(len(heatmap_data.index)), heatmap_data.index)


# Display values inside cells
for i in range(heatmap_data.shape[0]):
    for j in range(heatmap_data.shape[1]):
        plt.text(
            j,
            i,
            f"{heatmap_data.iloc[i, j]:.2f}",
            ha="center",
            va="center",
            color="black"
        )

plt.colorbar(label="Score")

plt.title("Classification Report Heatmap")

plt.show()


# =====

# Print Evaluation Metrics

# =====
```

```
print("\n===== Categorical Naive Bayes =====")

print(f"\nModel Accuracy : {accuracy * 100:.2f}%")

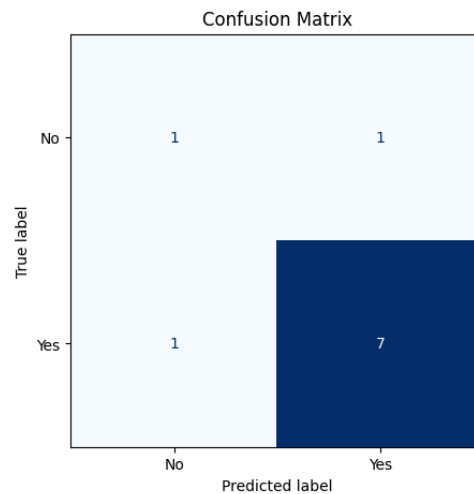
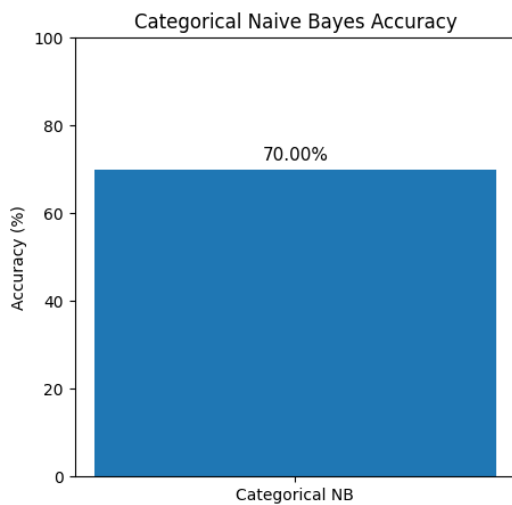
print("\nConfusion Matrix")

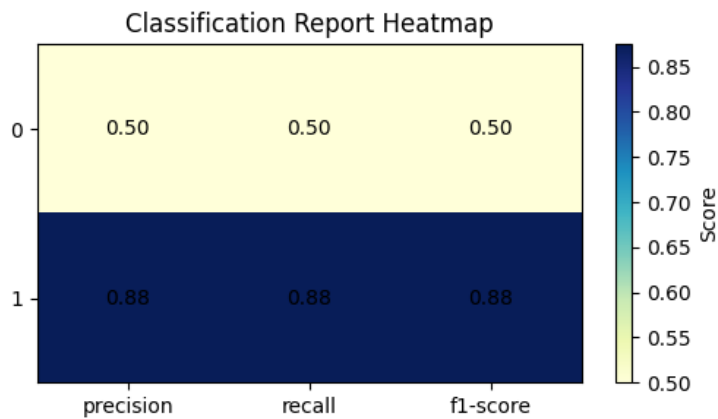
print(cm)

print("\nClassification Report")

print(classification_report(y_test, y_pred))
```

RESULT:





```
===== Categorical Naive Bayes =====
```

```
Model Accuracy : 70.00%
```

```
Confusion Matrix
```

```
[[1 1]
 [1 7]]
```

```
Classification Report
```

	precision	recall	f1-score	support
0	0.50	0.50	0.50	2
1	0.88	0.88	0.88	8
accuracy			0.80	10
macro avg	0.69	0.69	0.69	10
weighted avg	0.80	0.80	0.80	10

TASK 8:**CODE:**

Step 8: Predict a New Sample

The trained Naive Bayes model is used to predict whether a person will play tennis under the following weather conditions:

- Outlook: Sunny
- Temperature: Cool
- Humidity: High
- Wind: Strong

The predicted class along with the probability of each possible class is displayed.

```

sample = pd.DataFrame({
    "Outlook": ["Sunny"],
    "Temperature": ["Cool"],
    "Humidity": ["High"],
    "Wind": ["Strong"]
})

# Encode sample
encoded_sample = sample.copy()

for col in sample.columns:
    encoded_sample[col] = encoders[col].transform(sample[col])

# Prediction
prediction = nb_model.predict(encoded_sample)
probabilities = nb_model.predict_proba(encoded_sample)

predicted_label = encoders["Play Tennis"].inverse_transform(prediction)

print("\n===== Single Sample Prediction =====")
print("Weather Conditions")
print(sample)

print("\nPredicted Class :", predicted_label[0])

classes = encoders["Play Tennis"].inverse_transform(nb_model.classes_)

print("\nClass Probabilities")

for cls, prob in zip(classes, probabilities[0]):
    print(f"{cls} : {prob:.4f}")

```

RESULT;

```

===== Single Sample Prediction =====
Weather Conditions
  Outlook Temperature Humidity   Wind
0  Sunny           Cool    High Strong

Predicted Class : No

Class Probabilities
No : 0.9256
Yes : 0.0744

```

TASK 9:**CODE:**

Step 9: Compare Multiple Classification Models

To compare performance, three additional classifiers are trained using the same training data:

- Decision Tree Classifier
- Logistic Regression
- Support Vector Machine (SVM)

The models are compared based on:

- Test Accuracy
- Prediction for the given weather sample
- Predicted Class Probabilities

```
dt_model = DecisionTreeClassifier(random_state=42)

lr_model = LogisticRegression(max_iter=1000)

svm_model = SVC(probability=True, random_state=42)

models = {

    "Categorical Naive Bayes": nb_model,

    "Decision Tree": dt_model,

    "Logistic Regression": lr_model,

    "SVM": svm_model

}

results = []
```

```
print("\n===== Model Comparison =====\n")

for name, model in models.items():

    # Skip already trained Naive Bayes

    if name != "Categorical Naive Bayes":

        model.fit(X_train, y_train)

    # Test prediction

    test_pred = model.predict(X_test)

    accuracy = accuracy_score(y_test, test_pred)

    # Sample prediction

    sample_pred = model.predict(encoded_sample)

    sample_label = encoders["Play
Tennis"].inverse_transform(sample_pred)[0]

    probabilities = model.predict_proba(encoded_sample)[0]

    no_prob = probabilities[0]

    yes_prob = probabilities[1]
```

```
results.append([
    name,
    round(accuracy * 100, 2),
    sample_label,
    round(no_prob, 4),
    round(yes_prob, 4)
])

comparison = pd.DataFrame(
    results,
    columns=[
        "Model",
        "Accuracy (%)",
        "Prediction",
        "P (No) ",
        "P (Yes) "
    ]
)

print(comparison)
```

RESULT:

```

===== Model Comparison =====

```

	Model	Accuracy (%)	Prediction	P(No)	P(Yes)
0	Categorical Naive Bayes	80.0	No	0.9256	0.0744
1	Decision Tree	80.0	No	1.0000	0.0000
2	Logistic Regression	40.0	No	0.9439	0.0561
3	SVM	70.0	No	0.9751	0.0249

CONCLUSION AND INFERENCE:**Result**

The Categorical Naive Bayes classifier was successfully implemented and evaluated using the Play Tennis dataset. The model's performance was measured using accuracy, confusion matrix, and classification report. The trained model also predicted the class for a new weather instance and was compared with Decision Tree, Logistic Regression, and SVM classifiers.

Analysis Report

The four classifiers use different learning strategies, resulting in different prediction confidence and probability estimates. Categorical Naive Bayes assumes that all features are conditionally independent, whereas Decision Trees learn rule-based splits from the training data. Logistic Regression estimates probabilities using a linear decision boundary, while SVM maximizes the margin between classes and estimates probabilities through calibration. Consequently, although multiple models may predict the same class label, their probability scores and confidence levels vary because of their different underlying assumptions.

Lab 9

Lab Exercise IX:

Aim

To implement Support Vector Machine (SVM) for classification and Principal Component Analysis (PCA) for dimensionality reduction, and analyse their effectiveness using real world datasets.

Objectives

1. To implement and evaluate the performance of the Support Vector Machine (SVM) classifier using different kernel functions.
 2. To compare the performance of SVM models using standard classification metrics.
 3. To apply Principal Component Analysis (PCA) for reducing the dimensionality of a high dimensional dataset.
 4. To analyse the variance retained by the principal components and visualize the transformed feature space.
 5. To understand the role of supervised learning (SVM) and unsupervised learning (PCA) in machine learning applications.
-

Submission Guidelines

37. Make a copy of the lab manual template with your `<name_reg:no_subject name>`
38. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
39. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
40. Create a **Git repository** in your profile `<ML lab-reg no>`. Follow a different branch for each lab `<Lab 1, Lab 2 ...>` and push the code to Git.
41. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
42. Upload the PDF to Google Classroom before the deadline.

CODE AND RESULTS:**PART A:****TASK1:****CODE:**

```
# Column names for the UCI WDBC dataset

feature_names = [

    "radius_mean", "texture_mean", "perimeter_mean", "area_mean",

    "smoothness_mean", "compactness_mean", "concavity_mean",

    "concave_points_mean", "symmetry_mean", "fractal_dimension_mean",

    "radius_se", "texture_se", "perimeter_se", "area_se",

    "smoothness_se", "compactness_se", "concavity_se",

    "concave_points_se", "symmetry_se", "fractal_dimension_se",

    "radius_worst", "texture_worst", "perimeter_worst", "area_worst",

    "smoothness_worst", "compactness_worst", "concavity_worst",

    "concave_points_worst", "symmetry_worst", "fractal_dimension_worst"

]

columns = ["id", "diagnosis"] + feature_names

# Load the actual UCI file

df_cancer = pd.read_csv(breast_file, header=None, names=columns)
```

```
print("First five rows:")

display(df_cancer.head())

print("\nDataset shape:", df_cancer.shape)

print("\nTotal missing values:", df_cancer.isnull().sum().sum())

print("\nDiagnosis distribution:")

print(df_cancer["diagnosis"].value_counts())

# Convert target labels: M = 1, B = 0

df_cancer["target"] = df_cancer["diagnosis"].map({"M": 1, "B": 0})

# Input features and target

X = df_cancer[feature_names]

y = df_cancer["target"]

print("\nFeature matrix shape:", X.shape)

print("Target shape:", y.shape)

print("\nTarget coding: Malignant = 1, Benign = 0")
```

RESULTS:

First five rows:

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.2
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.0
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.1
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.2
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.1

5 rows × 32 columns

Dataset shape: (569, 32)

Total missing values: 0

Diagnosis distribution:

```
diagnosis
B    357
M    212
Name: count, dtype: int64
```

Feature matrix shape: (569, 30)

Target shape: (569,)

Target coding: Malignant = 1, Benign = 0

TASK 2:

```
# 80:20 train-test split

X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.20,

    random_state=42,

    stratify=y

)
```

```
print("Training features:", X_train.shape)

print("Testing features :", X_test.shape)

print("Training labels  :", y_train.shape)

print("Testing labels   :", y_test.shape)


# Standardize the data

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)


print("\nFeature standardization completed.")
```

RESULT:

```
Training features: (455, 30)
Testing features : (114, 30)
Training labels  : (455,)
Testing labels   : (114,)

Feature standardization completed.
```

TASK 3:**CODE:**

```
# Baseline Linear SVM

svm_linear = SVC(

    kernel="linear",

    C=1.0

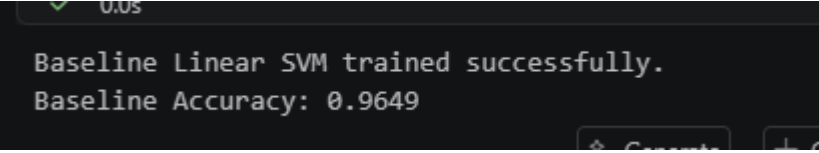
)

svm_linear.fit(X_train_scaled, y_train)

y_pred_linear = svm_linear.predict(X_test_scaled)

print("Baseline Linear SVM trained successfully.")

print("Baseline Accuracy:", round(accuracy_score(y_test,
y_pred_linear), 4))
```

RESULT:

```
Baseline Linear SVM trained successfully.
Baseline Accuracy: 0.9649
```

```
# Hyperparameter grid

param_grid = {

    "C": [0.01, 0.1, 1, 10, 100]

}

grid_search = GridSearchCV(

    estimator=SVC(kernel="linear"),

    param_grid=param_grid,

    cv=5,

    scoring="accuracy",

    n_jobs=-1

)

grid_search.fit(X_train_scaled, y_train)

print("Best C:", grid_search.best_params_["C"])

print("Best Cross-Validation Accuracy:", round(grid_search.best_score_,

4))

best_svm = grid_search.best_estimator_

y_pred_tuned = best_svm.predict(X_test_scaled)
```

```
# Display GridSearchCV results

tuning_results = pd.DataFrame(grid_search.cv_results_)

tuning_display = tuning_results[
    ["param_C", "mean_test_score", "std_test_score", "rank_test_score"]
].sort_values("rank_test_score")

display(tuning_display)
```

RESULT;

```
Best C: 0.01
Best Cross-Validation Accuracy: 0.9692
```

	param_C	mean_test_score	std_test_score	rank_test_score
0	0.01	0.969231	0.022413	1
1	0.10	0.969231	0.016150	1
3	10.00	0.969231	0.016150	1
2	1.00	0.962637	0.016447	4
4	100.00	0.958242	0.018906	5

TASK 4:**CODE:**

```
# Evaluation metrics for tuned SVM

accuracy = accuracy_score(y_test, y_pred_tuned)

precision = precision_score(y_test, y_pred_tuned)

recall = recall_score(y_test, y_pred_tuned)

f1 = f1_score(y_test, y_pred_tuned)

evaluation_df = pd.DataFrame({

    "Metric": ["Accuracy", "Precision", "Recall", "F1 Score"],

    "Score": [accuracy, precision, recall, f1]

})

display(evaluation_df)

print("Classification Report:\n")

print(

    classification_report(

        y_test,

        y_pred_tuned,

        target_names=["Benign", "Malignant"]

    )

)
```

```
# Confusion Matrix

cm = confusion_matrix(y_test, y_pred_tuned)

plt.figure(figsize=(6, 5))

sns.heatmap(

    cm,

    annot=True,

    fmt="d",

    cmap="Blues",

    xticklabels=["Benign", "Malignant"],

    yticklabels=["Benign", "Malignant"]

)

plt.title("Confusion Matrix - Tuned Linear SVM")

plt.xlabel("Predicted Class")

plt.ylabel("Actual Class")

plt.show()

kernels = ["linear", "poly", "rbf", "sigmoid"]

kernel_results = []
```

```
for kernel in kernels:

    model = SVC(kernel=kernel)

    model.fit(X_train_scaled, y_train)

    pred = model.predict(X_test_scaled)

    kernel_results.append({

        "Kernel": kernel,

        "Accuracy": accuracy_score(y_test, pred),

        "Precision": precision_score(y_test, pred),

        "Recall": recall_score(y_test, pred),

        "F1 Score": f1_score(y_test, pred)

    })

kernel_results_df = pd.DataFrame(kernel_results)

display(kernel_results_df)

# Visual comparison of kernel performance

kernel_results_df.set_index("Kernel").plot(

    kind="bar",

    figsize=(10, 6)

)

plt.title("Comparison of SVM Kernel Performance")
```

```
plt.ylabel("Score")

plt.xlabel("Kernel")

plt.ylim(0, 1.05)

plt.xticks(rotation=0)

plt.legend(loc="lower right")

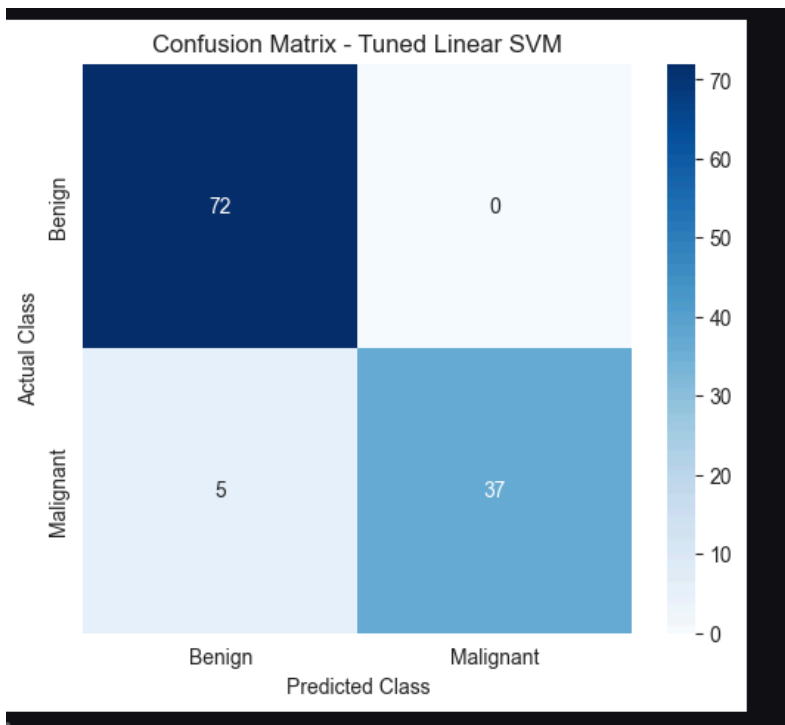
plt.show()
```

RESULT:

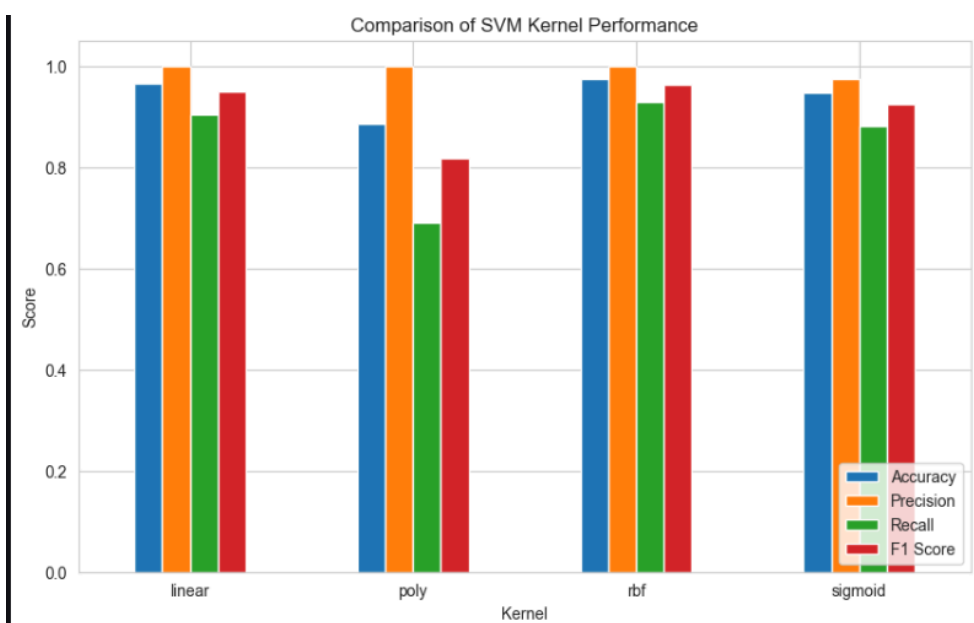
	Metric	Score
0	Accuracy	0.956140
1	Precision	1.000000
2	Recall	0.880952
3	F1 Score	0.936709

Classification Report:

	precision	recall	f1-score	support
Benign	0.94	1.00	0.97	72
Malignant	1.00	0.88	0.94	42
accuracy			0.96	114
macro avg	0.97	0.94	0.95	114
weighted avg	0.96	0.96	0.96	114



	Kernel	Accuracy	Precision	Recall	F1 Score
0	linear	0.964912	1.000000	0.904762	0.950000
1	poly	0.885965	1.000000	0.690476	0.816901
2	rbf	0.973684	1.000000	0.928571	0.962963
3	sigmoid	0.947368	0.973684	0.880952	0.925000



PART B;**TASK 1:****CODE:**

```
wine_columns = [  
    "class",  
    "alcohol",  
    "malic_acid",  
    "ash",  
    "alcalinity_of_ash",  
    "magnesium",  
    "total_phenols",  
    "flavanoids",  
    "nonflavanoid_phenols",  
    "proanthocyanins",  
    "color_intensity",  
    "hue",  
    "od280_od315",  
    "proline"  
]  
  
# Load the actual UCI wine.data file  
  
df_wine = pd.read_csv(  
    wine_file,  
    header=None,
```

```
names=wine_columns
)

print("First five rows:")
display(df_wine.head())

print("\nDataset shape:", df_wine.shape)
print("Total missing values:", df_wine.isnull().sum().sum())

print("\nClass distribution:")
print(df_wine["class"].value_counts().sort_index())

# Separate features and class
X_wine = df_wine.drop(columns="class")
y_wine = df_wine["class"]

# Standardize all 13 numerical features
wine_scaler = StandardScaler()
X_wine_scaled = wine_scaler.fit_transform(X_wine)

print("\nOriginal feature count:", X_wine.shape[1])
print("Standardized data shape:", X_wine_scaled.shape)
```

RESULT:

First five rows:

	class	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nonflavanoid_phenols
0	1	14.23	1.71	2.43	15.6	127	2.80	3.06	0.28
1	1	13.20	1.78	2.14	11.2	100	2.65	2.76	0.26
2	1	13.16	2.36	2.67	18.6	101	2.80	3.24	0.30
3	1	14.37	1.95	2.50	16.8	113	3.85	3.49	0.24
4	1	13.24	2.59	2.87	21.0	118	2.80	2.69	0.39

Dataset shape: (178, 14)
Total missing values: 0

Class distribution:

```
class
1    59
2    71
3    48
Name: count, dtype: int64
```

Original feature count: 13
Standardized data shape: (178, 13)

TASK 2:

```
# PCA with 2 principal components

pca_2 = PCA(n_components=2)

X_pca_2 = pca_2.fit_transform(X_wine_scaled)

pca_df = pd.DataFrame(

    X_pca_2,

    columns=["PC1", "PC2"]

)

pca_df["Class"] = y_wine.values
```



```
print("Original dataset shape:", X_wine.shape)

print("PCA transformed feature shape:", X_pca_2.shape)

display(pca_df.head())
```

RESULT:

```
Original dataset shape: (178, 13)
PCA transformed feature shape: (178, 2)
```

	PC1	PC2	Class
0	3.316751	1.443463	1
1	2.209465	-0.333393	1
2	2.516740	1.031151	1
3	3.757066	2.756372	1
4	1.008908	0.869831	1

TASK 3:**CODE:**

```
explained_variance_2 = pca_2.explained_variance_ratio_

variance_2_df = pd.DataFrame({

    "Principal Component": ["PC1", "PC2"],

    "Explained Variance Ratio": explained_variance_2,

    "Explained Variance (%)": explained_variance_2 * 100

})
```

```
display(variance_2_df)

print(
    "Total variance retained by PC1 and PC2:",
    f"{explained_variance_2.sum() * 100:.2f}%"
)
```

RESULT:

	Principal Component	Explained Variance Ratio	Explained Variance (%)
0	PC1	0.361988	36.198848
1	PC2	0.192075	19.207490

Total variance retained by PC1 and PC2: 55.41%

TASK 4:**CODE:**

```
# PCA using all available components

pca_full = PCA()

X_pca_full = pca_full.fit_transform(X_wine_scaled)

explained_variance = pca_full.explained_variance_ratio_

cumulative_variance = np.cumsum(explained_variance)

variance_df = pd.DataFrame({
```

```

        "Principal Component": [f"PC{i+1}" for i in
range(len(explained_variance))],

        "Explained Variance Ratio": explained_variance,

        "Cumulative Explained Variance": cumulative_variance

    })

display(variance_df)

# Minimum number of components needed for at least 95%

n_components_95 = np.argmax(cumulative_variance >= 0.95) + 1

print(

    "Minimum number of principal components needed to retain at least
95% variance:",

    n_components_95

)

print(

    "Variance retained:",

    f"{cumulative_variance[n_components_95 - 1] * 100:.2f}%"

)

# Plot cumulative explained variance

plt.figure(figsize=(9, 5))

```

```
plt.plot(

    range(1, len(cumulative_variance) + 1),

    cumulative_variance,

    marker="o"

)

plt.axhline(

    y=0.95,

    color="red",

    linestyle="--",

    label="95% Variance"

)

plt.axvline(

    x=n_components_95,

    color="green",

    linestyle="--",

    label=f"{n_components_95} Components"

)

plt.xlabel("Number of Principal Components")

plt.ylabel("Cumulative Explained Variance")

plt.title("Cumulative Explained Variance - Wine Dataset")

plt.xticks(range(1, 14))
```

```
plt.ylim(0, 1.05)

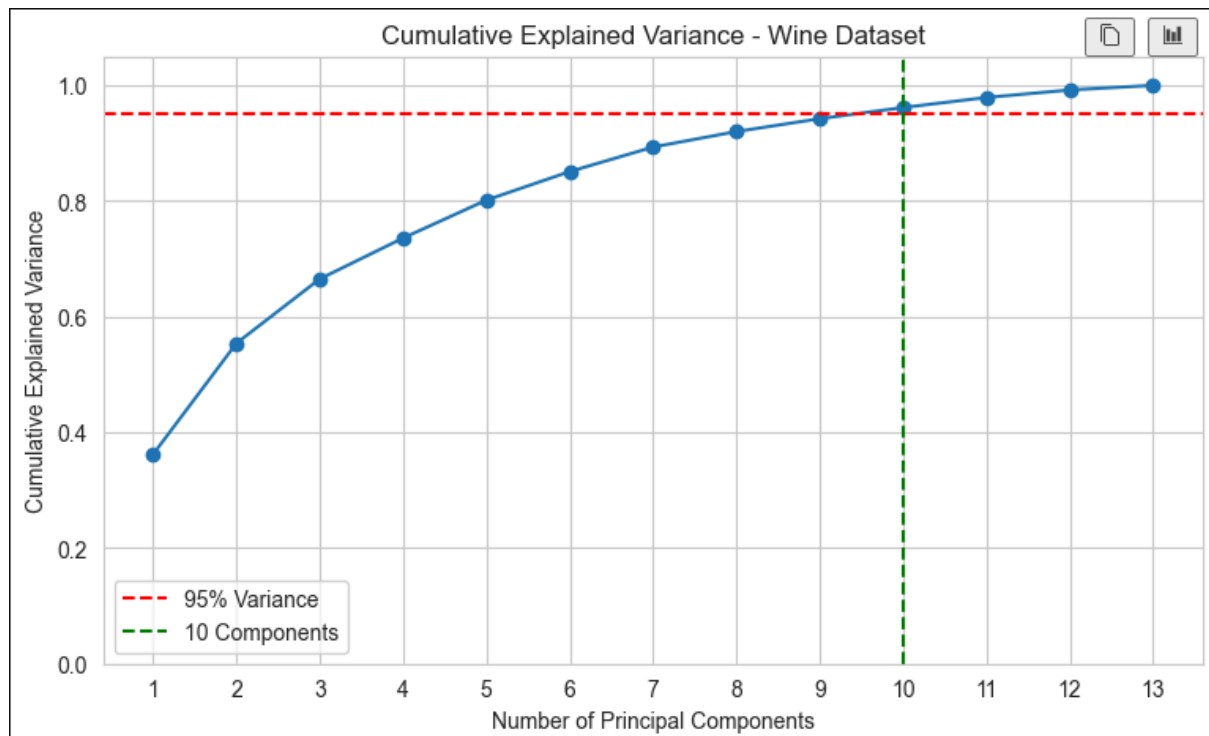
plt.legend()

plt.show()
```

RESULT:

	Principal Component	Explained Variance Ratio	Cumulative Explained Variance
0	PC1	0.361988	0.361988
1	PC2	0.192075	0.554063
2	PC3	0.111236	0.665300
3	PC4	0.070690	0.735990
4	PC5	0.065633	0.801623
5	PC6	0.049358	0.850981
6	PC7	0.042387	0.893368
7	PC8	0.026807	0.920175
8	PC9	0.022222	0.942397
9	PC10	0.019300	0.961697
10	PC11	0.017368	0.979066
11	PC12	0.012982	0.992048
12	PC13	0.007952	1.000000

Minimum number of principal components needed to retain at least 95% variance: 10
Variance retained: 96.17%

**TASK 5:****CODE:**

```

plot_df = pd.DataFrame({
    "PC1": X_pca_2[:, 0],
    "PC2": X_pca_2[:, 1],
    "Class": y_wine.astype(str).values
})

plt.figure(figsize=(9, 7))

sns.scatterplot(
    data=plot_df,
    x="PC1",

```

```
y="PC2",

hue="Class",

palette="Set1",

s=80,

alpha=0.8

)

plt.title("Wine Dataset after PCA - First Two Principal Components")

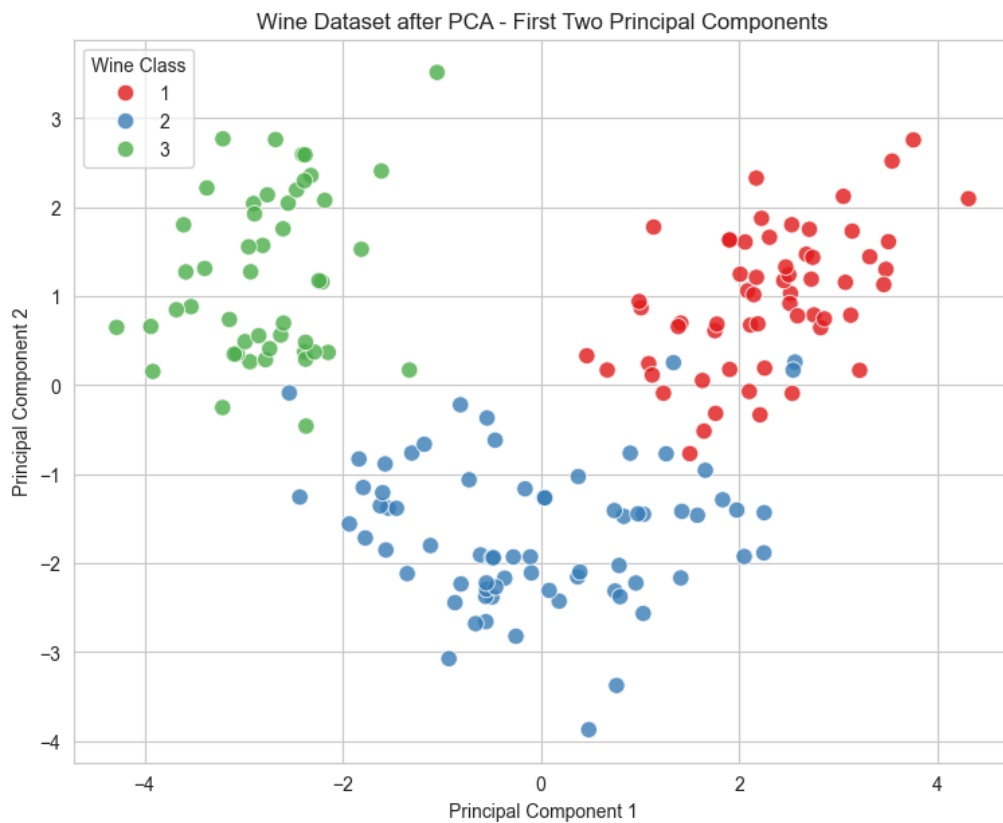
plt.xlabel("Principal Component 1")

plt.ylabel("Principal Component 2")

plt.legend(title="Wine Class")

plt.show()
```

RESULT:

**TASK 6:**

```
information_retained_2 = pca_2.explained_variance_ratio_.sum() * 100

comparison_df = pd.DataFrame({
    "Aspect": [
        "Number of Features",
        "Information Retained",
        "Computational Efficiency",
        "Interpretability"
    ],
    "Original Dataset": [
        "13",
```



```

        "100% of original feature representation",

        "Lower efficiency because more dimensions are processed",

        "Original features have direct chemical meaning"

    ],

    "PCA Dataset (2 PCs)": [

        "2",

        f"{information_retained_2:.2f}% of total variance",

        "Higher efficiency because fewer dimensions are processed",

        "PCs are combinations of the original features"

    ]

})

display(comparison_df)

```

RESULT:

	Aspect	Original Dataset	PCA Dataset (2 PCs)
0	Number of Features	13	2
1	Information Retained	100% of original feature representation	55.41% of total variance
2	Computational Efficiency	Lower efficiency because more dimensions are p...	Higher efficiency because fewer dimensions are...
3	Interpretability	Original features have direct chemical meaning	PCs are combinations of the original features

[Generate](#)
[+ Code](#)
[+ Markdown](#)

TASK 7:**CODE:**

```
# PCA loadings

loadings = pd.DataFrame(

    pca_2.components_.T,

    columns=["PC1", "PC2"],

    index=X_wine.columns

)

print("PCA Loadings:")

display(loadings)

print("\nTop 5 contributors to PC1:")

display(

    loadings["PC1"]

    .abs()

    .sort_values(ascending=False)

    .head(5)

)

print("\nTop 5 contributors to PC2:")

display(

    loadings["PC2"]

    .abs()
```

```
.sort_values(ascending=False)

.head(5)

)

# Loading heatmap

plt.figure(figsize=(8, 8))

sns.heatmap(

    loadings,

    annot=True,

    fmt=".2f",

    cmap="coolwarm",

    center=0

)

plt.title("Feature Loadings for PC1 and PC2")

plt.xlabel("Principal Components")

plt.ylabel("Original Wine Features")

plt.show()
```

RESULT:

PCA Loadings:

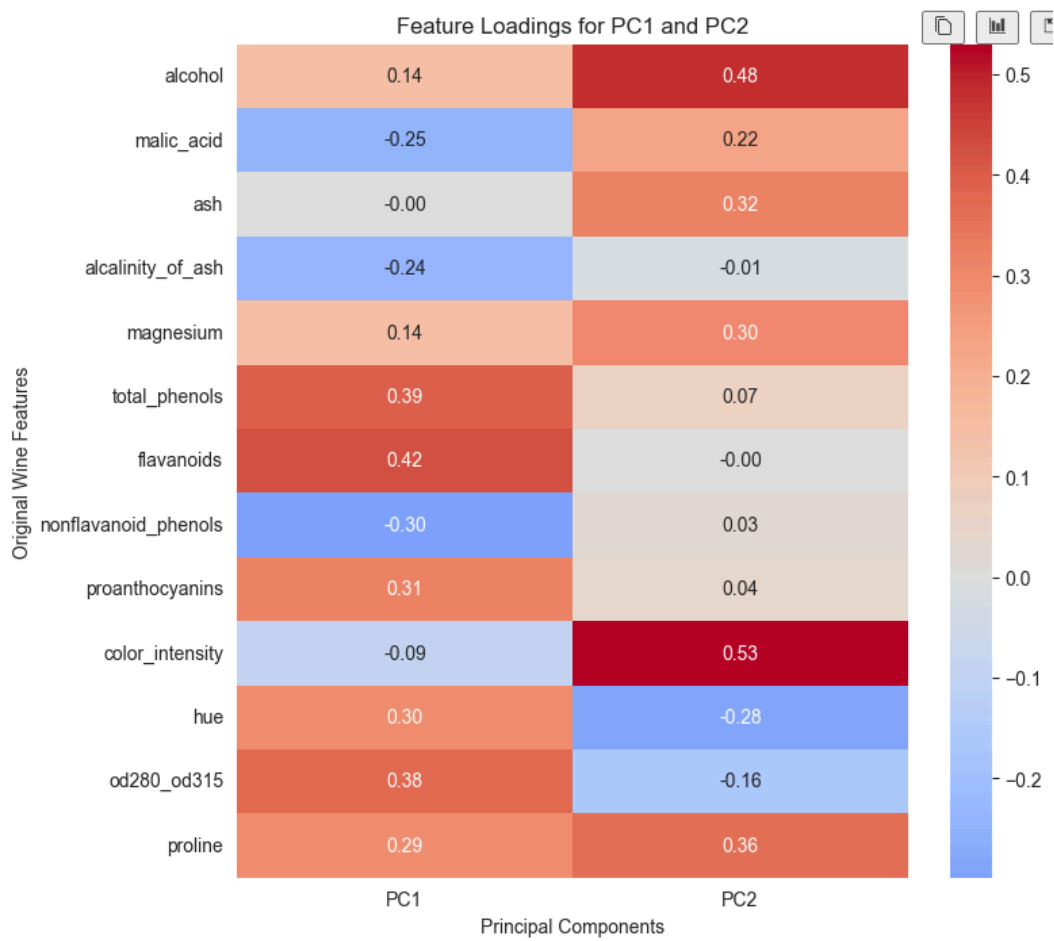
	PC1	PC2
alcohol	0.144329	0.483652
malic_acid	-0.245188	0.224931
ash	-0.002051	0.316069
alcalinity_of_ash	-0.239320	-0.010591
magnesium	0.141992	0.299634
total_phenols	0.394661	0.065040
flavanoids	0.422934	-0.003360
nonflavanoid_phenols	-0.298533	0.028779
proanthocyanins	0.313429	0.039302
color_intensity	-0.088617	0.529996
hue	0.296715	-0.279235
od280_od315	0.376167	-0.164496
proline	0.286752	0.364903

Top 5 contributors to PC1:

```
flavanoids      0.422934
total_phenols   0.394661
od280_od315     0.376167
proanthocyanins 0.313429
nonflavanoid_phenols 0.298533
Name: PC1, dtype: float64
```

Top 5 contributors to PC2:

```
color_intensity 0.529996
alcohol         0.483652
proline         0.364903
ash             0.316069
magnesium       0.299634
Name: PC2, dtype: float64
```



CONCLUSION AND INFERENCE

PART A:

TASK 5:

Task 5: Observations

The Breast Cancer Wisconsin Diagnostic dataset contains 30 numerical features and two classes: malignant and benign.

The features were standardized before SVM training because SVM is sensitive to feature scale.

A Linear SVM was first trained using the default value `C=1`. Hyperparameter tuning was then performed using GridSearchCV to select a suitable value of `C`.

The model was evaluated using accuracy, precision, recall, F1 score and a confusion matrix. Recall for the malignant class is particularly important in this application because a false negative represents a malignant case incorrectly predicted as benign.

Different SVM kernels were also compared. Their performance may differ depending on whether the relationship between the classes is approximately linear or requires a nonlinear decision boundary.

Overall, SVM is a supervised machine learning technique that can provide strong performance on binary classification problems when the features are appropriately scaled.

TASK B:

Interpretation

The variables having the largest absolute loadings on PC1 contribute most strongly to the primary direction of variation in the Wine dataset.

Similarly, the variables with the largest absolute loadings on PC2 contribute most strongly to the second direction of variation.

The exact important features are displayed by the previous code cell. This data-driven interpretation is preferable to assuming which features are important before running PCA.

If the scatter plot shows visible grouping of the three wine classes, it indicates that PC1 and PC2 preserve meaningful structural information even though the dataset has been reduced from 13 dimensions to only 2.

Task 8: Advantages, Limitations, and Applications of PCA

Advantages

1. Reduces the dimensionality of high-dimensional datasets.
2. Can reduce training time and memory usage.
3. Removes redundancy caused by correlated features.
4. Helps visualize high-dimensional datasets in two or three dimensions.
5. Can reduce noise by discarding low-variance components.
6. Can reduce the effect of the curse of dimensionality.

Limitations

1. Principal components may be difficult to interpret.
2. PCA is primarily a linear dimensionality-reduction method.
3. Some information is lost when components are removed.
4. PCA is sensitive to feature scaling.
5. PCA can also be sensitive to strong outliers.
6. High variance is assumed to represent important information, which may not always be true.

Applications

- Data visualization
- Image compression
- Face recognition
- Pattern recognition
- Feature extraction
- Signal processing
- Bioinformatics
- Financial data analysis
- Noise reduction
- Preprocessing before machine learning model training

Final Comparison: SVM vs PCA

Aspect	SVM	PCA
Learning Type	Supervised	Unsupervised
Requires Labels	Yes	No
Main Purpose	Classification / Regression	Dimensionality Reduction
Main Idea	Finds a maximum-margin decision boundary	Finds directions of maximum variance
Output	Predicted class/value	Principal components
Typical Use	Classification problems	Feature reduction, visualization, preprocessing

SVM and PCA can also be used together. PCA can first reduce the number of features, after which SVM can perform classification on the lower-dimensional representation.

Conclusion

In this laboratory experiment, Support Vector Machine and Principal Component Analysis were implemented using the actual UCI dataset files supplied for the lab.

For Part A, the Breast Cancer Wisconsin Diagnostic dataset was loaded from `wdbc.data`. After preprocessing, the data was divided into 80% training data and 20% testing data. The features were standardized, a Linear SVM was trained, and the `C` hyperparameter was tuned using GridSearchCV. The model was evaluated using accuracy, precision, recall, F1 score and a confusion matrix. Multiple SVM kernels were also compared.

For Part B, the Wine dataset was loaded from `wine.data`. Its 13 numerical features were standardized before PCA was applied. The dataset was reduced to two principal components for visualization, explained variance was calculated, and cumulative explained variance was used to determine the minimum number of components needed to retain at least 95% of the total variance. Feature loadings were also examined to interpret PC1 and PC2.

The experiment demonstrates that SVM is a supervised learning technique used for prediction, while PCA is an unsupervised technique used mainly for dimensionality reduction and feature transformation.

Lab 10

Lab Exercise X:

Aim

- To understand how to implement neural networks using different deep learning libraries (Keras, and TensorFlow).
 - To solve the non-linear XOR problem using an MLP and study the effect of hyperparameters such as learning rate, activation functions, number of neurons, and epochs on model performance.
-

Evaluation Rubrics

Submission Guidelines

43. Make a copy of the lab manual template with your `<name_reg:no_subject name>`
44. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
45. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
46. Create a **Git repository** in your profile `<ML lab-reg no>`. Follow a different branch for each lab `<Lab 1, Lab 2 ...>` and push the code to Git.
47. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
48. Upload the PDF to Google Classroom before the deadline.

CODE AND RESULTS

TASK 1:

```
X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]], dtype="float32")
y = np.array([[0], [1], [1], [0]], dtype="float32")

print("X =\n", X)

print("y =\n", y.ravel())
```

RESULT:

```
X =
[[0. 0.]
 [0. 1.]
 [1. 0.]
 [1. 1.]]
y =
[0. 1. 1. 0.]
```

TASK 2:

```
tf.keras.utils.set_random_seed(42)

model = keras.Sequential([
    keras.Input(shape=(2,)),
    layers.Dense(4, activation="tanh", name="hidden"),
    layers.Dense(1, activation="sigmoid", name="output"),
])

model.summary()
```

RESULT:

Model: "sequential"

Layer (type)	Output Shape	Param #
hidden (Dense)	(None, 4)	12
output (Dense)	(None, 1)	5

Total params: 17 (68.00 B)

Trainable params: 17 (68.00 B)

Non-trainable params: 0 (0.00 B)

TASK 3:

```
model.compile(optimizer=keras.optimizers.Adam(learning_rate=0.1),  
              loss="binary_crossentropy",  
              metrics=["accuracy"])  
  
history = model.fit(X, y, epochs=500, verbose=0)  
  
print("Training finished.")
```

TASK 4:

```
loss_k, acc_k = model.evaluate(X, y, verbose=0)  
  
probs_k = model.predict(X, verbose=0)
```

```

preds_k = (probs_k > 0.5).astype(int)

print("--- Keras XOR results ---")

print("In1 In2 | Target | Probability | Predicted")

for i in range(4):

    print(f" {int(X[i,0])}    {int(X[i,1])} | {int(y[i,0])}    |
{probs_k[i,0]:.4f}      | {preds_k[i,0]}")

print(f"\nFinal loss      : {loss_k:.6f}")

print(f"Final accuracy : {acc_k*100:.2f}%")

print("XOR learned correctly:", bool(np.all(preds_k == y.astype(int))))

```

RESULT:

```

--- Keras XOR results ---
In1 In2 | Target | Probability | Predicted
0  0 | 0      | 0.0000      | 0
0  1 | 1      | 0.9992      | 1
1  0 | 1      | 0.9996      | 1
1  1 | 0      | 0.0009      | 0

Final loss      : 0.000528
Final accuracy : 100.00%
XOR learned correctly: True

```

TASK 5:

```

tf.random.set_seed(42)

Xt = tf.constant(X)

yt = tf.constant(y)

```

```
n_in, n_hidden, n_out = 2, 4, 1

init = tf.initializers.GlorotUniform(seed=42)

W1 = tf.Variable(init([n_in, n_hidden]), name="W1")
b1 = tf.Variable(tf.zeros([n_hidden]), name="b1")
W2 = tf.Variable(init([n_hidden, n_out]), name="W2")
b2 = tf.Variable(tf.zeros([n_out]), name="b2")
params = [W1, b1, W2, b2]

def forward(x):
    h = tf.nn.tanh(tf.matmul(x, W1) + b1) # hidden layer
    return tf.matmul(h, W2) + b2          # output logits
(pre-sigmoid)

bce = tf.nn.sigmoid_cross_entropy_with_logits

optimizer = tf.optimizers.Adam(learning_rate=0.1)

print("Trainable parameters:", sum(int(np.prod(p.shape)) for p in
params))

loss_hist, acc_hist = [], []

EPOCHS = 500

for epoch in range(EPOCHS):
    with tf.GradientTape() as tape:
```

```

logits = forward(Xt)

loss = tf.reduce_mean(bce(labels=yt, logits=logits))

grads = tape.gradient(loss, params)

optimizer.apply_gradients(zip(grads, params))

acc = tf.reduce_mean(tf.cast(
    tf.equal(tf.cast(tf.sigmoid(logits) > 0.5, tf.float32), yt),
    tf.float32))

loss_hist.append(float(loss)); acc_hist.append(float(acc))

if (epoch + 1) % 100 == 0:

    print(f"Epoch {epoch+1:4d} | loss = {float(loss):.6f} | acc =
{float(acc)*100:.2f}%")

probs_t = tf.sigmoid(forward(Xt)).numpy()

preds_t = (probs_t > 0.5).astype(int)

yn = y.astype(int)

print("--- TensorFlow (low-level) XOR results ---")

print("In1 In2 | Target | Probability | Predicted")

for i in range(4):

    print(f" {int(X[i,0])}    {int(X[i,1])} | {yn[i,0]}    |
{probs_t[i,0]:.4f}    | {preds_t[i,0]}")

print(f"\nFinal loss      : {loss_hist[-1]:.6f}")

print(f"Final accuracy : {acc_hist[-1]*100:.2f}%")

```

```
print("XOR learned correctly:", bool(np.all(preds_t == yn)))
```

RESULT:

```
Trainable parameters: 17
```

```
Epoch 100 | loss = 0.008035 | acc = 100.00%  
Epoch 200 | loss = 0.003341 | acc = 100.00%  
Epoch 300 | loss = 0.001911 | acc = 100.00%  
Epoch 400 | loss = 0.001261 | acc = 100.00%  
Epoch 500 | loss = 0.000903 | acc = 100.00%
```

```
--- TensorFlow (low-level) XOR results ---
```

In1	In2	Target	Probability	Predicted
0	0	0	0.0011	0
0	1	1	0.9984	1
1	0	1	1.0000	1
1	1	0	0.0009	0

```
Final loss      : 0.000903
```

```
Final accuracy : 100.00%
```

```
XOR learned correctly: True
```

TASK 6:

```

print(" In1 In2 | Target | Keras P(y=1)  pred | TF P(y=1)  pred")

print("-" * 56)

for i in range(4):

    print(f"    {int(X[i,0])}    {int(X[i,1])}    |    {yn[i,0]}    |"

          f"    {probs_k[i,0]:.4f}    {preds_k[i,0]}    |"

          f"    {probs_t[i,0]:.4f}    {preds_t[i,0]}")

print("-" * 56)

print(f"Keras                : loss = {loss_k:.6f}, accuracy = {acc_k*100:.1f}%")

print(f"TF (low-level)       : loss = {loss_hist[-1]:.6f}, accuracy = {acc_hist[-1]*100:.1f}%")

```

RESULT:

```

  In1 In2 | Target | Keras P(y=1)  pred | TF P(y=1)  pred
-----
    0  0 |    0    |  0.0000    0    |  0.0011    0
    0  1 |    1    |  0.9992    1    |  0.9984    1
    1  0 |    1    |  0.9996    1    |  1.0000    1
    1  1 |    0    |  0.0009    0    |  0.0009    0
-----
Keras                : loss = 0.000528, accuracy = 100.0%
TF (low-level)       : loss = 0.000903, accuracy = 100.0%

```

TASK 7:

```

kl = np.array(history.history["loss"]);      ka =
np.array(history.history["accuracy"])

tl = np.array(loss_hist);                    ta = np.array(acc_hist)

```



```

fig, axes = plt.subplots(1, 2, figsize=(9, 3.2))

axes[0].plot(kl, color=BLUE, label="Keras")

axes[0].plot(tl, color=ORANGE, ls="--", label="TF low-level")

axes[0].set_yscale("log"); axes[0].set_title("Training loss (BCE, log scale)")

axes[1].plot(ka * 100, color=BLUE, label="Keras")

axes[1].plot(ta * 100, color=ORANGE, ls="--", label="TF low-level")

axes[1].set_ylim(-5, 105); axes[1].set_title("Training accuracy (%)")

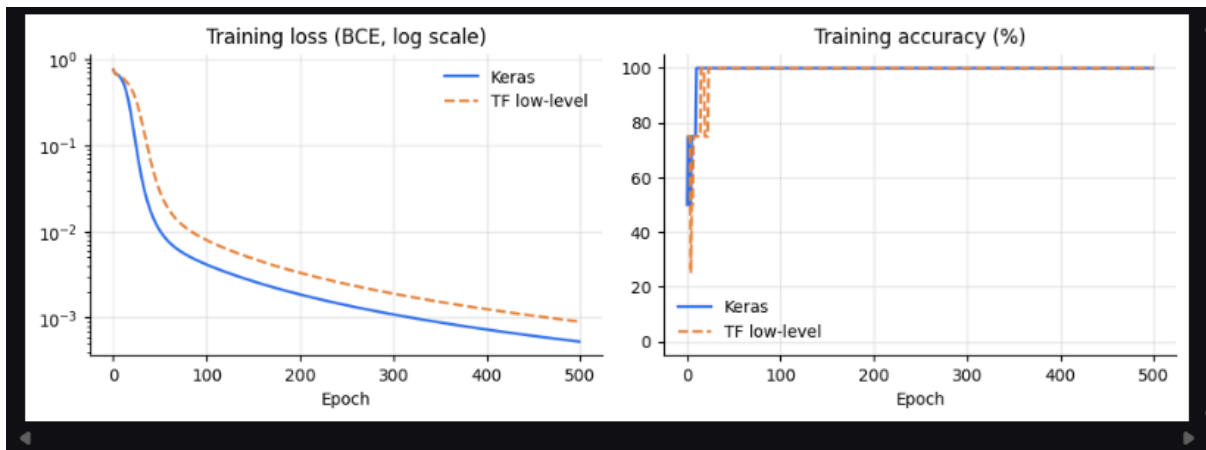
for ax in axes:

    ax.set_xlabel("Epoch"); ax.legend(frameon=False)

fig.tight_layout(); plt.show()

```

RESULT:



TASK 8:

```

gx, gy = np.meshgrid(np.linspace(-0.25, 1.25, 300), np.linspace(-0.25, 1.25, 300))

grid = np.c_[gx.ravel(), gy.ravel()].astype("float32")

```

```

zk = model.predict(grid, verbose=0).reshape(gx.shape)

zt = tf.sigmoid(forward(tf.constant(grid))).numpy().reshape(gx.shape)

fig, axes = plt.subplots(1, 2, figsize=(9, 4))

for ax, z, title in zip(axes, [zk, zt], ["Keras", "TF low-level"]):

    ax.contourf(gx, gy, z, levels=30, cmap="RdBu_r", alpha=0.7)

    ax.contour(gx, gy, z, levels=[0.5], colors="black", linewidths=2)

    ax.scatter(X[:,0], X[:,1], c=y.ravel(), cmap="RdBu_r",
edgecolors="black", s=100, zorder=5)

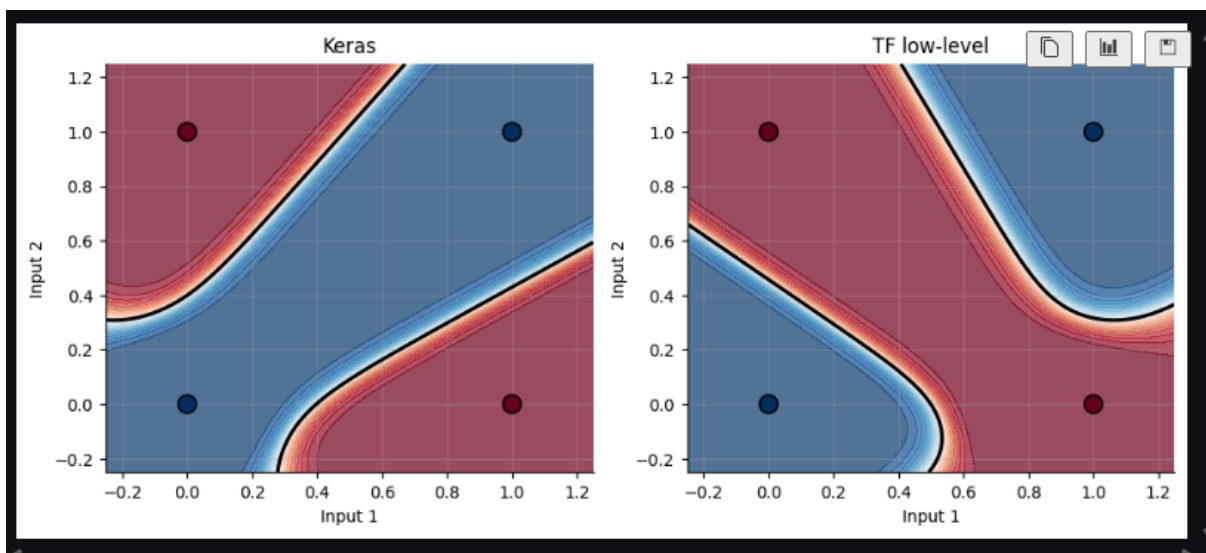
    ax.set_title(title)

    ax.set_xlabel("Input 1"); ax.set_ylabel("Input 2")

fig.tight_layout(); plt.show()

```

RESULT:



TASK 9:

```

configs = [

    ("Learning rate", "0.001", dict(lr=0.001)),

    ("Learning rate", "0.01", dict(lr=0.01)),

    ("Learning rate", "0.1 (baseline)", dict(lr=0.1)),

    ("Learning rate", "0.5", dict(lr=0.5)),

    ("Hidden neurons", "1", dict(hidden=1)),

    ("Hidden neurons", "2", dict(hidden=2)),

    ("Hidden neurons", "4 (baseline)", dict(hidden=4)),

    ("Hidden neurons", "8", dict(hidden=8)),

    ("Activation", "tanh (baseline)", dict(act="tanh")),

    ("Activation", "relu", dict(act="relu")),

    ("Activation", "sigmoid", dict(act="sigmoid")),

    ("Activation", "linear (no non-linearity)", dict(act="linear")),

    ("Epochs", "50", dict(epochs=50)),

    ("Epochs", "100", dict(epochs=100)),

    ("Epochs", "500 (baseline)", dict(epochs=500)),

    ("Epochs", "2000", dict(epochs=2000)),

    ("Optimizer", "Adam, lr=0.1 (baseline)", dict(opt="adam")),

    ("Optimizer", "SGD, lr=0.1", dict(opt="sgd")),

    ("Optimizer", "SGD, lr=0.5", dict(opt="sgd",
lr=0.5)),

]

```

```
SEEDS = 5

print(f"{'Hyperparameter':15s} {'Setting':26s} {'Mean loss':>10s}
{'Mean acc':>9s} {'Solved':>8s}")

print("-" * 73)

rows = []

for group, label, kw in configs:

    ls, accs = zip(*(run(seed=s, **kw) for s in range(SEEDS)))

    row = (group, label, float(np.mean(ls)), float(np.mean(accs)),

          f"{sum(a == 100 for a in accs)}/{SEEDS}")

    rows.append(row)

    print(f"{row[0]:15s} {row[1]:26s} {row[2]:10.5f} {row[3]:8.1f}%
{row[4]:>8s}")
```

RESULT:

Hyperparameter	Setting	Mean loss	Mean acc	Solved
Learning rate	0.001	0.59650	70.0%	1/5
Learning rate	0.01	0.01982	100.0%	5/5
Learning rate	0.1 (baseline)	0.06983	90.0%	4/5
Learning rate	0.5	0.20804	85.0%	2/5
Hidden neurons	1	0.47764	75.0%	0/5
Hidden neurons	2	0.20848	70.0%	2/5
Hidden neurons	4 (baseline)	0.06983	90.0%	4/5
Hidden neurons	8	0.00025	100.0%	5/5
Activation	tanh (baseline)	0.06983	90.0%	4/5
Activation	relu	0.09586	95.0%	4/5
Activation	sigmoid	0.14028	85.0%	3/5
Activation	linear (no non-linearity)	0.69315	55.0%	0/5
Epochs	50	0.12036	95.0%	4/5
Epochs	100	0.07425	90.0%	4/5
Epochs	500 (baseline)	0.06983	90.0%	4/5
Epochs	2000	0.06937	90.0%	4/5
Optimizer	Adam, lr=0.1 (baseline)	0.06983	90.0%	4/5
Optimizer	SGD, lr=0.1	0.41260	65.0%	1/5
Optimizer	SGD, lr=0.5	0.01359	100.0%	5/5

CONCLUSION AND INFERENCE

5.1 Discussion

Learning rate. At 0.001 the network has simply not moved far enough in 500 epochs — mean loss stays near 0.60 and only 1 of 5 runs solves XOR. At 0.01 all 5 succeed. At 0.5 the updates overshoot, the loss oscillates, and only 2 of 5 converge. The learning rate has a sweet spot: too small underfits within the epoch budget, too large is unstable.

Number of hidden neurons. With 1 hidden neuron the network is mathematically incapable of XOR — no run exceeds 75 % (3 of 4 points), exactly what a linear boundary achieves. Two neurons is the theoretical minimum and does work, but only 2 of 5 seeds find it because the loss surface has narrow basins. With 8 neurons every seed converges and the loss drops to ~ 0.0002 : extra capacity gives the optimiser many more redundant paths to a correct solution, which is why over-parameterisation helps *optimisation* even when it is not needed for *representation*.

Activation function. The most instructive row. With a linear hidden activation the mean loss sits at exactly $0.69315 = \ln 2$ — the loss of a model that outputs 0.5 for everything — and no run ever solves XOR, because stacking linear layers collapses into a single linear layer at any width or depth. Non-linearity is not an optimisation detail; it is what makes an MLP more expressive than a perceptron. Among the non-linear choices tanh and ReLU are comparable (ReLU can occasionally lose a unit to the "dying ReLU" problem on so tiny a dataset), while sigmoid learns more slowly because its gradients are smaller.

Epochs. Most of the learning is over quickly at this learning rate — even 50 epochs reaches the same accuracy, and going from 500 to 2000 epochs barely moves the mean loss. Extra epochs sharpen confidence (probabilities move closer to 0 and 1) but do not rescue a run stuck in a bad minimum. Overfitting cannot be observed here: the 4 points are the entire input space, so memorising them *is* the correct behaviour.

Optimizer. Plain SGD at 0.1 is much weaker than Adam at the same rate (1/5 vs 4/5 solved), but SGD at 0.5 solves all 5. Adam adapts its effective step size per parameter, so it is far less sensitive to the initial learning-rate choice — which is why it is the usual default.

6. Conclusion

An MLP with a single hidden layer was implemented in two ways — the Keras Sequential API and the TensorFlow low-level API with `tf.Variable` weights and a manual `GradientTape` loop — and both learned the XOR function correctly, reproducing all four rows of the truth table with **100 % accuracy** and a final binary cross-entropy loss below 0.001.

The exercise confirms the central theoretical point: XOR is not linearly separable, so a single-layer perceptron cannot solve it, but one hidden layer with a non-linear activation suffices. This was shown empirically twice over — a network with 1 hidden neuron never exceeded 75 % accuracy, and a network with a linear hidden activation stayed pinned at a loss of $\ln 2$ regardless of width. The plotted decision boundary shows the resulting non-linear surface separating the two diagonals of the input square.

The hyperparameter study showed that the learning rate has a usable middle range, that extra hidden neurons improve the *reliability* of convergence rather than representational capability, that Adam is far less sensitive to the learning rate than plain SGD, and that beyond a point additional epochs only sharpen confidence. Comparing the two implementations makes clear that Keras is a convenience layer over exactly the operations written out by hand in the low-level version: the high-level API is shorter and less error-prone, while the low-level API exposes the forward pass, loss and gradient update that are otherwise hidden.